

SESSION 1 · AUTUMN WEEK 1

# Foundations of Medical Microbiology

*Introduction to Microbiology · The Bacterial Cell and Its Structures*

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**PART 1**      The Microbial World

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**PART 2**      How We Learned It — Germ Theory to Antibiotics

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**PART 3**      The Bacterial Cell Envelope

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**PART 4**      External and Internal Structures

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**PART 5**      Putting It Together in the Laboratory

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**PART 6**      Review Questions, Master Tables and Glossary

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SYNTHESIZED FROM

**Lecture 1 — “Medical Microbiology: a brief introduction”**

*Prof. Chao Chen (陈潮) · Zhejiang University School of Medicine · 59 slides + in-class MCQ set*

**Lecture 2 — “Bacterial cell structure”**

*Prof. Chao Chen (陈超) · Zhejiang University School of Medicine · 60 slides + in-class MCQ set*

**Jawetz, Melnick & Adelberg’s Medical Microbiology, 29e (Ch. 1–2) · Lippincott Illustrated  
Reviews: Microbiology · Brock Biology of Microorganisms · First Aid for the USMLE Step 1**

*The three texts cited on the course slides*

# How to Use This Guide

Microbiology is not learned the way pathology is. Pathology asks *what process is happening*; microbiology asks *which organism is this, and how do I prove it*. That difference shapes every page of this guide.

## The organizing idea

Almost everything in this session reduces to one sentence: **structure determines staining, staining determines classification, classification determines treatment**. A bacterium has a thick peptidoglycan layer → it retains crystal violet → we call it Gram-positive →  $\beta$ -lactams that block peptidoglycan cross-linking will probably work on it. Follow that chain and the subject stops being a list of facts.

Part 2 may look like history, but it is not decoration. The course leans on it heavily and so does the exam — and more importantly, every historical experiment in it is a piece of reasoning you still use. Koch's postulates are how we still attribute a disease to an organism; Pasteur's swan-necked flask is why we sterilise the way we do; Semmelweis is why you wash your hands.

## The seven kinds of boxes

| Box                  | What it contains                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------|
| High-Yield           | Repeatedly examined. If you memorize nothing else, memorize these.                                      |
| Exam Trap            | Where two concepts are confused, or the intuitive answer is wrong. This is where marks are lost.        |
| Lab Pearl            | Practical bench knowledge: which medium, which stain, what the plate actually looks like, and why.      |
| Mnemonic             | Used <i>only</i> for genuinely arbitrary lists. Anything with a mechanism should be understood instead. |
| Clinical Correlation | How the structure or mechanism shows up at the bedside.                                                 |
| Historical Anchor    | One line on who discovered this and why it mattered — the thread this course is built on.               |
| Mechanism chain      | Green numbered steps. Each carries a <i>Why</i> line explaining why that step forces the next one.      |

## Gram tags

Throughout this guide, organisms carry a coloured tag the moment they are named, so that classification becomes a visual reflex rather than something you look up:

- *Staphylococcus aureus* **GRAM +** — violet, the colour it stains
- *Neisseria meningitidis* **GRAM -** — pink, the colour of the safranin counterstain
- *Mycobacterium tuberculosis* **ACID-FAST** — neither; it needs its own stain

## Suggested study sequence

1. **First pass (≈2 h)**. Parts 1–2 straight through. This is the easy half; read it like a story.
2. **Second pass (≈2.5 h)**. Parts 3–4, slowly, with the mechanism chains. Do *not* skip the green boxes — the envelope is the conceptual core of the whole course.

3. **Third pass ( $\approx 45$  min).** Part 5. **Run the flowcharts actively** — cover the answers and walk an imaginary specimen down each branch. Re-reading a flowchart teaches nothing; using one teaches everything.
4. **Fourth pass ( $\approx 1.5$  h).** All 30 vignettes in Part 6, *closed-book*, then every explanation.
5. **Night before ( $\approx 40$  min).** High-Yield boxes + Exam Traps + master tables + glossary.

#### A NOTE ON THE SOURCES

This guide is built from both lecture decks in full, plus the textbooks the slides themselves cite. Where the lecture gives a number, **the lecture's number is the one used here** — including the two that differ from some textbooks: the Gram-positive wall is **20–80 nm thick containing 15–50 peptidoglycan layers**, and the Gram-negative wall is **10–15 nm with 1–2 layers**. Thickness and layer count are different quantities, and mixing them up is the single commonest slip on this topic.

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## PART ONE

# 01 The Microbial World

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Four kinds of organism can infect a human being, and they are not variations on a theme — a virus and a tapeworm differ more from each other than a tapeworm differs from you. Before any of the detail can be useful, you need a map of what lives where on that spectrum, because the group an organism belongs to already tells you how it reproduces, what drug class might touch it, and how a laboratory will look for it.

### BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Define a microbe and state the scale at which microbiology operates.
- Place viruses, bacteria, fungi and parasites on a structural spectrum, and say which are prokaryotic, which eukaryotic, and which are neither.
- Write a binomial name correctly and say why the convention matters.
- Explain what the normal flora is and why it is not merely harmless but necessary.

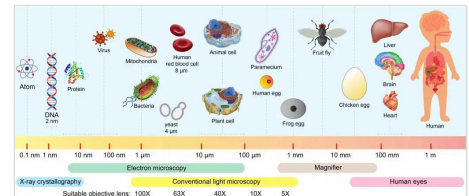
## 1.1 What Is a Microbe, and What Is Microbiology?

### DEFINITION

**Microbiology** breaks into three Greek parts: *micro* — too small to be seen with the naked eye; *bio* — life; *-ology* — the study of. A **microbe** (microorganism) is therefore any organism too small to be resolved without a microscope; most, though not all, are single cells.

Four facts frame the subject, and the lecture opens with them deliberately:

- **Microbes are the oldest life on Earth.** Microbial fossils date back more than **3.5 billion years**, to a time when the oceans regularly reached boiling point — hundreds of millions of years before any animal existed.
- **They are almost unimaginably numerous.** Millions fit in the eye of a needle.
- **Without microbes we could not eat or breathe.** They fix nitrogen, drive the carbon and sulphur cycles, generate atmospheric oxygen, digest our food and produce vitamin K in our gut.
- **Without us they would be fine.** The dependence is entirely one-way, which is a useful corrective to thinking of microbes primarily as pathogens.



**Fig 1.1** The scale of microbiology. Viruses occupy the tens-of-nanometre range, bacteria roughly 0.2–10  $\mu\text{m}$ , and eukaryotic microbes from a few  $\mu\text{m}$  upwards. The limit of the unaided human eye is about 100  $\mu\text{m}$  — which is precisely why none of this was known before the microscope.

### What counts as "alive"?

Cellular life — bacteria, fungi, parasites — shares a defined set of capabilities: it is bounded by a membrane, metabolises, grows, responds to its environment, and reproduces itself using its own machinery. **Viruses do none of these independently.** They possess genetic information but no metabolic apparatus, so they must borrow a host cell's ribosomes, enzymes and energy to replicate. This is why viruses sit awkwardly at the edge of the definition of life, and why *no antibiotic works on a virus* — antibacterial drugs attack structures and enzymes that viruses simply do not have.

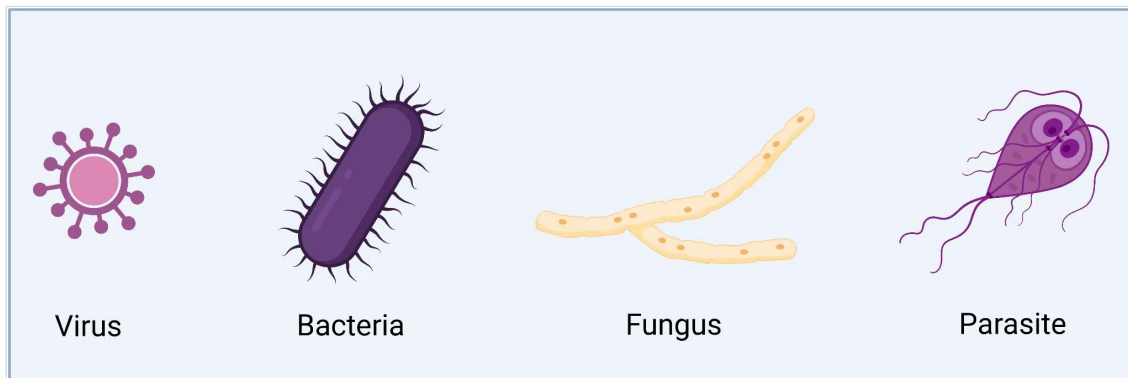
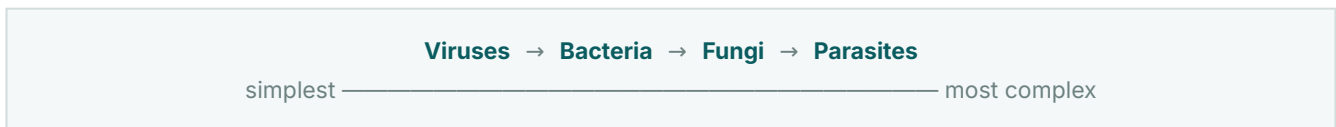
### DEFINITION

**Medical microbiology** is the subset of microbiology dealing with microorganisms that infect humans — viruses, bacteria, fungi and parasites. Its aims are to **identify** the organism, **diagnose** the disease it causes, **determine** its pathogenic mechanism, and **treat and prevent** recurrence.

Identification rests largely on **phenotypic features** — shape, structure, mode of reproduction, physiology and metabolism. Those five words are the reason Parts 3–5 of this guide exist.

## 1.2 The Four Groups of Medical Microbes

The lecture arranges the four groups along a single axis — **from the structurally simplest to the most complex**. Learn them in that order, because complexity predicts almost everything else: how the organism reproduces, whether it has its own metabolism to poison, and how big a target it presents to the immune system.



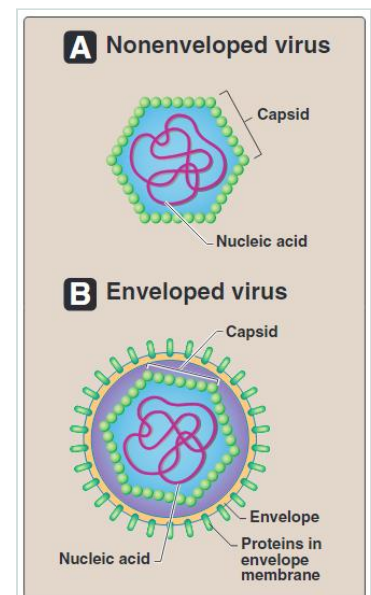
**Fig 1.2** The four groups drawn to relative scale. Note that the size difference between a virus and a parasite spans five orders of magnitude — from tens of nanometres to, in the case of some tapeworms, ten metres.

## Viruses

- **No cellular structure at all** — no membrane-bounded cytoplasm, no ribosomes, no metabolism.
- Genome is **either DNA or RNA, never both**, surrounded by a protein coat (capsid) and sometimes a lipid envelope.
- They carry genetic information but **require the host's cellular structures and enzymes** to express and replicate it — obligate intracellular parasites in the strictest sense.

## Bacteria

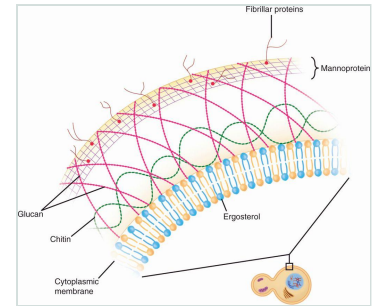
- **Prokaryotic** — no nucleus, no membrane-bound organelles. **All bacteria of medical importance are eubacteria** ("true bacteria"); the archaea, the other prokaryotic domain, contain no human pathogens.
- **Shapes:** sphere (coccus), rod (bacillus), corkscrew (spirochaete), and several intermediates — covered in detail in §3.3.
- **Cell wall and Gram staining** — the single most important classification tool in the entire subject.
- Divide by **binary fission** — one cell becomes two identical daughters. No mitosis, no meiosis, no sexual reproduction.
- May carry **plasmids**: small circular DNA molecules separate from the chromosome, often carrying antibiotic-resistance or virulence genes (§4.4).



**Fig 1.3** **A** Non-enveloped ("naked") virus: nucleic acid within a protein capsid only. **B** Enveloped virus: the same core, wrapped in a host-derived lipid membrane studded with viral proteins. Enveloped viruses are the more fragile of the two — lipid is dissolved by soap, alcohol and drying, which is why alcohol hand rub works well against influenza and SARS-CoV-2 but poorly against norovirus.

## Fungi

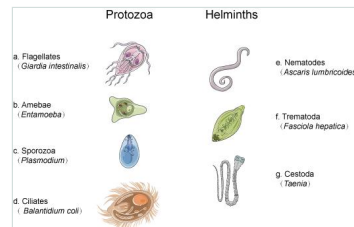
- **Eukaryotic** — true nucleus, mitochondria, endoplasmic reticulum. Saprophytic (living on dead organic matter) or parasitic.
- Roughly **200,000 fungal species** are known; about **100** have pathogenic potential; only a **few** are clinically important. The narrowing is worth noticing — fungi are overwhelmingly harmless.
- Disease is classified by **depth of invasion**: cutaneous, subcutaneous, and systemic. Systemic mycoses split into **true (primary) pathogens**, which infect healthy hosts, and **opportunists**, which require immunosuppression.
- Two non-infectious routes to harm: **toxic metabolic products** (mycotoxins such as aflatoxin) and **allergenic spores** (asthma, allergic bronchopulmonary aspergillosis).



**Fig 1.4** The fungal cell wall: **chitin** and  **$\beta$ -glucan** over a membrane containing **ergosterol** rather than cholesterol. All three are absent from human cells, which is exactly why they are the targets of antifungal drugs — echinocandins block  $\beta$ -glucan synthesis, azoles and amphotericin B target ergosterol.

## Parasites

- **Eukaryotic**, either unicellular (protozoa) or multicellular (helminths).
- Size spans **4  $\mu$ m to 10 m** — the widest range of any group in medicine.
- Bodies range from amorphous (amoeba) to complex, with differentiated organs.
- Distribution runs from remote regions to common community settings; transmission may be **person to person** or require **complex multi-host life cycles** with intermediate hosts and vectors.
- **Millions of prevalent cases** worldwide, with substantial mortality — malaria alone causes several hundred thousand deaths a year.



**Fig 1.5** The two halves of medical parasitology. **Protozoa** (left) are single eukaryotic cells — flagellates, amoebae, sporozoa, ciliates. **Helminths** (right) are multicellular worms — nematodes (roundworms), trematodes (flukes), cestodes (tapeworms). The distinction matters clinically: protozoa multiply inside the host, so a small inoculum can become overwhelming disease; most helminths do not, so worm burden is set by exposure.

**TABLE 1.1 THE FOUR GROUPS COMPARED — THE TABLE TO KNOW COLD**

| Feature         | Viruses                      | Bacteria                                | Fungi                                    | Parasites         |
|-----------------|------------------------------|-----------------------------------------|------------------------------------------|-------------------|
| Cell type       | <b>Acellular</b>             | <b>Prokaryotic</b>                      | <b>Eukaryotic</b>                        | <b>Eukaryotic</b> |
| Nucleus         | None                         | Nucleoid, no membrane                   | True nucleus                             | True nucleus      |
| Nucleic acid    | <b>DNA or RNA</b>            | DNA + RNA                               | DNA + RNA                                | DNA + RNA         |
| Ribosomes       | None                         | <b>70S</b> (50S + 30S)                  | 80S                                      | 80S               |
| Cell wall       | None (capsid $\pm$ envelope) | <b>Peptidoglycan</b>                    | <b>Chitin, <math>\beta</math>-glucan</b> | Usually none      |
| Membrane sterol | —                            | <b>None</b> (except <i>Mycoplasma</i> ) | <b>Ergosterol</b>                        | Cholesterol       |

| Feature        | Viruses                             | Bacteria                                              | Fungi                             | Parasites                              |
|----------------|-------------------------------------|-------------------------------------------------------|-----------------------------------|----------------------------------------|
| Own metabolism | <b>No</b> — obligate intra-cellular | Yes                                                   | Yes                               | Yes                                    |
| Reproduction   | Host-directed assembly              | <b>Binary fission</b>                                 | Budding, spores, hyphal extension | Simple division to complex life cycles |
| Treated with   | <b>Antivirals</b>                   | <b>Antibacterials</b>                                 | <b>Antifungals</b>                | <b>Antiparasitics</b>                  |
| Example        | Influenza, HIV, <i>Varicella</i>    | <i>S. aureus</i> <b>G+</b> , <i>E. coli</i> <b>G-</b> | <i>Candida albicans</i>           | <i>Plasmodium falciparum</i>           |

**HIGH-YIELD**

The classic exam question is simply "**which of these is eukaryotic?**" — and the answer is **fungi** (or parasites). Bacteria and archaea are prokaryotic; viruses are neither, because they are not cells at all. A second favourite: **ribosome size**. Bacteria are **70S**, human and fungal cells are **80S**; that difference is the entire reason aminoglycosides, tetracyclines and macrolides can kill a bacterium without killing you.

### 1.3 Naming Organisms — Binomial Nomenclature

**Carl Linnaeus** is the father of modern taxonomy, and his two-part naming system is still in use. Every organism gets a **genus** and a **species**:

*Staphylococcus aureus* (genus — capitalised) *aureus* (species — lower case)  
both italic, or underlined when handwritten

- After first mention, the genus may be abbreviated to its initial: *S. aureus*. Use the full name the first time.
- **Names are informative.** *Staphylo-* = grape cluster, *-coccus* = berry, *aureus* = golden — so the name literally describes grape-like clusters of spheres forming golden colonies. *Streptococcus* = chains of spheres. *Helicobacter pylori* = spiral rod living at the pylorus. Reading names as descriptions is free revision.
- Do **not** italicise informal or group names: staphylococci, streptococci, enterobacteria, the pneumococcus.

**EXAM TRAP**

Writing *Staphylococcus Aureus* or *staphylococcus aureus* loses marks in short-answer questions where the examiner is checking nomenclature. **Genus capitalised, species not, both italic.** Always.

## 1.4 The Normal Flora and the Human Microbiome

### DEFINITION

The **normal flora** (normal microbiota) is the community of microorganisms that permanently or transiently colonises body surfaces in health, without causing disease. The **microbiome** strictly means the collective *genomes* of that community, although the two words are now often used interchangeably.

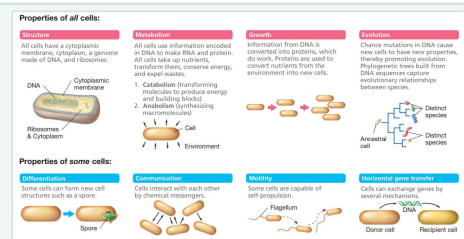
**Where it lives:** skin, mouth, nasopharynx, gut (the overwhelming majority, in the colon), and the vagina. **Where it does not:** blood, cerebrospinal fluid, the lower respiratory tract, the bladder and urine, and deep tissue — these are **normally sterile**, and that fact is the foundation of clinical microbiology. A single colony of *S. epidermidis* **G+** from a skin swab is meaningless; the same organism from a blood culture is either a contaminant or an infected line, and you must decide which.

### Why it is beneficial, not merely tolerated:

- **Colonisation resistance** — the resident flora occupies attachment sites and consumes nutrients, physically denying a foothold to incoming pathogens.
- **Nutrient synthesis** — gut flora produce vitamin K and several B vitamins.
- **Immune education** — early microbial exposure drives development of gut-associated lymphoid tissue and regulatory T cells.
- **Metabolic work** — fermentation of dietary fibre into short-chain fatty acids that nourish colonocytes.

### CLINICAL CORRELATION — WHAT HAPPENS WHEN YOU REMOVE IT

Broad-spectrum antibiotics destroy colonisation resistance. Two consequences appear constantly on wards and in exams: *Clostridioides difficile* **G+** **colitis** — a spore-forming anaerobe that survives the antibiotic (spores are not killed) and then overgrows the emptied colon, causing pseudomembranous colitis; and *Candida albicans* **overgrowth** — oral thrush or vulvovaginal candidiasis. Both are diseases of a *missing* flora rather than a newly acquired organism. Note also that this is the clearest argument against unnecessary antibiotic prescription.



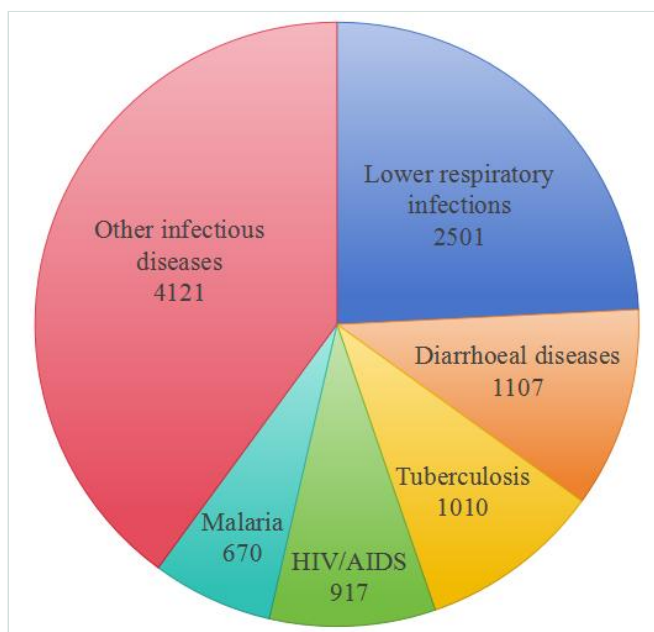
**Fig 1.6** The proportion of microbial to human cells in the body. Older textbooks quote 10:1; careful recounting puts the ratio closer to **1.3:1**

— roughly 38 trillion bacteria to 30 trillion human cells. Either way, you are at best half human by cell count, and the microbial genome dwarfs your own in gene number.

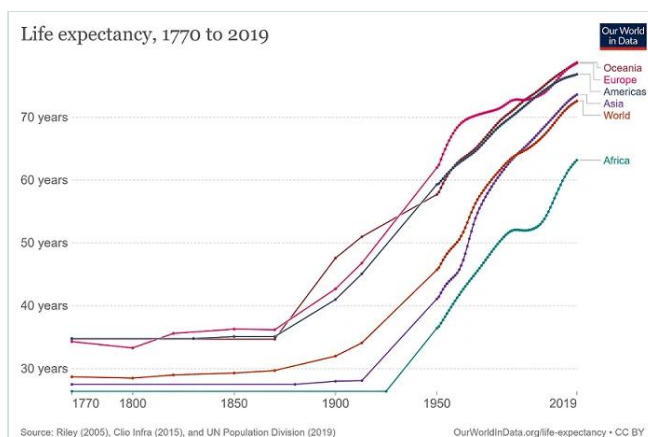
## 1.5 Infectious Disease Today — Why This Course Exists

### DEFINITION

An **infectious disease** is a disease condition caused by the presence or growth of infectious microbes.



**Fig 1.7** Global deaths by infectious cause. Lower respiratory infections, diarrhoeal disease, tuberculosis, HIV/AIDS and malaria dominate — and every one of them is preventable, treatable, or both.



**Fig 1.8** Life expectancy, 1770–2019. Medieval life expectancy was about **30 years**; the current world average is about **71**. Most of that gain came not from treating disease but from preventing infection — clean water, sanitation, vaccination.

In 2023, communicable diseases caused **10.33 million deaths — 17.2% of all deaths worldwide**. For scale, that is roughly equal to all cancers (10.57 million, 17.6%) and second only to cardiovascular disease (19.16

million, 31.9%). And the figure *excludes* infection-driven cancers and liver disease — hepatitis-related liver cancer and cirrhosis add a further 1.07 million.

**COVID-19** is the recent reminder that this is not a solved problem: an estimated **4.80 million** deaths globally in 2020 and **7.89 million** in 2021.

#### HISTORICAL ANCHOR

The lecture frames the whole historical section with a line from Charles Williams: *"I think in order to move forward into the future, you need to know where you've been."* That is the argument for Part 2 — every technique you will use in the laboratory was somebody's hard-won insight, and knowing the insight makes the technique impossible to forget.

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## PART TWO

# 02

## How We Learned It

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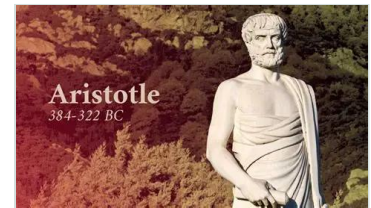
Three hundred years separate the first glimpse of a bacterium from the first antibiotic. What happened in between was not a series of lucky discoveries but a sequence of *arguments* — each one killing a belief that had seemed obvious. Read this Part as a chain of reasoning, because the reasoning is what you keep: Koch's postulates are still how we attribute a disease to an organism, and Semmelweis is still why you wash your hands between patients.

### BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Explain how Redi and Pasteur disproved spontaneous generation, and why controls were the decisive element in both experiments.
- State Koch's four postulates in order and give two situations in which they fail.
- Distinguish variolation, attenuated vaccines and inactivated vaccines, with the person and disease attached to each.
- Match each pioneer to the field they founded — a guaranteed MCQ.

## 2.1 Spontaneous Generation and Its Defeat

**Aristotle** introduced **spontaneous generation** — the idea that living things arise from non-living matter. He considered it "readily observable" that *aphids arise from the dew which falls on plants, fleas from putrid matter, mice from dirty hay*. The belief survived unchallenged for more than **2,000 years**, which is itself the lesson: an idea can feel obvious, fit everyday observation, and still be completely wrong.

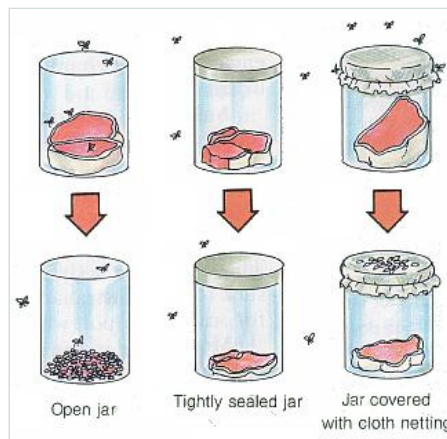


**Fig 2.1 Aristotle** (384–322 BC), who introduced the belief that life arises spontaneously from non-living matter. It went unchallenged for over 2,000 years.

### Francesco Redi (1626–1697) — the power of a control

Redi, an Italian physician, asked a deliberately small question: **where do maggots come from?** The prevailing answer was that rotting meat generated them. He set up jars of meat — one open, one sealed, one covered with cloth netting — and observed which produced maggots.

**Why the netting mattered so much:** a sealed jar alone proves nothing, because a defender of spontaneous generation can reply that sealing excluded the air or "vital force" needed for life to form. The netted jar admits air but excludes flies. When the netted meat stayed maggot-free, that objection died. **Redi's contribution was not the observation but the control** — and that is why the exam asks what process Redi studied: *spontaneous generation*.

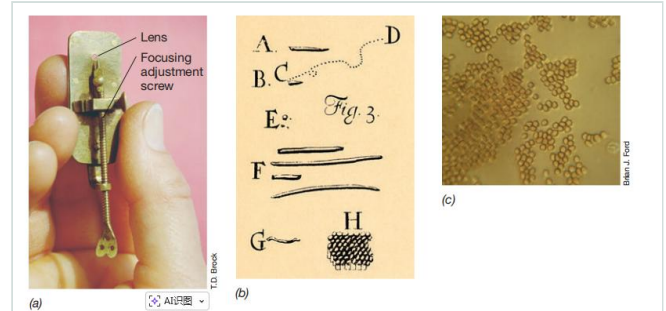


**Fig 2.2 Redi's experiment.** Three jars of meat: open, sealed, and covered with cloth netting. Maggots appeared only in the open jar. The **netted jar is the key** — air (and therefore any "vital force") still reached the meat, but flies did not. That single control eliminated the alternative explanation and made the result decisive.

## Antonie van Leeuwenhoek (1632–1723) — the first sight



**Fig 2.3 Antonie van Leeuwenhoek**, a Dutch draper — not a scientist by training. His trade required him to inspect cloth through magnifying lenses, and that interest in lens-making produced the first practical microscopes.



**Fig 2.4** His single-lens microscope and his drawings of what he called "**animalcules**" — the first human sight of bacteria, published to the Royal Society from 1676. He is the **grandfather of microbiology**.

### HIGH-YIELD

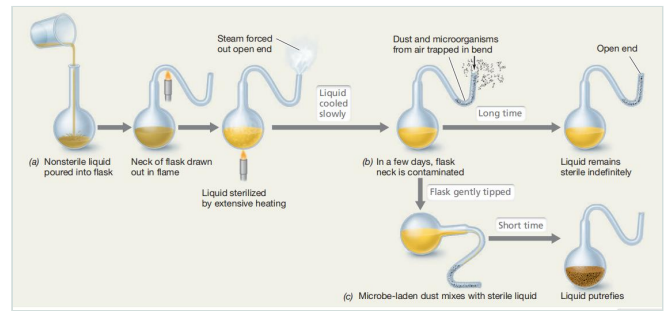
**Leeuwenhoek was the first person to see bacterial cells with a microscope.** This is a near-certain MCQ, and the distractors are always Pasteur, Koch and Snow. Note the distinction in titles the course uses: Leeuwenhoek is the **grandfather** of microbiology (he saw them), Pasteur the **father** (he explained them), Koch the **founder of modern bacteriology** (he proved they cause disease).

## Louis Pasteur (1822–1895) — the swan-necked flask

Pasteur, a French *chemist* rather than a physician, delivered the decisive blow. His problem was the same as Redi's but harder: microbes, unlike flies, are invisible and everywhere, so no netting will exclude them.



**Fig 2.5** The swan-necked flask. The long S-bend is open to the atmosphere at its tip, so air passes freely — but dust and airborne microbes settle in the bend and never reach the broth.



**Fig 2.6** The full sequence: broth is boiled (sterilising it), then left open through the swan neck — and stays clear indefinitely. Break or tilt the neck so that the trapped dust reaches the broth, and growth appears within days.

### ○ MECHANISM — WHY THE SWAN NECK SETTLED THE ARGUMENT

- 1 Nutrient broth is **boiled** in the flask, killing every organism present.  
*WHY: You must start from a genuinely sterile state, or any later growth could be from survivors rather than newcomers.*
- 2 The neck is drawn into a long **S-shaped bend** that remains **open to the air** at its tip.  
*WHY: This is the whole design. It grants the opponent's premise — air, and any hypothetical "vital force" it carries, reaches the broth freely — so the experiment can no longer be dismissed on that ground.*
- 3 Dust particles and the microbes riding on them **settle by gravity in the curve** and cannot climb up the far side.  
*WHY: Microbes are not gaseous; they travel on particles, and particles obey gravity. The bend is a physical trap that air passes but dust does not.*
- 4 The broth stays **sterile indefinitely** — months, even years.  
*WHY: If life arose spontaneously from nutrients plus air, this flask should teem with it. It does not.*
- 5 **Break the neck**, or tip the flask so broth touches the trapped dust — and growth appears within days.  
*WHY: This is the positive control that completes the proof. The broth was always capable of supporting life; it simply had not been seeded. Life comes only from pre-existing life — **biogenesis**.*

**NET RESULT** Spontaneous generation is dead and **biogenesis** is established. Every sterile technique in every laboratory and operating theatre descends from this one flask.

## Pasteurization

Pasteur's commercial problem was wine and beer spoiling. He showed that **spoilage organisms could be killed by heat that was not hot enough to evaporate the alcohol** — a high temperature for a short time. That compromise is **pasteurization**: not sterilisation (spores survive), but enough to destroy vegetative pathogens and spoilage organisms while preserving the product. Modern milk pasteurization — 72 °C for 15 seconds — is the same idea, and the reason milk-borne tuberculosis and brucellosis largely vanished from developed countries.

### EXAM TRAP

**Pasteurization ≠ sterilization.** Pasteurization reduces the microbial load and kills vegetative pathogens; sterilization kills *everything, including endospores*. Only autoclaving (121 °C at 15 psi for 15–20 min), dry heat, or chemical sterilants achieve true sterility. This distinction returns in §4.6 and in Lab 3.

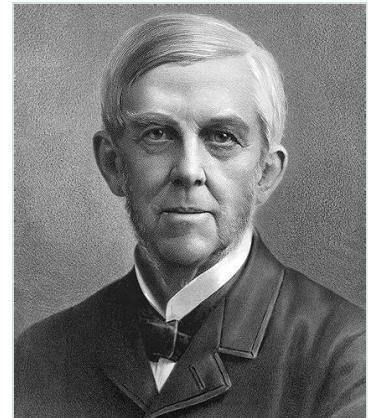
## 2.2 The Birth of Germ Theory

Knowing that microbes exist is one thing; proving they cause disease is another. Three physicians made that leap, and notably **none of them could see the organism they were arguing about** — each reasoned from pattern alone.

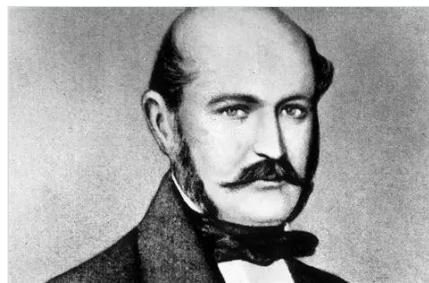
### Oliver Wendell Holmes (1809–1894)

**Puerperal (childbed) fever** is an infection of the female reproductive organs, usually contracted as a result of childbirth or miscarriage. Its mortality was **10–35%**. In 1843 Holmes published "The Contagiousness of Puerperal Fever", stating that the disease passes from patient to patient **via contact with their physician**. He was largely ignored and, in places, ridiculed.

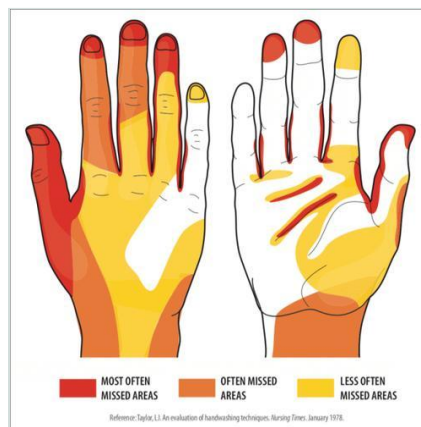
*The organism responsible is Streptococcus pyogenes* **GRAM +** — Group A Streptococcus. The same organism also causes strep throat, scarlet fever, streptococcal pneumonia and necrotizing fasciitis, a range you will meet again in the pyogenic cocci lecture.



**Fig 2.7 Oliver Wendell Holmes**, American physician. His 1843 paper "The Contagiousness of Puerperal Fever" argued that the disease was carried from patient to patient by the physician.



**Fig 2.8 Ignaz Semmelweis**, Hungarian physician working in Vienna.



**Fig 2.9** Areas most often missed during hand hygiene — fingertips, thumbs and between the fingers. Technique matters as much as frequency.



**Fig 2.10** Culture plates from an uncleaned hand, a washed hand, and a properly scrubbed hand. The visual argument that Semmelweis could not make in 1847.

#### MECHANISM — SEMMELWEIS'S REASONING, STEP BY STEP

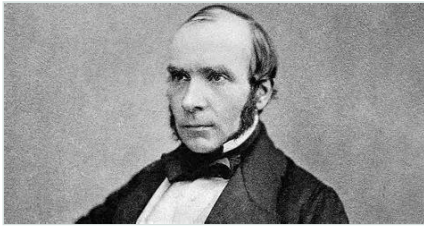
- 1 He observed that the maternity ward staffed by **medical students** had a far higher death rate from puerperal fever than the ward staffed by **midwives**.  
*WHY: A difference this large between two wards in the same hospital, with the same patients, cannot be explained by the disease itself. Something about the attendants must differ.*
- 2 He identified the one systematic difference: the students performed **postmortem examinations** before attending deliveries. The midwives did not.  
*WHY: This isolates the variable. Everything else — ward, season, patient population — was shared.*
- 3 He proposed that "cadaverous material" was being transferred on the students' hands from the dissection room to the birth canal.  
*WHY: He had no germ theory and no microscope evidence — he inferred an invisible transmissible agent purely from the epidemiological pattern.*
- 4 He introduced a policy of **handwashing with chlorinated lime** between the autopsy room and the ward.  
*WHY: Chlorinated lime removes the odour of decomposition — his crude marker for the invisible material. It also happens to be an effective disinfectant, which is why it worked.*

### 5 Mortality on the students' ward fell dramatically, to match the midwives'.

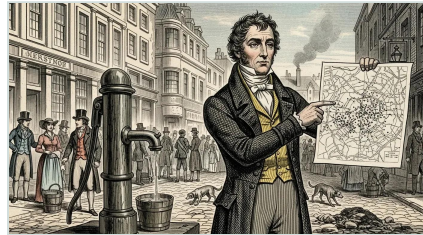
**WHY:** *The intervention confirmed the hypothesis. Yet Semmelweis was rejected by his colleagues — he could not explain why it worked, and the claim that doctors were killing their own patients was unwelcome.*

**NET RESULT** **Hand hygiene remains the single most effective measure for preventing healthcare-associated infection.** The WHO codifies it as the **"Five Moments"**: before patient care · before an aseptic task · after exposure to blood or body fluids · after patient care · after contact with patient surroundings.

## John Snow (1813–1858) — the father of epidemiology



**Fig 2.11** John Snow, British physician, founder of **epidemiology**.



**Fig 2.12** The Broad Street pump, London, 1854. Snow mapped cholera deaths, found they clustered around this one pump, and had the handle removed. The outbreak ended.



**Fig 2.13** *Vibrio cholerae* **GRAM -**, colorized scanning electron micrograph. A comma-shaped rod with a single polar flagellum.

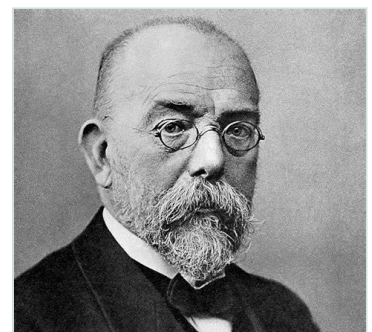
In 1854, Snow investigated a cholera outbreak in Soho, London. He **mapped every death** and traced the cluster to a single water pump on **Broad Street**. He persuaded the authorities to remove the pump handle, and the outbreak ended. He did this **29 years before Koch isolated *Vibrio cholerae*** — he never saw the organism, and reasoned entirely from the geography of disease. His work drove reform of water and waste systems worldwide.

### CLINICAL CORRELATION — CHOLERA

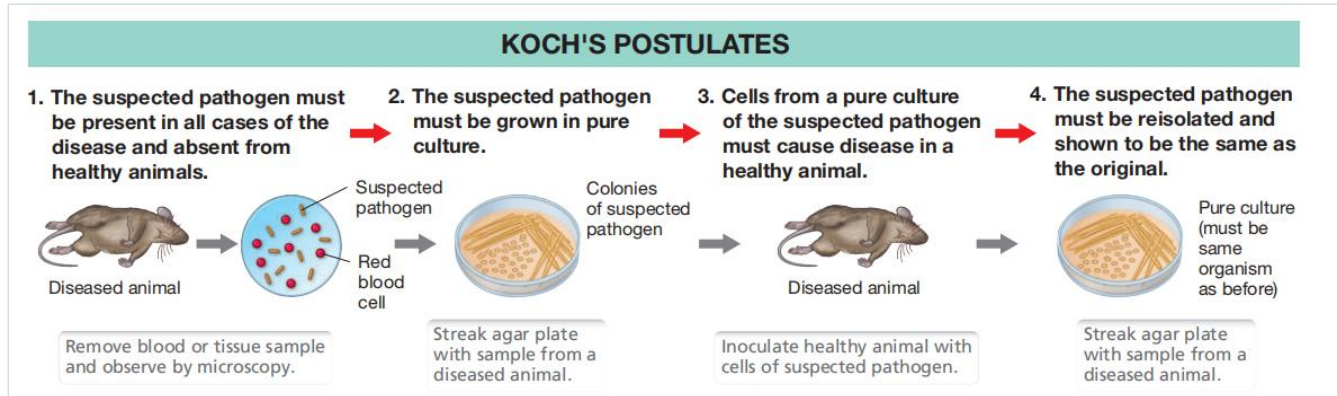
*Vibrio cholerae* **GRAM -** causes infectious gastroenteritis. Its enterotoxin damages the mucosal epithelium of the small intestine, producing **massive watery diarrhoea** ("rice-water stool") and **rapid dehydration** — patients can lose litres per hour and die of hypovolaemic shock within a day. Transmission is by **contaminated water or food**, exactly as Snow deduced. It killed tens of millions in 19th- and 20th-century epidemics and still affects **3–5 million people annually**, with about **100,000 deaths**. The treatment that saves most lives is not an antibiotic but **oral rehydration solution**.

## 2.3 Koch's Postulates — and Where They Break

Koch supplied what Holmes, Semmelweis and Snow lacked: a **formal method** for proving that a *particular* organism causes a *particular* disease. Before this, "germs cause disease" was a hypothesis; after it, causation became demonstrable one organism at a time.



**Fig 2.14 Robert Koch (1843–1910)**, German physician, **founder of modern bacteriology**. He identified the causative agents of **anthrax, tuberculosis and cholera**, and introduced **agar** and the **Petri dish** for purifying colonies — techniques you will use in every lab session of this course.



**Fig 2.15** Koch's postulates as a closed loop: isolate from the diseased animal → grow in pure culture → inoculate a healthy animal → recover the same organism from the newly diseased animal. The loop closing is the proof.

### ○ THE FOUR POSTULATES — AND WHY EACH IS NECESSARY

- 1 The same microorganism must be present in every case of the disease.**  
**WHY:** *If the organism is absent from even one genuine case, it cannot be the necessary cause. This is the screening step.*
- 2 The microorganism must be isolated from the diseased host and grown in pure culture.**  
**WHY:** *"Pure" is the load-bearing word. A mixed culture proves nothing, because any of several organisms present could be the culprit. This is exactly why Koch had to invent solid agar media and the Petri dish first — a liquid culture cannot be separated into single colonies.*
- 3 The pure-cultured organism must cause the same disease when inoculated into a healthy, susceptible animal.**  
**WHY:** *This converts correlation into causation. Presence alone could be coincidence or consequence; reproducing the disease from a pure culture makes the organism sufficient.*
- 4 The organism must be re-isolated from the experimentally infected animal and shown to be identical to the original.**  
**WHY:** *This closes the loop and excludes contamination — it proves the disease was caused by the organism you inoculated, not by something introduced during the procedure.*

### EXAM TRAP — THE MCQ THEY ALWAYS ASK

"Which statement is **NOT** part of Koch's postulates?" The answer is always the one claiming the organism can be identified from a **mixed culture**. Postulate 2 requires a **pure** culture — that is the entire point, and the reason Koch had to invent agar plates before he could invent the postulates.

## Where the postulates fail

The postulates were formulated for culturable bacteria in the 1880s, and modern microbiology has found their limits. Knowing these exceptions is what separates a memorised answer from an understood one:

- **Some organisms cannot be grown in pure culture.** *Mycobacterium leprae* **ACID-FAST** and *Treponema pallidum* **GRAM -** have never been cultured on artificial media; all viruses require living cells. Postulate 2 fails outright.
- **Asymptomatic carriers exist.** Many healthy people carry *Neisseria meningitidis* **GRAM -** or *S. aureus* **GRAM +** without disease, so the organism's presence is not sufficient — host factors matter.
- **No suitable animal model.** Some human pathogens infect no other species, so postulate 3 cannot be tested ethically or practically.
- **One organism, many diseases; one disease, many organisms.** *S. aureus* causes boils, endocarditis, osteomyelitis and toxic shock; pneumonia is caused by dozens of different species.
- **Polymicrobial and microbiome-associated disease.** Periodontitis and bacterial vaginosis arise from community shifts, not a single agent.

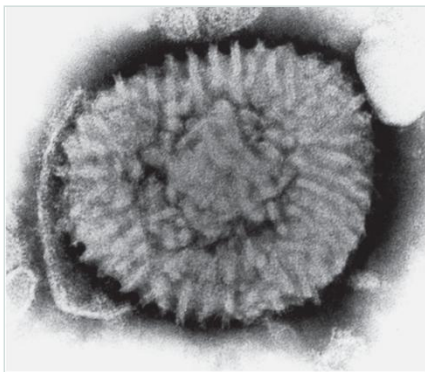
**Molecular Koch's postulates** (Falkow, 1988) update the logic for the genomic era: the *gene* associated with virulence should be present in pathogenic strains and absent from avirulent ones; inactivating it should abolish virulence; and restoring it should restore virulence. Same reasoning, applied to a gene instead of an organism.

### CLINICAL CORRELATION — ANTHRAX, THE ORGANISM KOCH PROVED IT ON

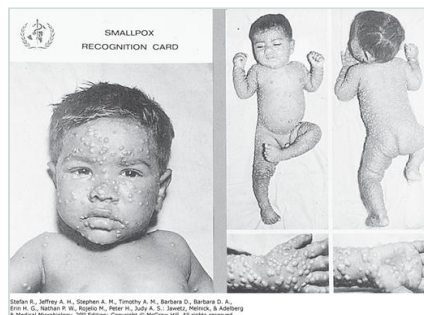
*Bacillus anthracis* **GRAM +** is a large spore-forming rod common in soil. Three clinical forms follow the route of entry: **cutaneous** (through a skin lesion — a painless black eschar), **gastrointestinal** (eating infected meat), and **pulmonary/inhalational** (the most lethal, with haemorrhagic mediastinitis). Its **endospores survive for centuries**, which is what makes it both a persistent agricultural problem and a biological weapon. Pasteur developed the first anthrax vaccine; Koch used it to prove the postulates. You will meet its spores again in §4.6.

## 2.4 Vaccination — Variolation to Modern Vaccines

### Smallpox and variolation



**Fig 2.16** Variola virus, electron micrograph. Smallpox caused **300–500 million deaths** in the 20th century alone, with a **30–35% mortality rate**.



**Fig 2.17** WHO smallpox recognition card. Survivors were commonly left with **disfiguring scars and blindness**.



**Fig 2.18** Chinese text on inoculation. Variolation was developed in the **Song dynasty** and well established by the **Ming and Qing** — centuries before Jenner.

#### DEFINITION

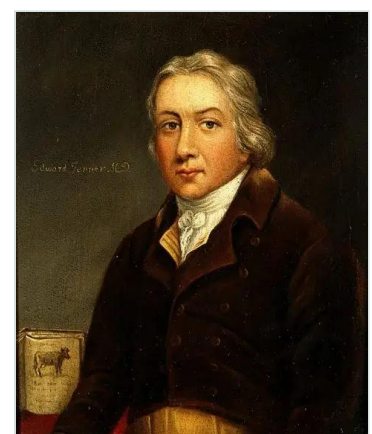
**Variolation** is deliberate inoculation with material from a **smallpox** lesion — dried scabs or vesicle fluid — placed into the nostril or under the skin, in the hope of producing mild disease and lasting immunity. **Vaccination** (from *vacca*, cow) uses a **different, milder organism** to induce cross-protective immunity.

The Chinese technique transferred **patients' pox vesicle fluid or crusts to a child's naris**. Success depended heavily on the physician's skill, and the practice carried real mortality — but far less than natural smallpox. It reached **Russia in 1688**, then Turkey and Britain.

### Edward Jenner (1749–1823) — the father of immunology

Jenner was aware of a belief among farm workers: **if you had caught cowpox, you would not get smallpox**. He tested it by inoculating cowpox — which causes only mild symptoms — and demonstrating protection against smallpox. The conceptual leap is that **a different, related organism can teach the immune system** without the danger of the real pathogen.

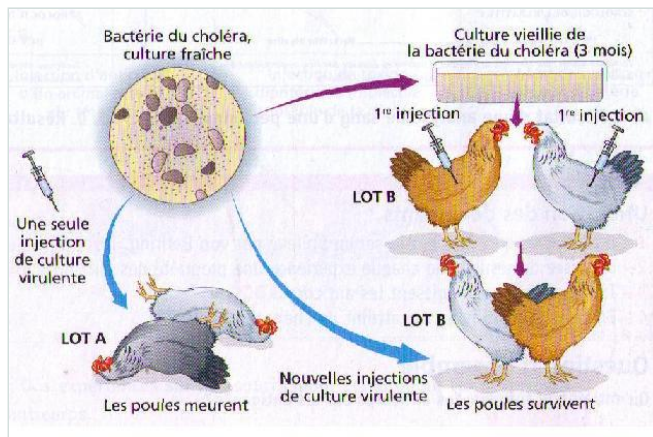
**The payoff:** through a worldwide vaccination programme, the WHO declared smallpox **eradicated in 1979** — the only human infectious disease ever eliminated from the planet.



**Fig 2.19** Edward Jenner, British physician. At the time, an estimated

20% of the population died of small-pox.

## Pasteur and the two classes of vaccine



**Fig 2.20** The chicken cholera experiment. **Fresh culture kills; old (attenuated) culture does not — and the survivors are then immune to fresh culture.** The discovery was an accident: an assistant's neglect left cultures to age over a holiday.



**Fig 2.21** Pasteur and the rabbits used to produce rabies vaccine. He **desiccated the spinal cords of infected rabbits** until the preparation became almost non-virulent — the first **inactivated** vaccine.

**MECHANISM — HOW ATTENUATION WAS DISCOVERED, AND WHY IT WORKS**

- 1 Pasteur was working on chicken cholera using **fresh cultures**, which reliably killed the birds.  
*WHY: A fresh culture is full of fully virulent organisms, so it reproduces the disease exactly.*
- 2 Through **an assistant's neglect**, old cultures were used instead.  
*WHY: The accident is the point — this was not a planned experiment, and Pasteur's contribution was recognising what the failure meant.*
- 3 The chickens **fell ill but recovered**. The bacteria had become **weakened (attenuated)** during prolonged storage.  
*WHY: Repeated subculture or ageing outside the host selects for variants adapted to laboratory conditions rather than to invading tissue; virulence genes are lost or downregulated.*
- 4 When those same recovered chickens were later given **fresh, fully virulent culture, they survived**.  
*WHY: The attenuated organism still carried the same surface antigens, so the immune system had already built memory B and T cells against them. Antigenic identity is preserved while virulence is lost — that is the entire principle of a live attenuated vaccine.*
- 5 Pasteur named these artificially weakened pathogens **"vaccines" (1879)**, in deliberate homage to Jenner's cowpox.  
*WHY: He generalised a one-off observation about cowpox into a manufacturable principle applicable to any organism — and went on to apply it to anthrax and rabies.*

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**NET RESULT** Two classes of vaccine, both from Pasteur: **live attenuated** (weakened but alive — chicken cholera, anthrax) and **inactivated/killed** (rabies, made by desiccating infected spinal cord).

**CLINICAL CORRELATION — RABIES AND POST-EXPOSURE PROPHYLAXIS**

Rabies is a viral infection of the brain caused by **lyssaviruses**, generally transmitted by bites from infected dogs or bats. It is **almost 100% fatal once symptoms begin** — but **preventable if vaccine or immunoglobulin is given within about 10 days of exposure**. That narrow window exists because the virus travels slowly up peripheral nerves to the CNS, and vaccination can outrun it. In 1885 Pasteur vaccinated **Joseph Meister**, a nine-year-old bitten by a rabid dog; the boy lived, and hundreds of bite victims worldwide were saved subsequently. Rabies still causes **26,000–55,000 deaths per year**.

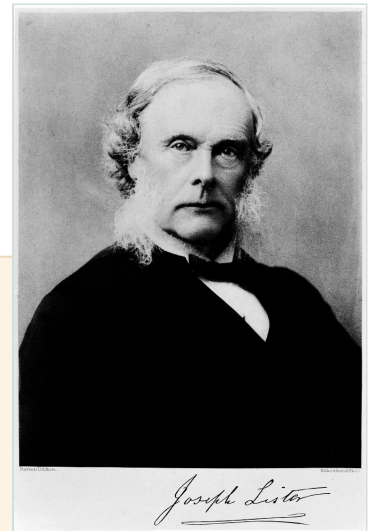
## 2.5 Antisepsis and the First Antimicrobials

### Joseph Lister (1827–1912) — antiseptic surgery

**Gangrene and wound rotting** were the dominant problem of surgery — operations frequently succeeded technically and then killed the patient by sepsis. Lister heard of Pasteur's work and reasoned that if microbes in the air spoil broth, microbes entering a wound must spoil tissue in the same way. He introduced **carbolic acid (phenol)** to clean wounds and sterilise instruments, and mortality in his wards collapsed.

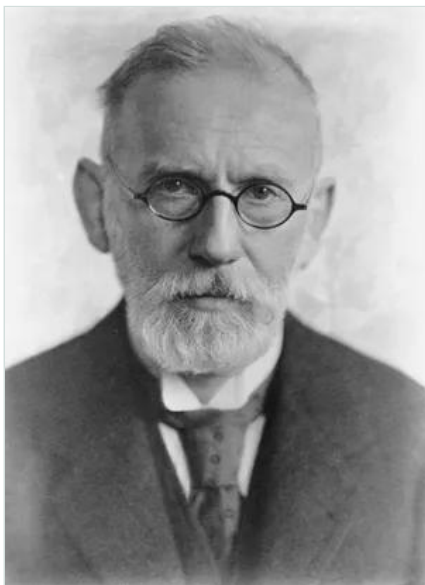
#### LAB PEARL — THREE WORDS THAT ARE NOT SYNONYMS

**Antisepsis** — killing or inhibiting microbes on *living tissue* (skin prep, hand rub). **Disinfection** — the same on *inanimate surfaces*, killing vegetative organisms but not necessarily spores. **Sterilisation** — destruction of *all* microbial life including endospores. Lister introduced antisepsis; Koch's autoclave gave sterilisation. Exams exploit the overlap relentlessly, and Lab 3 of this course is built on the distinction.

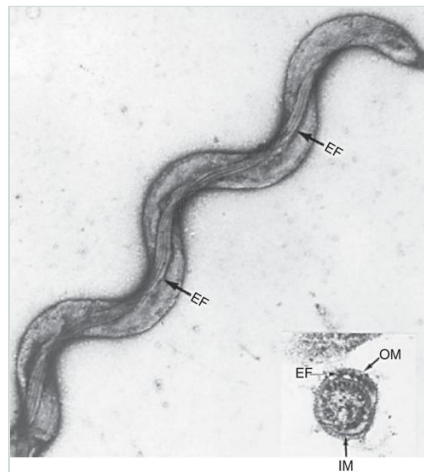


**Fig 2.22** Joseph Lister, British surgeon, **father of modern surgery**. Listerine mouthwash is named after him — though he did not invent it.

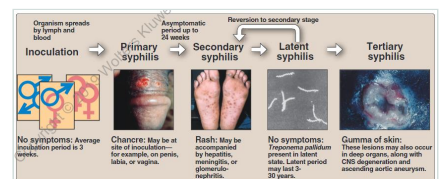
### Paul Ehrlich (1854–1915) — the magic bullet



**Fig 2.23** Paul Ehrlich, German scientist, who **initiated and named the concept of chemotherapy**.



**Fig 2.24** *Treponema pallidum* **GRAM –**, the spirochaete causing syphilis. Too thin to be seen on Gram stain — it requires dark-field microscopy or silver stain.



**Fig 2.25** Syphilis develops in **distinct stages** — primary chancre, secondary rash, latency, tertiary disease — separated by symptom-free intervals.

Ehrlich's idea was that a chemical could be found which binds a parasite selectively and spares the host — a "magic bullet". His laboratory synthesised **Arsphenamine** (Salvarsan, compound 606), the **first effective medical treatment for syphilis**. It was very unstable and oxygen-sensitive, and was later replaced by penicillin.

### Gerhard Domagk (1895–1964) — sulfonamides

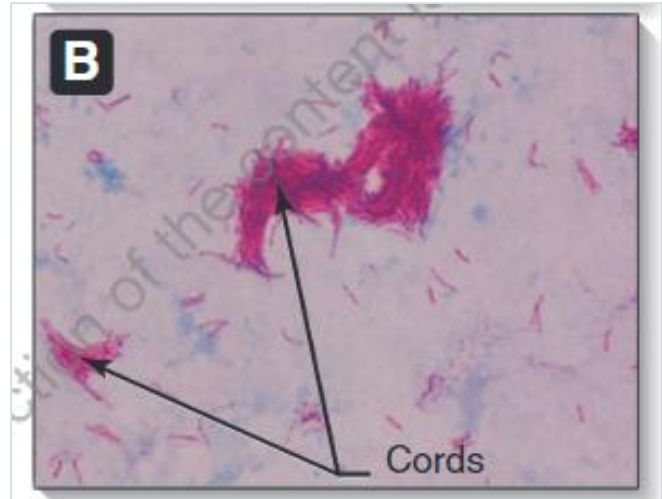
Domagk discovered that the **sulfanilamide portion of the dye Prontosil** was effective against bacteria. Sulfanilamide was later largely replaced by penicillin, but the chemical lineage mattered enormously: it led on to the antituberculous drugs **thiosemicarbazone and isoniazid**.

**CLINICAL CORRELATION — TUBERCULOSIS**

*Mycobacterium tuberculosis* **ACID-FAST** infects an estimated **one third of the world's population**. It is typically a lung disease but can affect almost any organ. During the **latent stage** it sits encapsulated and dormant within a granuloma, then **reactivates** when immunity falls — which is why HIV co-infection is so devastating and why latent TB is screened for before immunosuppressive therapy.



**Fig 2.26 Ziehl-Neelsen acid-fast stain:** *M. tuberculosis* appears as slender **red bacilli** against a **blue background**. A Gram stain would show almost nothing — see §3.10.

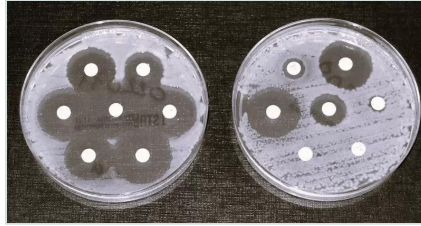


**Fig 2.27 Cord formation** — the parallel "serpentine cords" of *M. tuberculosis*, caused by cord factor (trehalose dimycolate), a virulence factor of the cell wall.

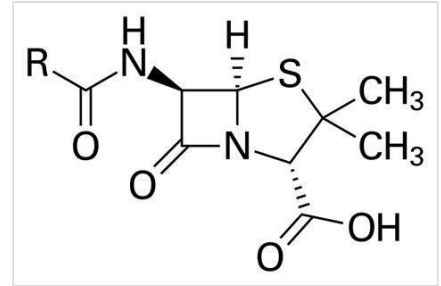
## Fleming, Florey and Chain — penicillin



**Fig 2.28** Alexander Fleming, Scottish bacteriologist, in his laboratory.



**Fig 2.29** Disk diffusion susceptibility testing — the descendant of Fleming's observation. Clear **zones of inhibition** around a disk mean the organism is susceptible to that drug.



**Fig 2.30** The penicillin nucleus, with its  **$\beta$ -lactam ring** — the four-membered ring that attacks peptidoglycan cross-linking, and the bond that  $\beta$ -lactamase enzymes cleave.

**Fleming (1881–1955)** observed that colonies of *Staphylococcus aureus* **GRAM +** **disappeared on plates contaminated with a mould**. He identified the mould as a *Penicillium* and named the active substance in its "mould juice" **penicillin**. But his clinical trials had only limited success, he found extraction to large scale too difficult, and he essentially abandoned it.

**Howard Florey** (Australian pathologist) and **Ernst Chain** (British biochemist) continued the work at Oxford. They **found a process for mass production** and **proposed the correct chemical structure**. Without them penicillin would have remained a curiosity — which is why all three shared the 1945 Nobel Prize.

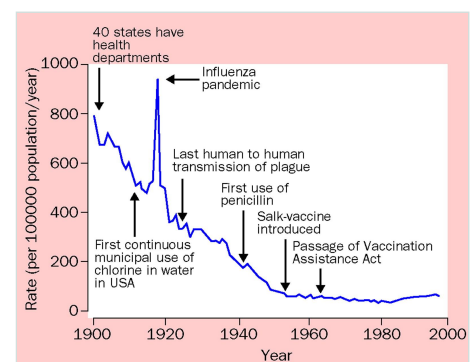
### HIGH-YIELD — DISCOVERY IS NOT ENOUGH

Fleming **discovered** penicillin in 1928; Florey and Chain **made it a medicine** in the 1940s. Exams ask for all three names, and the twelve-year gap is the lesson — a finding without a manufacturing route saves nobody. Note also that penicillin's target, **peptidoglycan cross-linking**, is the subject of §3.5: the drug works because that structure exists in bacteria and in nothing of ours.

## 2.6 The Three Determinants of Our Success

The lecture closes with the three main determinants of our success in limiting infectious disease. They are worth memorising in order, because an MCQ asks which item in a list is **not** one of them:

1. **Improved hygiene conditions** — clean water, sanitation, hand-washing, food safety.
2. **Vaccination** — from Jenner's cowpox to modern conjugate and mRNA vaccines.
3. **Antibiotics** — from Salvarsan and sulfonamides to the modern armamentarium.



**Fig 2.31** Infectious disease mortality in the USA, 1900–2000. Note **when** the curve falls:

**MNEMONIC**

**H-V-A — Hygiene, Vaccines, Antibiotics.** **Population** is the classic distractor and is *not* a determinant of success — if anything, dense populations make transmission easier.

the steep decline begins with municipal chlorination and public health departments, decades before the first use of penicillin. Antibiotics accelerated a fall that sanitation had already begun.

**EXAM TRAP — WHO FOUNDED WHAT**

The single most reliable MCQ in this lecture pairs a person with a field or process. Learn this list as a unit:

**Leeuwenhoek** — first to see bacteria · **Redi** — disproved spontaneous generation · **Jenner** — first Western vaccine, father of immunology · **Semmelweis** — handwashing · **Snow** — **epidemiology** · **Pasteur** — pasteurization, fermentation, attenuated vaccines, father of microbiology · **Koch** — postulates, modern bacteriology · **Lister** — **antisepsis** · **Ehrlich** — chemotherapy · **Domagk** — sulfonamides · **Fleming** — penicillin · **Linnaeus** — **taxonomy**.

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## PART THREE

# 03 The Bacterial Cell Envelope

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Everything that matters clinically about a bacterium is decided within about 50 nanometres of its surface. The envelope determines how it stains, therefore how it is classified; what toxins it releases when it dies; which antibiotics can reach and destroy it; and what the immune system sees. If you understand one thing in this course completely, make it this Part.

### BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Draw and label the Gram-positive and Gram-negative envelopes from memory.
- Explain the Gram stain *mechanistically* – why each of the four reagents does what it does.
- Describe peptidoglycan synthesis and name the drugs that block each step.
- Trace LPS from a dying Gram-negative cell to refractory septic shock.
- Explain why *Mycobacterium* and *Mycoplasma* break the Gram system, and what to do instead.

### 3.1 How We Tell Bacteria Apart

The lecture opens with the only question that matters at the bedside: **"How do I know what bacterium makes my patient ill?"** Everything in this Part and the next is an answer to it.

Bacterial species are differentiated by **four criteria**, and they are deliberately listed in order of how quickly they give an answer:

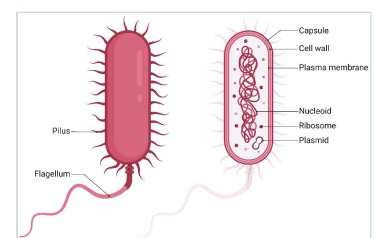
1. **Morphology** — shape, size and arrangement. Minutes, from a stained smear.
2. **Composition** — the cell envelope and the other structures. This is what the Gram stain actually reads, and it is the subject of the rest of Parts 3 and 4.
3. **Metabolism and growth characteristics** — what it eats, what it needs, what it makes. Hours to days, in culture.
4. **Genetics** — sequencing and molecular methods. Definitive, and increasingly fast.

#### HIGH-YIELD

This session deals with criteria **1 and 2** — morphology and composition. Criteria 3 and 4 (growth, metabolism, genetics) are the next session. That division is not arbitrary: morphology and composition are what you can have *in the first hour*, which is exactly when empirical therapy must be chosen.

#### The map of the cell — basic versus specific structures

The lecturer divides bacterial structures into two classes, and returns to this slide before every new structure. **Which class a structure belongs to is itself an exam question**, because it is the same as asking whether the structure is essential:



**Fig 3.1** The structural map of a bacterial cell. Everything in Parts 3 and 4 is one label on this diagram. Learn the picture first and the details will have somewhere to attach.

**TABLE 3.1** BASIC VERSUS SPECIFIC STRUCTURES

| Class                                                 | Structures                                                    | What the classification means                                                                                                                                                |
|-------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Basic</b> — present in essentially every bacterium | Cell membrane · cell wall · cytoplasm · chromosome · ribosome | Remove any one and the cell dies. This is also why they make the best <b>antibiotic targets</b> — the drug cannot be evaded by simply discarding the structure.              |
| <b>Specific</b> — present in some species only        | Plasmid · flagellum · pilus · capsule · inclusion · endospore | Not essential for viability — but this is where most <b>virulence factors</b> and most transferable <b>resistance</b> live. Optional for the cell, decisive for the patient. |

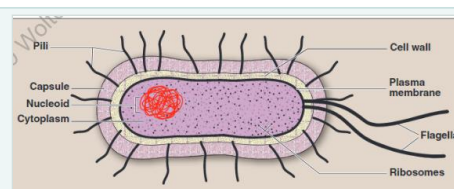
**EXAM TRAP**

"Not essential for viability" is **not** the same as "not important". A capsule-less *S. pneumoniae* grows perfectly well on a plate and is **avirulent** in a patient. The structures that a bacterium can live without are precisely the ones that let it live *in you*.

## 3.2 Prokaryote versus Eukaryote

**DEFINITION**

A **prokaryote** (*pro* = before, *karyon* = nucleus) is a cell whose genetic material is **not enclosed in a nuclear membrane** and which lacks membrane-bound organelles. All bacteria are prokaryotes; **all bacteria of medical importance are eubacteria**.



**Fig 3.2** The bacterial cell. Note what is present — cell wall, plasma membrane, nucleoid, ribosomes, cytoplasm — and, just as importantly, what is absent: no nuclear membrane, no mitochondria, no endoplasmic reticulum, no Golgi, no lysosomes.

**TABLE 3.2** PROKARYOTIC VERSUS EUKARYOTIC CELLS

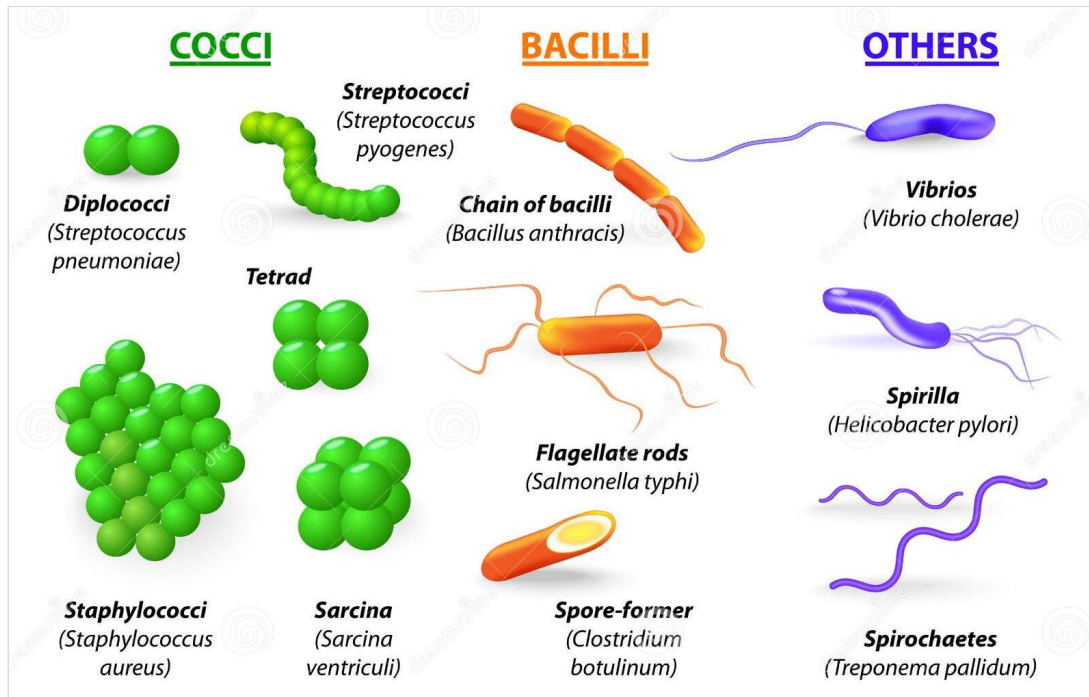
| Feature             | Prokaryote (bacteria)                                                                             | Eukaryote (fungi, parasites, human)                |
|---------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Size of cell        | Typically <b>0.2–2.0 µm</b> in diameter                                                           | Typically <b>10–100 µm</b> in diameter             |
| Nucleus             | <b>Nucleoid</b> — no membrane, no nucleolus                                                       | True nucleus with membrane and nucleolus           |
| Chromosome          | <b>One circular</b> double-stranded DNA molecule, haploid                                         | Multiple linear chromosomes, usually diploid       |
| Histones            | Absent (nucleoid-associated proteins instead)                                                     | Present                                            |
| Organelles          | <b>None</b> — no mitochondria, ER, Golgi, lysosomes                                               | All present                                        |
| Respiration site    | <b>Cytoplasmic membrane</b> (it does the mitochondrion's job)                                     | Mitochondrion                                      |
| Ribosomes           | <b>70S</b> = 50S + 30S subunits                                                                   | <b>80S</b> = 60S + 40S subunits                    |
| Cell wall           | <b>Peptidoglycan (murein)</b>                                                                     | Chitin/glucan (fungi); none (animal cells)         |
| Flagella            | Simple — <b>two protein building blocks</b> (flagellin plus motor)                                | Complex — <b>multiple microtubules</b> (9+2)       |
| Glycocalyx          | Present as a <b>capsule or slime layer</b>                                                        | Present in some cells that <b>lack a cell wall</b> |
| Cytoplasm           | <b>No cytoskeleton</b> , no cytoplasmic streaming                                                 | Cytoskeleton present; cytoplasmic streaming        |
| Membrane sterols    | <b>Absent</b> — except <i>Mycoplasma</i>                                                          | Present (cholesterol / ergosterol)                 |
| Cell division       | <b>Binary fission</b>                                                                             | Mitosis / meiosis                                  |
| Sexual reproduction | <b>No meiosis</b> — DNA fragments transferred instead (conjugation, transformation, transduction) | Involves <b>meiosis</b>                            |
| Plasmids            | <b>Common</b>                                                                                     | Rare                                               |

**HIGH-YIELD — "WHICH IS NOT FOUND IN A BACTERIAL CELL?"**

This exact MCQ appears in the Lecture 2 question set. Bacteria **do** have ribosomes, DNA and cytoplasm; they do **not** have **mitochondria**, **Golgi**, endoplasmic reticulum, lysosomes or a nuclear membrane. The underlying principle is simple: **a prokaryote has no internal membranes**, so any structure defined by a membrane is absent — and the cytoplasmic membrane must therefore do all their jobs, including respiration.

### 3.3 Size, Shape and Arrangement

Most medically important bacteria are **0.2–2 µm wide and 1–10 µm long**. They are described by **three independent features**, and a complete morphological description gives all three: **shape, arrangement, and Gram reaction**. "Gram-positive cocci in clusters" is a usable preliminary diagnosis; "cocci" alone is not.



**Fig 3.3** Bacterial shapes and arrangements with representative genera. **Cocci** (left): diplococci, chains, tetrads, sarcinae, clusters. **Bacilli** (centre): single rods, chains, coccobacilli, palisades. **Others** (right): vibrios, spirilla, spirochaetes.

#### Shape

- **Coccus** — spherical. Plural *cocci*.
- **Bacillus** — rod-shaped. Note the trap: lower-case *bacillus* is a shape; capitalised italic *Bacillus* is a specific genus. Rods **divide only across their short axis**, which is why you never see a tetrad or a cluster of rods — only single rods, pairs (**diplobacilli**), end-to-end chains (**streptobacilli**) and side-by-side **palisades**.
- **Coccobacillus** — a short, plump rod that can be mistaken for a coccus (e.g. *Haemophilus influenzae* **GRAM -**).
- **Vibrio** — comma-shaped, a single curve (*Vibrio cholerae* **GRAM -**).
- **Spirillum** — rigid corkscrew, moves with external flagella.
- **Spirochaete** — flexible corkscrew with **internal** axial filaments (endoflagella): *Treponema pallidum* **GRAM -**, *Borrelia*, *Leptospira*.

#### Arrangement — and why it happens

Arrangement is **determined entirely by the plane of division and whether daughter cells separate**. This is not a detail to memorise separately — it falls straight out of how the cell divides:

**TABLE 3.3** ARRANGEMENT FOLLOWS THE PLANE OF DIVISION

| Arrangement                 | How it arises                                                    | Classic organism                                                                                                   |
|-----------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Diplococci                  | Division in one plane, cells remain in pairs                     | <i>Neisseria</i> <b>GRAM -</b> (kidney-bean shaped); <i>Streptococcus pneumoniae</i> <b>GRAM +</b> (lancet shaped) |
| Chains                      | Division in <b>one plane</b> , cells <b>stay attached</b>        | <i>Streptococcus</i> <b>GRAM +</b> — the name means "twisted chain"                                                |
| Tetrads                     | Division in <b>two</b> perpendicular planes                      | <i>Micrococcus</i> <b>GRAM +</b>                                                                                   |
| Sarcinae (cubes of 8)       | Division in <b>three</b> perpendicular planes                    | <i>Sarcina</i> <b>GRAM +</b>                                                                                       |
| Clusters ("grapes")         | Division in <b>random planes</b>                                 | <i>Staphylococcus</i> <b>GRAM +</b> — the name means "grape-cluster"                                               |
| Palisades / Chinese letters | <b>Snapping division</b> — cells bend at the point of separation | <i>Corynebacterium diphtheriae</i> <b>GRAM +</b>                                                                   |

**HIGH-YIELD**

**Spherical cells in chains** = *Streptococcus*. **Spherical cells in clusters** = *Staphylococcus*. Both appear verbatim in the lecture MCQs. Remember the etymology and you can never confuse them: *strepto-* = twisted chain, *staphylo-* = bunch of grapes.

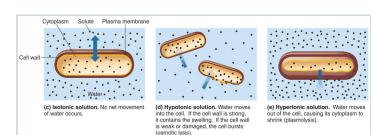
## 3.4 The Cytoplasmic Membrane

A standard **phospholipid bilayer with embedded proteins**, and — critically — **no sterols**, unlike every eukaryotic membrane. The sole exception is *Mycoplasma*, which incorporates cholesterol scavenged from the host. Because a bacterium has no organelles, this single membrane must perform **every** membrane-dependent function:

- **Selective permeability and transport** — the real cell boundary (the wall is a rigid mesh, not a barrier to small molecules).
- **Electron transport and oxidative phosphorylation** — it does the mitochondrion's job.
- **Secretion** of enzymes and toxins.
- **Synthesis of cell wall and membrane precursors**.
- **Anchorage** of DNA during replication and of flagellar motors.

### Osmosis — why the membrane needs the wall

Bacterial cytoplasm is crowded with solute, so a bacterium in tissue fluid or in water is almost always in a **hypotonic** environment: water is permanently trying to enter. The membrane cannot resist that pressure — it is a fluid bilayer, not a rigid shell. The **cell wall** is what holds the cell together against it.

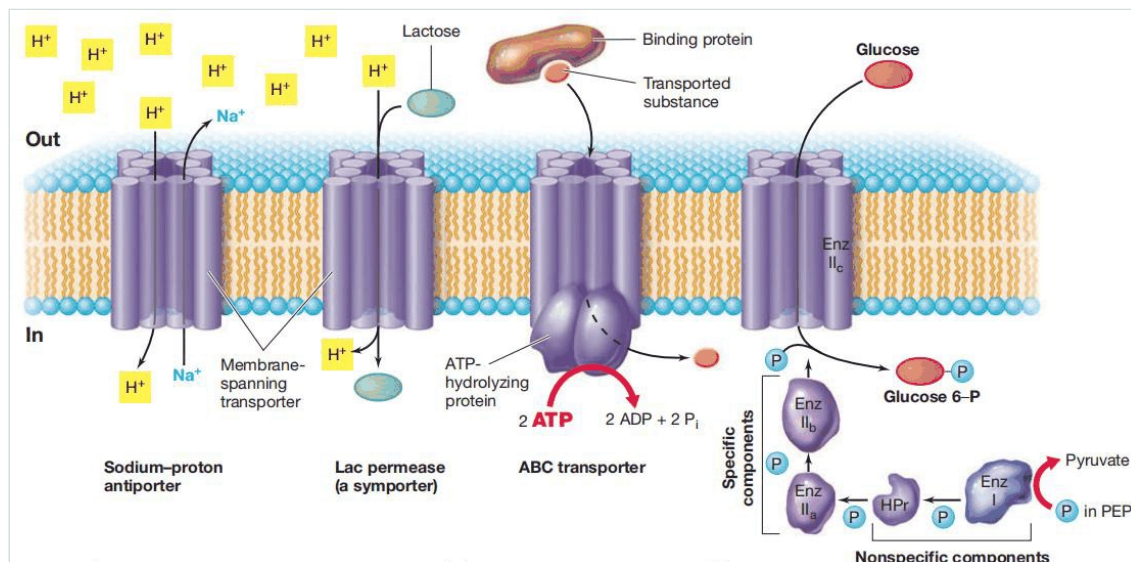


**Fig 3.4** The membrane is semi-permeable, so water follows solute. **Iso-tonic:** no net movement. **Hypotonic:** water enters — a strong wall contains the swelling, a weak or dam-

**HIGH-YIELD**

This single fact is the hinge of the whole antibiotic story. **Penicillin does not kill bacteria. Osmosis does.** The drug only stops the wall being repaired; the water pressure that was always there then does the killing. Every downstream fact follows — why  $\beta$ -lactams are bactericidal, why they work only on growing cells, and why an organism with no wall (*Mycoplasma*) is untouched.

aged wall lets the cell burst (**osmotic lysis**). **Hypertonic:** water leaves and the cytoplasm shrinks (**plasmolysis**).



**Fig 3.5** Membrane transport systems. A **sodium-proton antiporter** and the **lac permease** symporter run on the proton gradient; the **ABC transporter** spends ATP directly; the **phosphotransferase system** phosphorylates its substrate as it imports it. All four exist because the membrane must do the work that organelles do in a eukaryote.

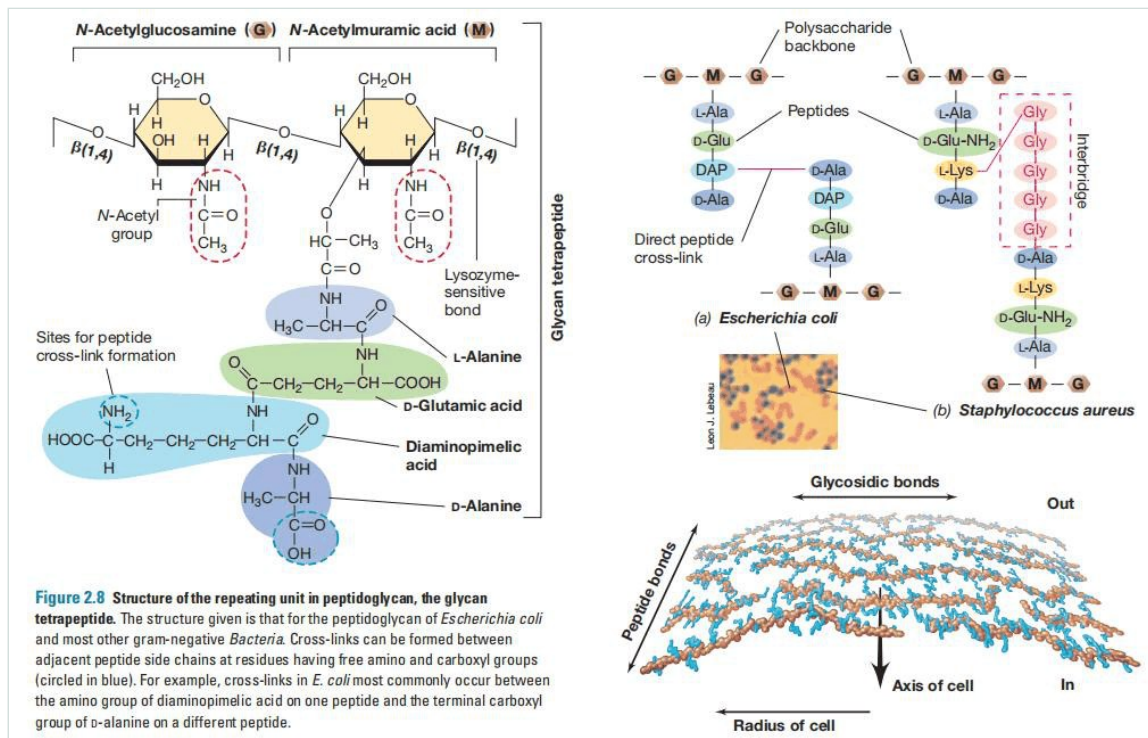
**CLINICAL CORRELATION**

Drugs that attack the membrane are powerful but **relatively toxic**, precisely because bacterial and human membranes are broadly similar in construction. **Polymyxins** (colistin) act as cationic detergents on Gram-negative membranes and are reserved for multidrug-resistant infection because of nephrotoxicity and neurotoxicity. **Daptomycin** inserts into Gram-positive membranes and depolarises them — but it is inactivated by pulmonary surfactant, so it must never be used for pneumonia. That single fact is a favourite exam item, and it follows directly from the drug's mechanism.

## 3.5 Peptidoglycan — Structure and Synthesis

**DEFINITION**

**Peptidoglycan** (murein) is a **single, bag-shaped, highly cross-linked macromolecule** that surrounds the cytoplasmic membrane — one molecule, running to **billions in molecular weight**, enclosing the whole cell like a mesh string bag. It is made of **glycan chains cross-linked by short peptides**, is **unique to bacteria**, and is present in all eubacteria **except *Mycoplasma* and *Ureaplasma***. Its rigidity is what **decides the shape of the bacterium** and what resists osmotic lysis.



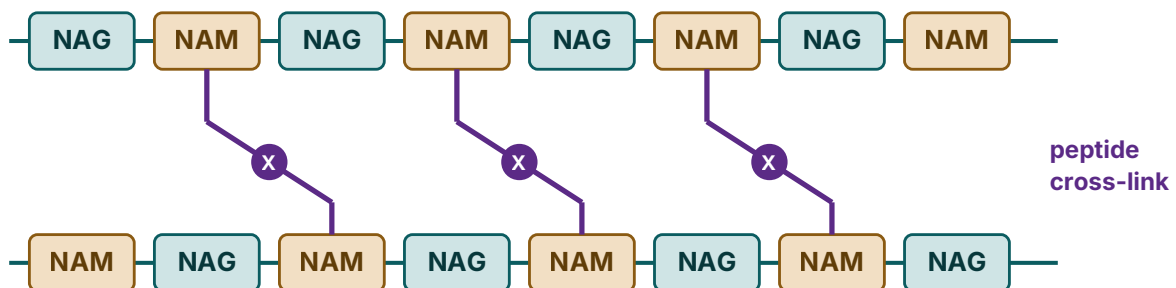
**Fig 3.6** The repeating unit, at the chemical level. NAG–NAM joined by the  $\beta$ -1,4 bond that lysozyme cuts, with the tetrapeptide hanging off NAM. Note the two cross-linking styles the course contrasts: *E. coli* makes a **direct** bond from diaminopimelic acid to D-alanine, while *S. aureus* inserts a **five-glycine interbridge**. Same chemistry, different geometry — and it is why a wall can be thick or thin without changing its building blocks.

## What the cell wall is for

The lecture gives four functions. They are worth learning as a set, because **three of the four are lost the moment a  $\beta$ -lactam works**, and the fourth explains why the drug is bactericidal rather than merely inhibitory:

1. **Maintains the characteristic shape.** A coccus is a coccus because its wall is built that way — strip the wall and every bacterium becomes a sphere.
2. **Provides resistance to osmotic change.** The point of §3.4 — it is the pressure vessel.
3. **Anchors surface appendages** such as flagella and pili. No wall, no motility and no adhesion.
4. **Assists in cell division.** The septum that separates two daughter cells is new wall. This is why a cell that cannot build wall *also* cannot divide — and why  $\beta$ -lactams only harm cells that are actively growing.

### Glycan backbone — alternating NAG / NAM, $\beta$ -1,4 linked



Transpeptidase (PBP) forms the cross-link — **blocked by  $\beta$ -lactams**  
 $\beta$ -1,4 glycan backbone — **cut by lysozyme**

**Fig 3.7** Peptidoglycan architecture. Alternating **N-acetylglucosamine (NAG)** and **N-acetylmuramic acid (NAM)** form glycan chains joined by  **$\beta$ -1,4 glycosidic bonds**. A short peptide hangs from every NAM, and adjacent peptides are cross-linked (X) into a single continuous mesh. Two attack points follow directly from this picture: **lysozyme** hydrolyses the  $\beta$ -1,4 backbone, and  **$\beta$ -lactam antibiotics** block the enzyme that forms the cross-link.

### ○ MECHANISM — PEPTIDOGLYCAN SYNTHESIS AND WHERE EACH DRUG STRIKES

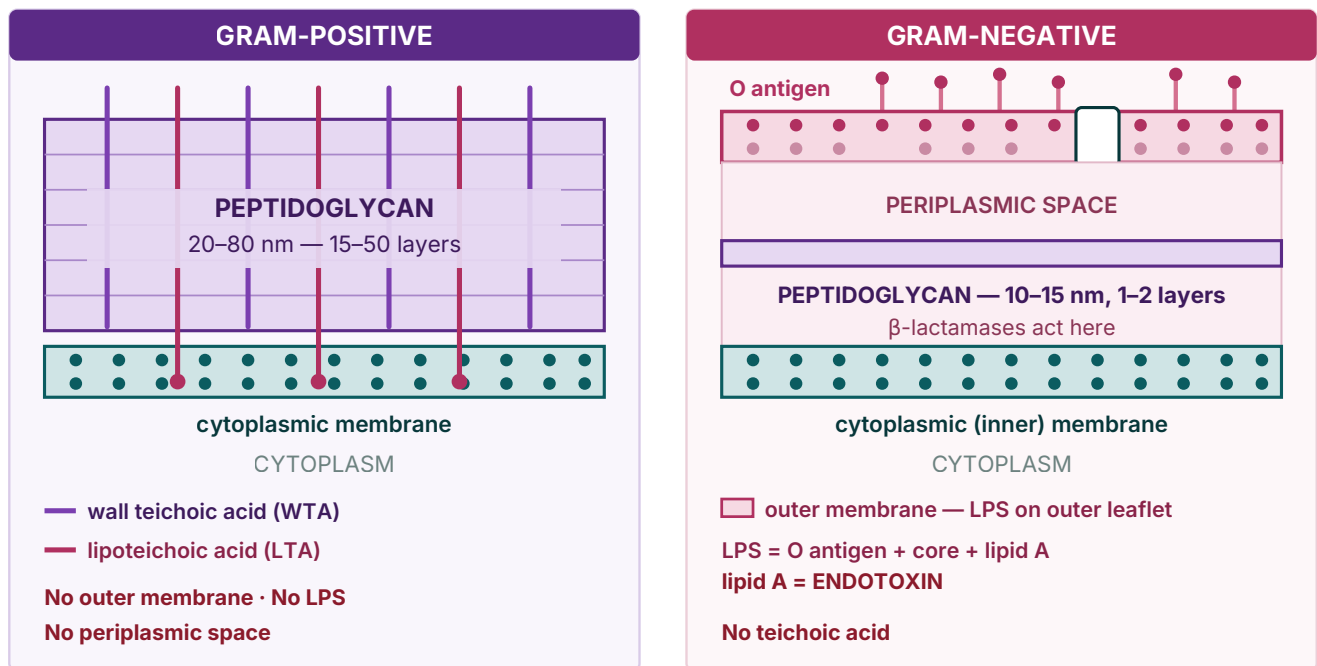
- 1 **Cytoplasm.** NAM and NAG precursors are made, and a **pentapeptide** ending in **D-alanyl-D-alanine** is attached to NAM.  
*WHY: The terminal D-Ala-D-Ala is the handle everything later grabs. Vancomycin binds this dipeptide directly, and the fact that bacteria use the D-isomer — which human enzymes cannot make or cleave — is why the pathway is selectively attackable. Fosfomycin and cycloserine block precursor formation here.*
- 2 **Membrane.** The precursor is handed to **bactoprenol**, a lipid carrier, which ferries it across the hydrophobic membrane to the outside.  
*WHY: A large hydrophilic sugar-peptide cannot cross a lipid bilayer unaided; it needs a lipid escort. Bacitracin blocks the recycling of bactoprenol, stalling the conveyor belt.*
- 3 **Outside the membrane — transglycosylation.** The new disaccharide unit is added to a growing glycan chain via a  $\beta$ -1,4 bond.  
*WHY: This lengthens the chains but does not yet strengthen the wall; long parallel strands without cross-links are like uncemented bricks.*
- 4 **Transpeptidation — the cross-link.** **Transpeptidase enzymes**, also called **penicillin-binding proteins (PBPs)**, cleave the terminal D-Ala and use the energy released to form a peptide bond to the neighbouring chain.  
*WHY: This converts thousands of separate strands into one covalently continuous molecule encasing the whole cell. It is the step that makes the wall strong — and it is the step  $\beta$ -lactams block.*
- 5  **$\beta$ -lactams bind the PBP irreversibly.** Penicillins, cephalosporins, carbapenems and monobactams all carry a  **$\beta$ -lactam ring that structurally mimics D-Ala-D-Ala**, so the transpeptidase binds the drug instead of its true substrate and is permanently acylated.  
*WHY: Molecular mimicry is the whole trick — the enzyme cannot tell the difference until it is too late.*
- 6 **The cell lyses.** Cross-linking stops, but autolysins — the enzymes that normally cut the wall to allow growth — keep working. The wall weakens, and because bacterial cytoplasm is strongly hypertonic relative to its surroundings, water rushes in and the cell **bursts**.  
*WHY: This explains three exam facts at once:  $\beta$ -lactams are bactericidal not bacteriostatic; they only work on actively growing cells (a dormant cell is not building wall, so there is nothing to block); and they are useless against organisms with no peptidoglycan, such as Mycoplasma.*

**NET RESULT** A single, covalently continuous mesh that resists an internal osmotic pressure of up to **25 atmospheres** in Gram-positives. Remove it and the cell dies by osmotic lysis within minutes.

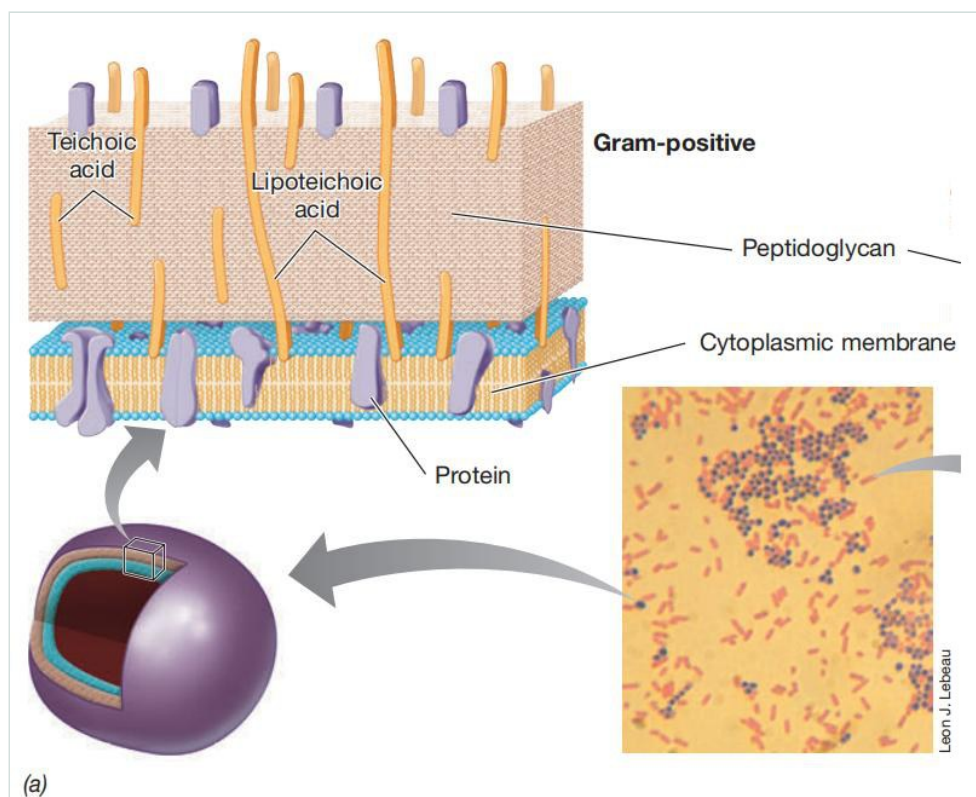
### LAB PEARL — LYSOZYME

**Lysozyme**, present in tears, saliva, mucus and egg white, hydrolyses the  **$\beta$ -1,4 bond between NAG and NAM**. It is far more effective against **Gram-positives**, whose peptidoglycan is exposed at the surface, than against Gram-negatives, whose outer membrane shields the thin peptidoglycan beneath. This is one of the reasons the conjunctiva and oral mucosa — bathed in lysozyme — resist Gram-positive colonisation so well. Discovered, incidentally, by **Alexander Fleming** in 1922, six years before penicillin.

### 3.6 The Gram-Positive Envelope



**Fig 3.8** The two envelopes side by side — the single most important diagram in this course. The Gram-positive cell has **one membrane and a very thick peptidoglycan layer** exposed at the surface. The Gram-negative cell has **two membranes**, a **thin** peptidoglycan layer sandwiched in a **periplasmic space**, and an **outer membrane whose outer leaflet is LPS**. Everything else — staining, toxin biology, antibiotic access, resistance — follows from this one structural difference.



**Fig 3.9** The Gram-positive envelope as the course draws it. One membrane, then a deep block of peptidoglycan with **teichoic acid** threaded through it and **lipoteichoic acid** reaching all the way down into the membrane. Nothing above the wall — and that absence is the whole of the Gram-positive story.

### Wall teichoic acid and lipoteichoic acid

Teichoic acids are **polymers of glycerol or ribitol phosphate** found **only in Gram-positive bacteria**. There are two, distinguished by where they anchor:

- **Wall teichoic acid (WTA)**, also written *cell wall teichoic acid* — covalently bound to the **peptidoglycan** itself.
- **Lipoteichoic acid (LTA)** — anchored in the **cytoplasmic membrane** by a lipid tail and extending out through the wall.

**Functions:** they carry a strong negative charge that binds and regulates cations (especially  $Mg^{2+}$ ), control autolysin activity during growth, mediate **adherence** to host cells, and act as important **surface antigens**. LTA is a potent inducer of inflammatory cytokines through **TLR2** — making it, functionally, the Gram-positive counterpart of endotoxin, although it is considerably less potent.

#### HIGH-YIELD

**Teichoic acid is unique to Gram-positives. LPS is unique to Gram-negatives.** Neither ever appears in the other. This single pair of facts answers a remarkable number of exam questions.

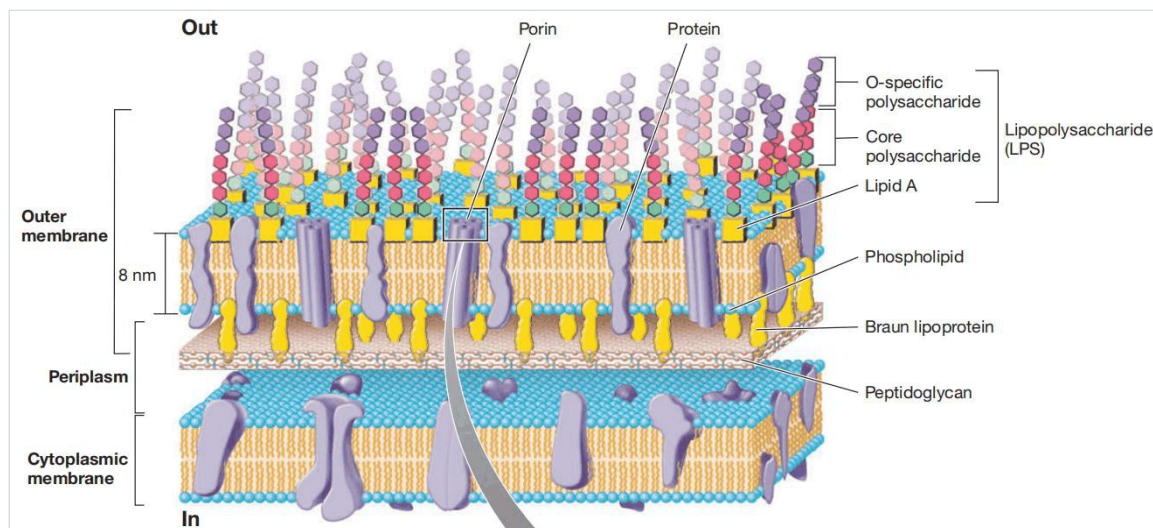
## 3.7 The Gram-Negative Envelope and LPS

The defining feature is the **outer membrane** — a second lipid bilayer outside the peptidoglycan, which no Gram-positive possesses. It is **asymmetric**: the inner leaflet is ordinary phospholipid, but the **outer leaflet is lipopolysaccharide (LPS)**.

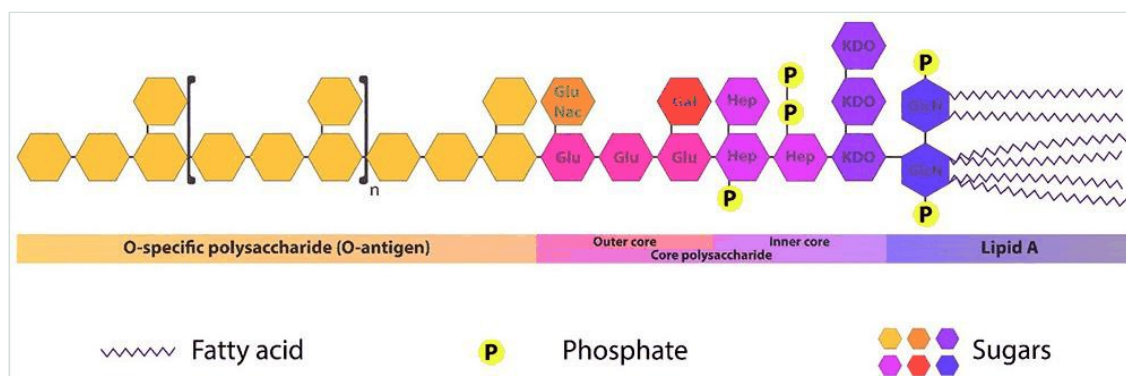
### Three components of LPS

TABLE 3.4 LPS, FROM OUTSIDE IN

| Component                         | What it is                                                                                                                                                                                    | Why it matters                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| O antigen (O-specific side chain) | Outermost; a repeating polysaccharide, highly variable between strains                                                                                                                        | <b>Serotyping</b> — this is the "O" in <i>E. coli</i> O157:H7. Antigenic variation helps evade antibody. |
| Core polysaccharide               | Middle; <b>conserved within a genus</b>                                                                                                                                                       | Structural bridge; its conservation makes it a target for broad-spectrum vaccine research                |
| Lipid A                           | Innermost; a phosphorylated glucosamine disaccharide bearing <b><math>\beta</math>-hydroxy fatty acids</b> — a bacteria-specific feature found in no host lipid. Anchors LPS in the membrane. | <b>This is the endotoxin.</b> The toxic activity of LPS resides entirely in lipid A — see §3.9.          |



**Fig 3.10** The Gram-negative envelope as the course draws it. Read it from the top: **O-specific polysaccharide → core polysaccharide → lipid A** make up LPS, which occupies the outer leaflet only; the inner leaflet is ordinary phospholipid. **Braun lipoprotein** tethers the outer membrane to the peptidoglycan below, and a **porin** punches the only hydrophilic route through. The outer membrane is about **8 nm** thick.



**Fig 3.11** LPS laid out flat. The repeating O-antigen sugars run to the left, the core in the middle, and lipid A with its fatty-acid tails at the right — the only part embedded in the membrane, and the only part that is toxic.

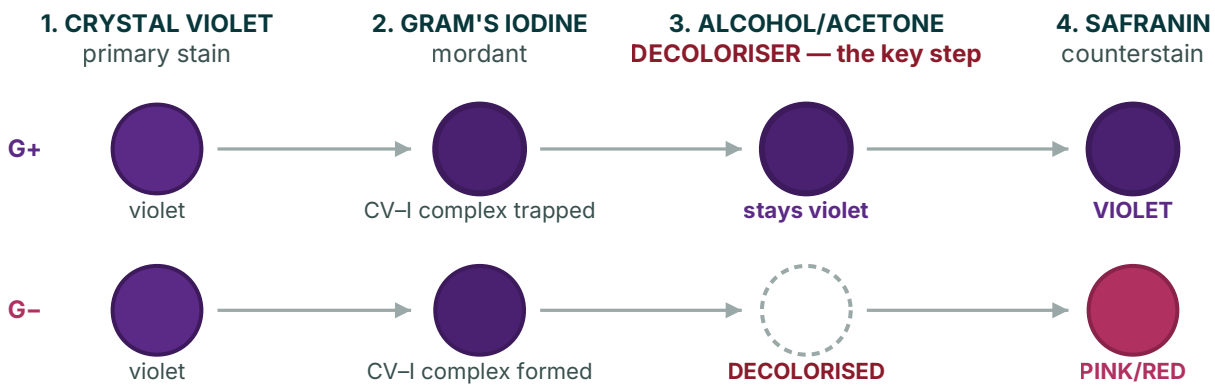
### Porins and the periplasmic space

**Porins** are trimeric  $\beta$ -barrel proteins forming water-filled channels through the outer membrane. Because the outer membrane is otherwise impermeable, **small hydrophilic molecules — including many antibiotics — can only enter through a porin**. Losing or mutating a porin is therefore a complete, single-step resistance mechanism, and is exactly how *Pseudomonas* and carbapenem-resistant Enterobacteriaceae achieve much of their resistance.

The **periplasmic space** between the two membranes is not empty. It is a working compartment holding hydrolytic enzymes, binding proteins for transport, and — critically —  **$\beta$ -lactamases**. Concentrating  $\beta$ -lactamase in the periplasm means the drug is destroyed *after* it crosses the outer membrane but *before* it reaches the PBPs on the inner membrane. This is a far more efficient defence than secreting the enzyme into the surroundings, which is what Gram-positives must do.

## 3.8 The Gram Stain — Why It Works

Devised by **Hans Christian Gram** in 1884, this is the **single most important test in clinical bacteriology**: it takes three minutes, costs almost nothing, and narrows a differential from thousands of species to a handful — usually before any culture result exists.



**Fig 3.12** The four steps of the Gram stain. Both cell types look identical after steps 1 and 2. **Step 3 is where the two diverge**, and it is the only step that depends on the envelope — which is why decolorisation is both the mechanism and the commonest source of error.

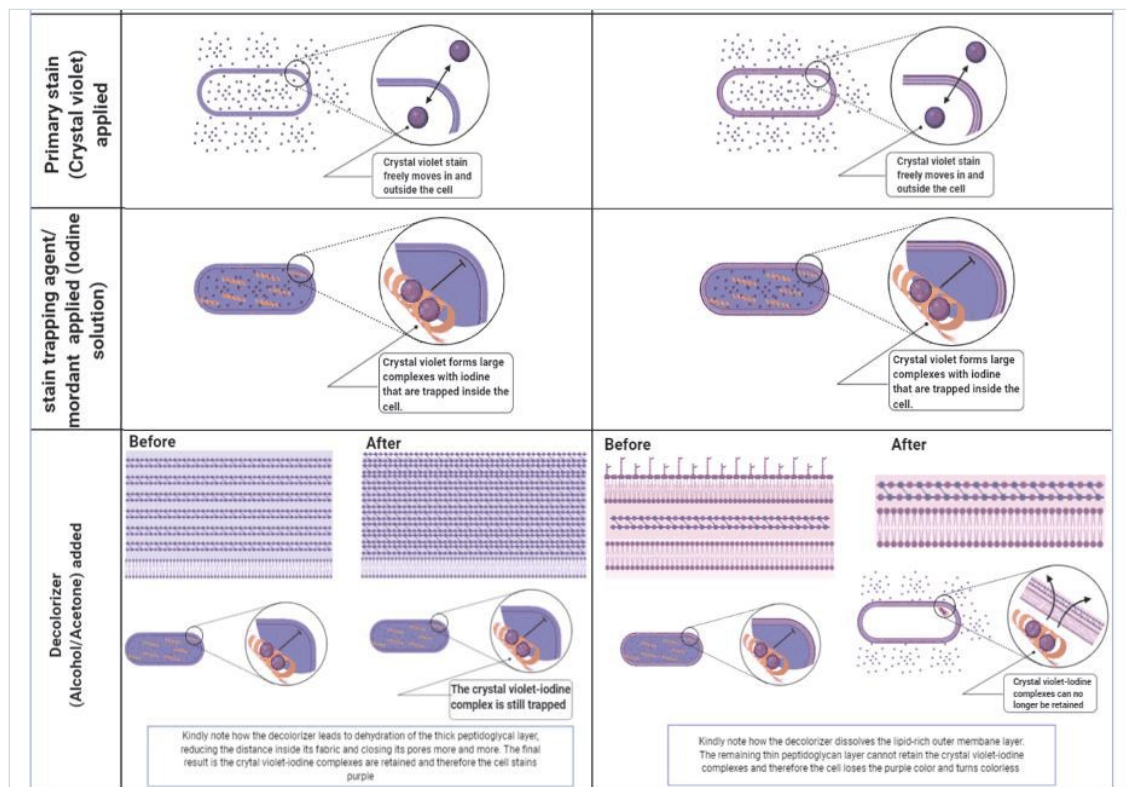
### ○ MECHANISM — WHY GRAM-POSITIVES KEEP THE VIOLET AND GRAM-NEGATIVES DO NOT

- 1 **Crystal violet** is applied. It is a positively charged dye, so it is drawn into **both** cell types, whose surfaces are negatively charged.  
*WHY: At this point every cell on the slide is violet. Nothing has been distinguished yet.*
- 2 **Gram's iodine** is applied as a **mordant**. Iodine binds crystal violet to form a large **crystal violet-iodine (CV-I) complex** inside the cell.  
*WHY: The purpose is purely to make the dye bigger. A small molecule diffuses out easily; a large complex is physically trapped — but only if the mesh holding it is fine enough.*
- 3 **Alcohol or acetone** is applied — **the decolorising step, and the only one that discriminates**.  
*WHY: Everything up to here was identical for both organisms. From here the envelope decides the outcome.*
- 4 **In the Gram-positive cell:** alcohol **dehydrates the thick peptidoglycan**, causing its pores to shrink and close. The large CV-I complex cannot escape and is **locked inside**.  
*WHY: Thickness alone is not the whole answer — it is thickness plus dehydration-induced pore closure. A 15–50 layer mesh that contracts becomes effectively impermeable to a large complex.*
- 5 **In the Gram-negative cell:** alcohol is a lipid solvent, so it **dissolves the lipid-rich outer membrane**, blowing a hole in the envelope. The remaining peptidoglycan is only 1–2 layers thick and cannot retain the complex, which **washes out**. The cell is now colourless.  
*WHY: Two independent features conspire: the outer membrane is destroyed and the peptidoglycan behind it is too thin to act as a sieve. Either alone would be insufficient.*
- 6 **Safranin** (or carbol fuchsin) is applied as a **counterstain**, colouring the now-colourless Gram-negative cells **pink/red**.  
*WHY: Without this step Gram-negatives would simply be invisible, and you could not distinguish "Gram-negative" from "no organisms present" — a clinically dangerous ambiguity. Gram-positives are already so darkly violet that the pink does not show.*

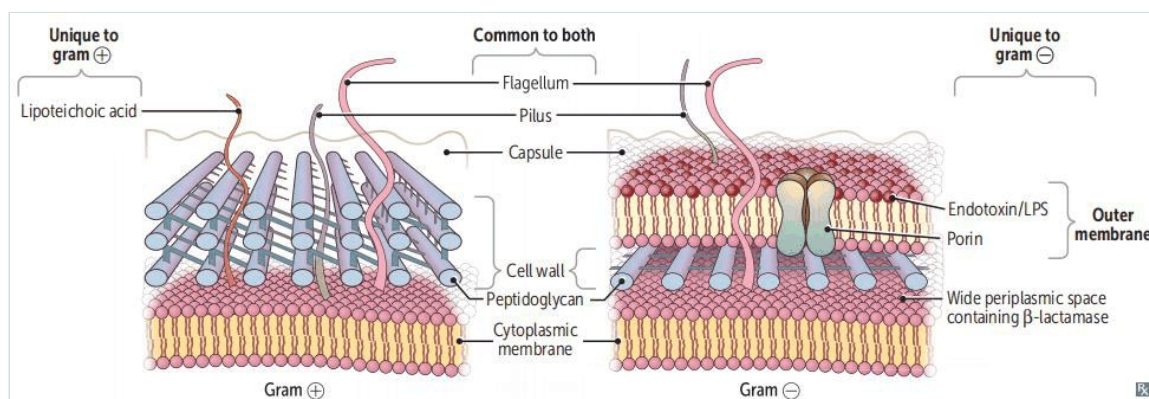
**NET RESULT** **Gram-positive = violet/purple. Gram-negative = pink/red.** The result is a direct readout of envelope architecture — you are, quite literally, seeing the thickness of the peptidoglycan.

## LAB PEARL — THE FOUR WAYS TO GET IT WRONG

- **Over-decolorisation** is the commonest error: leave the alcohol on too long and even Gram-positives lose the complex, reported falsely as Gram-negative.
- **Old cultures** of Gram-positives (>24 h) have damaged, autolysed walls and stain **Gram-variable** — always Gram stain a **fresh** culture.
- **Too thick a smear** traps dye and cannot be decolorised evenly.
- **Prior antibiotic therapy** damages the wall and produces atypical staining — always ask whether the patient has already been treated.



**Fig 3.13** The same mechanism drawn at the level of the wall, Gram-positive on the left, Gram-negative on the right. Row 1: crystal violet enters both freely. Row 2: iodine traps it as a large complex inside both. Row 3 is where they part — in the thick wall the decolouriser **dehydrates and tightens** the mesh so the complex cannot get out; in the thin-walled cell it **dissolves the outer membrane** and the complex escapes.



**Fig 3.14** The comparison sorted the way exam questions ask it. **Unique to Gram-positive:** lipoteichoic acid. **Common to both:** flagellum, pilus, capsule, cell wall, cytoplasmic membrane. **Unique to Gram-negative:** endotoxin/LPS, porins, outer membrane, and a wide periplasmic space containing  $\beta$ -lactamase. Learn the middle column as carefully as the outer two — most wrong answers are structures that are in fact common to both.

### 3.9 Endotoxin and Septic Shock

#### ○ MECHANISM — FROM A DYING GRAM-NEGATIVE CELL TO REFRACTORY SHOCK

- 1 A Gram-negative bacterium **dies and lyses** — spontaneously, through immune attack, or because you gave a bactericidal antibiotic. **LPS is shed from the outer membrane.**  
**WHY:** Endotoxin is a structural component, not a secreted product. It is released only on cell death, which is why killing large numbers of bacteria rapidly can transiently make a septic patient worse.
- 2 **Lipid A** — the toxic moiety — binds **LPS-binding protein** in plasma, which delivers it to **CD14** on macrophages and then to **Toll-like receptor 4 (TLR4)**.  
**WHY:** This is pattern recognition: TLR4 evolved specifically to detect lipid A as an unambiguous signal of Gram-negative invasion. The response is deliberately violent because, in a localised infection, violence is appropriate.
- 3 TLR4 signalling activates **NF-κB**, driving massive release of **TNF-α, IL-1 and IL-6**.  
**WHY:** These are the master inflammatory cytokines. Locally they recruit neutrophils and contain infection; released systemically in bulk, they become the disease.
- 4 **Fever** — IL-1 and TNF-α act on the hypothalamus and induce prostaglandin E<sub>2</sub>, raising the thermoregulatory set point.  
**WHY:** This is why endotoxin is also called a **pyrogen**, and why NSAIDs and paracetamol — which block prostaglandin synthesis — reduce the fever.
- 5 **Hypotension** — cytokines induce **inducible nitric oxide synthase**, flooding the circulation with NO, a powerful vasodilator.  
**WHY:** Widespread arteriolar vasodilation collapses systemic vascular resistance. This is **distributive shock**: the patient has a normal or high cardiac output and warm peripheries, unlike hypovolaemic or cardiogenic shock. Because the cause is NO rather than volume loss, fluids alone often fail and vasopressors are needed.
- 6 **Coagulation activation** — cytokines induce **tissue factor** on endothelium and monocytes, triggering widespread microvascular thrombosis and consuming platelets and clotting factors.  
**WHY:** This is **DIC**: simultaneous thrombosis and bleeding. Organs infarct from microthrombi while the patient bleeds from every puncture site.
- 7 **Multi-organ failure and death.** Hypoperfusion plus microthrombosis produces acute kidney injury, ARDS, hepatic failure and encephalopathy.  
**WHY:** The tragedy of septic shock is that almost all the damage is done by the host's own response, not by the bacterium. This is why source control and early antibiotics matter so much — once the cascade is fully running, it is very hard to stop.

**NET RESULT** **Fever + hypotension + DIC + multi-organ failure** = the endotoxin syndrome of Gram-negative sepsis.

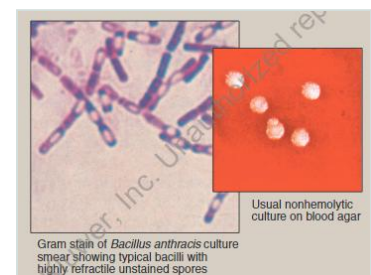
**TABLE 3.5** ENDOTOXIN VERSUS EXOTOXIN — A GUARANTEED EXAM COMPARISON

| Feature   | Endotoxin                                                         | Exotoxin                                                   |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------|
| Chemistry | <b>Lipopolysaccharide</b> (lipid A is the toxic part)             | <b>Protein</b>                                             |
| Source    | <b>Gram-negative only</b> — structural part of the outer membrane | <b>Gram-positive and Gram-negative</b> — actively secreted |
| Released  | On <b>cell lysis/death</b>                                        | During <b>active growth</b>                                |

| Feature              | Endotoxin                                                              | Exotoxin                                                                     |
|----------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Heat stability       | <b>Heat-stable</b> (250 °C for 30 min to destroy)                      | <b>Heat-labile</b> (usually destroyed at 60 °C)                              |
| Potency              | <b>Low</b> — needs large amounts                                       | <b>Very high</b> — botulinum toxin is among the most potent substances known |
| Specificity          | <b>Non-specific</b> — same syndrome whatever the organism              | <b>Highly specific</b> — each toxin has a defined target and effect          |
| Converted to toxoid? | <b>No</b>                                                              | <b>Yes</b> — the basis of tetanus and diphtheria vaccines                    |
| Antibody response    | Poor; antibodies are not protective                                    | Strong; antitoxin is protective                                              |
| Examples             | LPS of <i>E. coli</i> , <i>Neisseria</i> , <i>Salmonella</i> <b>G-</b> | Tetanospasmin, botulinum toxin, diphtheria toxin, cholera toxin              |

### 3.10 The Exceptions — Acid-Fast, Mycoplasma, L-forms

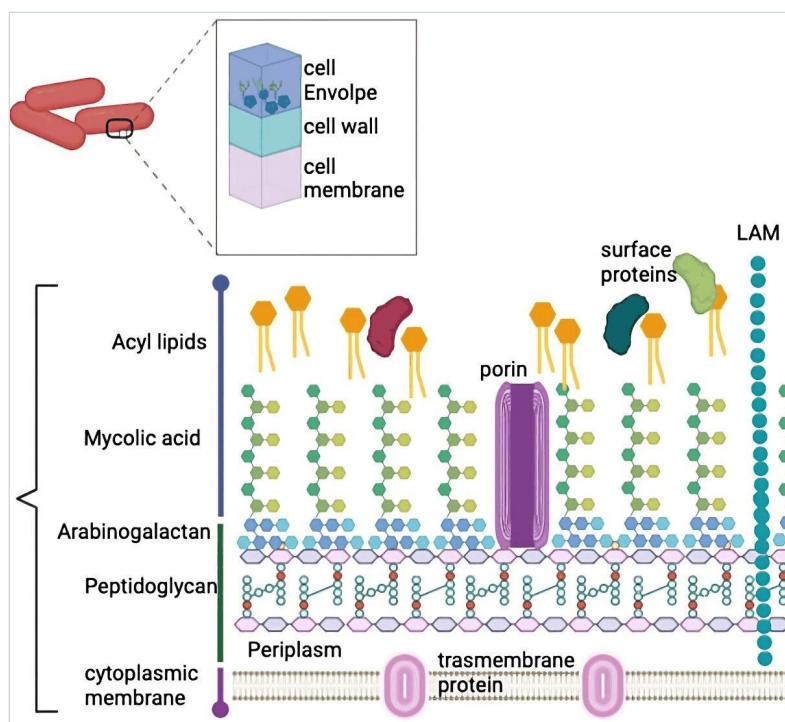
Everything so far has assumed the Gram stain works. For a handful of medically important organisms it does not, and each fails for its own structural reason. Knowing *which* reason matters clinically, because the reason dictates the replacement test: a wall too waxy for aqueous dye needs a stain driven in by heat, a cell with no wall at all cannot be stained by any wall-based method, and an organism too thin to resolve needs a different kind of microscopy rather than a different dye.



**Fig 3.15** Gram stain of *Bacillus anthracis* **GRAM +** showing typical bacilli with **tightly refractile unstained spores** (left), and the usual non-haemolytic colony on blood agar (right). Note that **spores do not take up the Gram stain** — they appear as clear gaps within the cell.

**TABLE 3.6** ORGANISMS THAT BREAK THE GRAM SYSTEM

| Organism group                           | Why Gram stain fails                                                                                                       | What to do instead                                                                                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mycobacterium</b><br><b>ACID-FAST</b> | Cell wall contains up to <b>60% mycolic acid</b> — very long-chain waxy lipids. The wax excludes aqueous crystal violet.   | <b>Ziehl–Neelsen</b> or <b>Kinyoun</b> acid-fast stain: carbol fuchsin driven in with heat or detergent, then acid-alcohol fails to remove it → <b>red bacilli on blue background</b> . <b>Auramine-rhodamine</b> fluorescence is used for screening. |
| <b>Mycoplasma</b>                        | <b>No cell wall at all</b> — the smallest free-living organism. Nothing to retain any stain.                               | Serology, PCR, or special media. <b>Inherently resistant to all <math>\beta</math>-lactams</b> — there is no peptidoglycan to attack. Treat with macrolides, tetracyclines or fluoroquinolones. Uniquely, it has <b>cholesterol</b> in its membrane.  |
| <b>Spirochaetes</b>                      | Technically Gram-negative, but <b>too thin</b> ( $<0.2\ \mu\text{m}$ ) to be seen by light microscopy after Gram staining. | <b>Dark-field microscopy</b> ( <i>T. pallidum</i> ), <b>silver stain</b> , immunofluorescence, or serology (VDRL/RPR, FTA-ABS).                                                                                                                       |
| <b>L-forms</b>                           | Cell-wall-deficient variants induced by $\beta$ -lactams or lysozyme. Pleomorphic, osmotically fragile.                    | Require hypertonic media to survive. Clinically relevant as a cause of <b>persistent or relapsing infection</b> after apparently adequate therapy.                                                                                                    |
| <b>Legionella, Chlamydia, Rickettsia</b> | Gram-negative but stain poorly — intracellular, and <i>Chlamydia</i> lacks classical muramic acid.                         | Silver stain ( <i>Legionella</i> ), Giemsa, urine antigen, PCR, serology.                                                                                                                                                                             |



**Fig 3.16** The mycobacterial envelope, which is why nothing about this genus behaves normally. Above an ordinary membrane and peptidoglycan sits **arabinogalactan**, and esterified to that, the **mycolic acid** layer with free **acyl lipids** packed alongside — a waxy barrier tens of molecules deep, pierced only by porins and threaded by **LAM** (lipoarabinomannan). That wax is what lets *Mycobacterium* survive acids, alkalis, detergents, the phagocyte's oxidative burst, immune lysis and most antibiotics.

**EXAM TRAP**

"Gram-negative" and "does not stain with Gram" are **not** the same claim. *Mycobacterium* is technically Gram-positive by genetics and wall chemistry but is **not visualised** by the Gram stain; *Treponema* is genuinely Gram-negative but too thin to see; *Mycoplasma* has no wall so the question does not apply. A blank Gram stain in a sick patient is a *finding*, not a negative result — it should prompt you to order a different stain.

### 3.11 Structure → Function → Drug Target

**TABLE 3.7** EVERY ENVELOPE COMPONENT, WHAT IT DOES, WHAT ATTACKS IT, AND WHAT HAPPENS IF IT IS LOST

| Component                     | Function                                                     | Targeted by                                                                                                                                           | If absent or mutated                                                                                     |
|-------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Peptidoglycan                 | Shape; resists osmotic lysis                                 | <b>β-lactams</b> (transpeptidase), <b>vancomycin</b> (D-Ala-D-Ala), <b>bacitracin</b> , <b>fosfomycin</b> , <b>cycloserine</b> , host <b>lysozyme</b> | Osmotic lysis; or survival as a fragile <b>L-form</b> in hypertonic surroundings                         |
| D-Ala-D-Ala terminus          | Substrate handle for cross-linking                           | <b>Vancomycin</b> binds it directly                                                                                                                   | Mutation to <b>D-Ala-D-Lactate</b> → <b>vancomycin resistance (VRE)</b>                                  |
| Penicillin-binding proteins   | Transpeptidase enzymes                                       | All <b>β-lactams</b>                                                                                                                                  | Altered <b>PBP2a</b> ( <i>mecA</i> gene) → <b>MRSA</b>                                                   |
| Teichoic acid <b>G+ ONLY</b>  | Cation binding; adherence; antigen; TLR2 ligand              | No drug in clinical use                                                                                                                               | Reduced adherence and virulence                                                                          |
| Outer membrane <b>G- ONLY</b> | Permeability barrier                                         | <b>Polymyxins</b> disrupt it                                                                                                                          | Its presence is why Gram-negatives are <b>intrinsically resistant</b> to vancomycin (too large to cross) |
| LPS / lipid A <b>G- ONLY</b>  | <b>Endotoxin</b> ; TLR4 ligand                               | No direct drug; managed supportively                                                                                                                  | Lipid A modification → <b>polymyxin resistance</b> ( <i>mcr-1</i> )                                      |
| Porins <b>G- ONLY</b>         | Entry of small hydrophilic solutes                           | The route by which β-lactams enter                                                                                                                    | Porin loss → <b>carbapenem resistance</b>                                                                |
| Mycolic acid <b>AFB</b>       | Waxy impermeable coat; acid-fastness; intracellular survival | <b>Isoniazid</b> blocks mycolic acid synthesis; <b>ethambutol</b> blocks arabinogalactan                                                              | Loss of acid-fastness and of resistance to drying and disinfectants                                      |
| Cytoplasmic membrane          | Transport, respiration, secretion                            | <b>Polymyxins</b> <b>G-</b> ; <b>daptomycin</b> <b>G+</b>                                                                                             | Immediate cell death                                                                                     |
| 70S ribosome                  | Protein synthesis                                            | <b>30S</b> : aminoglycosides, tetracyclines · <b>50S</b> : macrolides, clindamycin, chloramphenicol, linezolid                                        | Ribosomal methylation → <b>macrolide resistance</b> ( <i>erm</i> )                                       |

**MNEMONIC — 30S VS 50S INHIBITORS**

**"Buy AT 30, CCEL at 50"** — **A**minoglycosides and **T**etracyclines bind **30S**; **C**loramphenicol, **C**lindamycin, **E**rythromycin (macrolides) and **L**inezolid bind **50S**. This list is genuinely arbitrary, so a mnemonic is justified — unlike the envelope, which should be understood.

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## PART FOUR

# 04

# External and Internal Structures

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The envelope decides what a bacterium *is*. These structures decide what it can *do*: stick to a catheter, swim up a ureter, hide from a phagocyte, hand a resistance gene to a neighbour, or survive boiling for an hour. Almost every one of them is a virulence factor, and several are the reason a particular infection is hard to treat.

### BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Explain how the capsule defeats phagocytosis and how vaccines get round it.
- Distinguish flagella, pili and fimbriae on structure, function and composition.
- Describe sporulation step by step and explain each source of spore resistance.
- Say why plasmids matter clinically even though they carry no essential genes.

## 4.1 Capsule, Slime Layer and Biofilm

### DEFINITION

A **capsule** is a well-organised layer firmly attached outside the cell wall, usually made of **polysaccharide** and less frequently of polypeptide, glycoprotein or glycolipid. A **slime layer** is the same kind of material but loosely attached, unorganised and easily washed off. Both are collectively called the **glycocalyx**. Neither is **present in all bacteria** — and it varies even between strains of a capsule-forming species — because neither is **essential for viability**.

### MECHANISM — HOW A CAPSULE DEFEATS THE IMMUNE SYSTEM

- 1 The capsule is **highly hydrated and strongly negatively charged**.  
*WHY: Phagocyte surfaces are also negatively charged, so like charges repel. The bacterium is physically pushed away before contact is even attempted.*
- 2 It **masks the underlying cell wall antigens** — the PAMPs that phagocyte receptors recognise.  
*WHY: A phagocyte cannot ingest what it cannot recognise. The capsule is a smooth, poorly immunogenic coat hiding a highly immunogenic surface.*
- 3 It **prevents complement C3b from depositing** on the cell surface.  
*WHY: C3b is the principal opsonin: phagocytes carry C3b receptors and use them as handles. No C3b on the surface means no handle to grip.*
- 4 The organism therefore **survives in blood and tissue** and can cause invasive disease.  
*WHY: This is why encapsulated organisms cause bacteraemia, meningitis and pneumonia rather than staying at a mucosal surface.*
- 5 **Anticapsular antibody restores opsonisation**. Antibody binds the capsule, and phagocytes grip the Fc portion.  
*WHY: This is the entire basis of the pneumococcal, meningococcal and Hib vaccines. Because pure polysaccharide is a **T-independent antigen** and works poorly in children under 2, the polysaccharide is **conjugated to a protein carrier**, which recruits T-cell help and produces lasting memory. That single design decision is why Hib meningitis has nearly vanished.*

**NET RESULT** The capsule is the single most important anti-phagocytic virulence factor in bacteriology, and encapsulated organisms are the ones asplenic patients must be vaccinated against.

### MNEMONIC — THE ENCAPSULATED ORGANISMS

#### "Some Nasty Killers Have Some Capsule Protection"

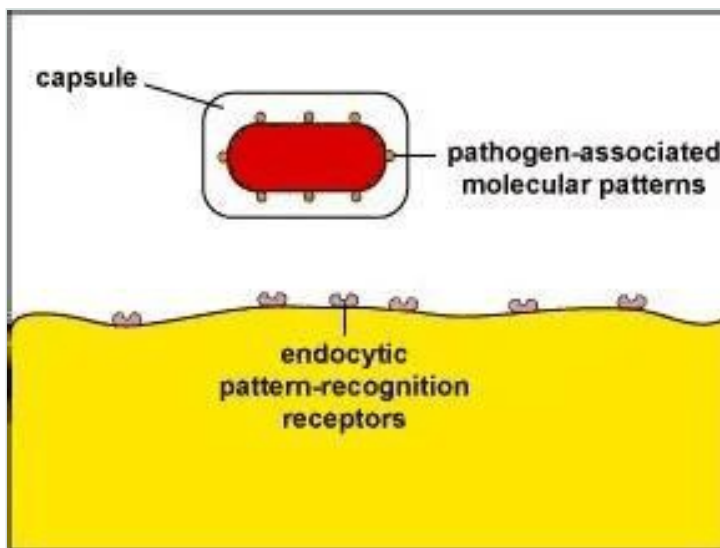
*Streptococcus pneumoniae* **G+** · *Neisseria meningitidis* **G-** · *Klebsiella pneumoniae* **G-** · *Haemophilus influenzae* type b **G-** · *Salmonella* **G-** · *Cryptococcus neoformans* (a fungus) · *Pseudomonas aeruginosa* **G-**.

This list is arbitrary, which is why a mnemonic is appropriate here. **Asplenic patients and those with sickle cell disease must be vaccinated against the first three**, because the spleen is the main site of clearing encapsulated organisms from blood.

**LAB PEARL — SEEING THE CAPSULE**

Capsules are **not stained by the Gram stain** — they appear as a clear halo around the cell. To demonstrate one: **India ink / nigrosin negative stain** (the background is stained, the capsule stays clear — classic for *Cryptococcus neoformans* in CSF), or the **quellung reaction**, in which anticapsular antiserum makes the capsule appear to swell — historically the definitive test for *S. pneumoniae*. On an agar plate, heavily encapsulated organisms such as *Klebsiella* give **mucoïd, stringy colonies** you can draw into a thread with a loop.

**Biofilm.** When bacteria attach to a surface and secrete an extracellular polymeric matrix, they form a **biofilm** — a structured community, not free-floating cells. Biofilms matter clinically for three reasons: antibiotics penetrate the matrix poorly; cells deep inside are metabolically dormant and so escape drugs that need active growth ( $\beta$ -lactams, aminoglycosides); and phagocytes cannot engulf a surface-attached mass. This is why **infected prosthetic joints, heart valves, and intravascular catheters usually have to be removed** rather than treated — the single most practical consequence of this whole section. *Staphylococcus epidermidis* **G+** is the classic biofilm former on plastic.



**Fig 4.1** Why the capsule works, in one picture. The wall underneath carries **pathogen-associated molecular patterns** that the phagocyte's pattern-recognition receptors are built to grip. The capsule simply covers them. The receptors are intact, the ligand is intact — they just never meet.

## 4.2 Flagella, Motility and Chemotaxis

A **flagellum** is a long, thin, helical appendage made of the protein **flagellin**, driven by a rotary motor embedded in the cell envelope and powered by the **proton motive force** — not by ATP directly. It has three parts: the **filament** (external), the **hook** (a flexible coupling), and the **basal body** (the motor).

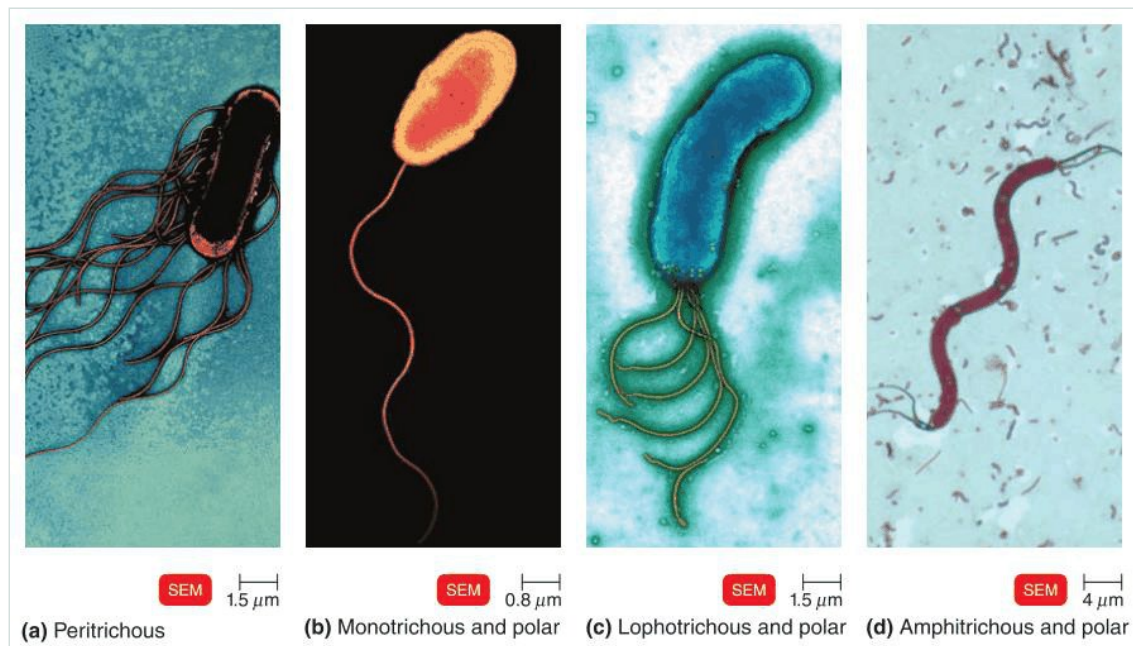
- **Monotrichous** — one polar flagellum (*Vibrio cholerae* **G-**).
- **Lophotrichous** — a tuft at one pole. **Amphitrichous** — one or a tuft at each pole.
- **Peritrichous** — distributed over the whole surface (*E. coli*, *Salmonella*, *Proteus* **G-**).

**Chemotaxis** is directed movement along a chemical gradient. Chemoreceptors detect attractants and repellents and bias the motor's behaviour: counter-clockwise rotation bundles peritrichous flagella and the cell **runs** smoothly forward; clockwise rotation breaks the bundle apart and the cell **tumbles** and reorients ran-

domly. The bacterium cannot steer — it simply **lengthens the runs when conditions are improving**, which over time carries it up the gradient. That is a biased random walk, not navigation.

#### CLINICAL CORRELATION

Flagellin is the **H antigen** used in serotyping (the "H7" of *E. coli* O157:H7) and a potent **TLR5** ligand. Motility is a genuine virulence factor: uropathogenic *E. coli* and *Proteus mirabilis* **G<sup>-</sup>** swim *up* the ureters to cause pyelonephritis, and *Helicobacter pylori* **G<sup>-</sup>** uses its flagella to burrow through gastric mucus to the epithelial surface where the pH is survivable. On a plate, *Proteus* is so motile it **swarms** across the whole agar surface in concentric waves instead of forming discrete colonies — an instantly recognisable lab finding.



**Fig 4.2** The four arrangements, in real cells. (a) **Peritrichous** — flagella all over. (b) **Monotrichous** — one, at a pole. (c) **Lophotrichous** — a tuft at one pole. (d) **Amphitrichous** — at both poles. Arrangement is a fixed species trait, so it is a taxonomic character, not a variable state.

#### LAB PEARL — YOU CANNOT SEE A FLAGELLUM ON A ROUTINE SLIDE

A flagellum is about 20 nm across — far below the resolution of the light microscope. It becomes visible only after a **flagellar stain**, which deposits mordant along the filament until it is thick enough to see. So "non-motile on the Gram stain" is meaningless: motility is assessed by a **hanging-drop preparation** or by spreading growth in semi-solid motility medium, never by looking at a stained smear.

## 4.3 Pili and Fimbriae

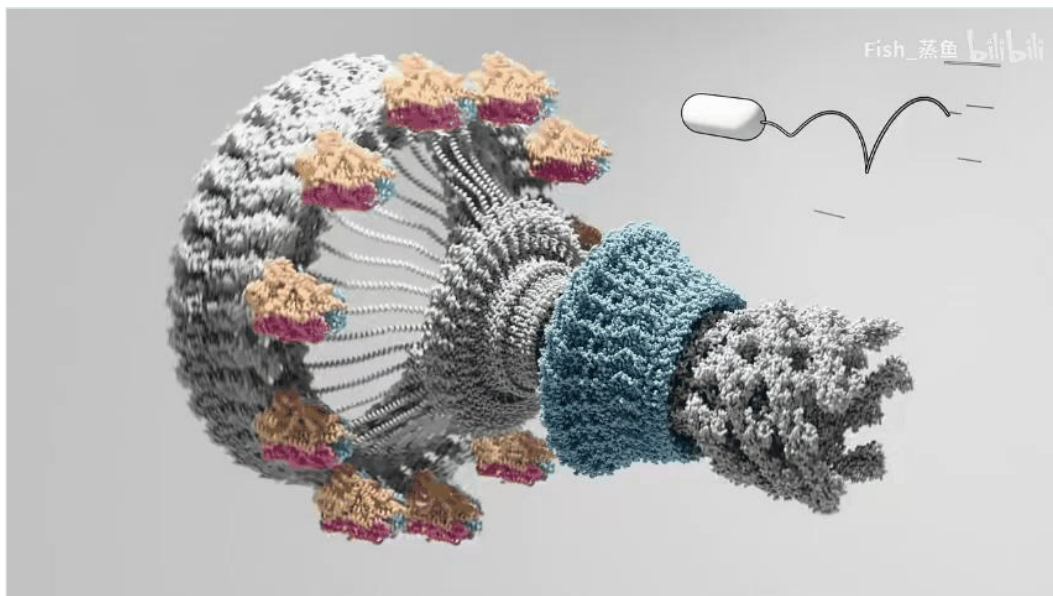
Both terms describe **short, straight, hair-like protein appendages** that are **shorter and thinner than flagella** and **not involved in motility**. Usage varies: many texts use *fimbriae* for the numerous short attachment structures and *pilus* for the single long conjugative one; others use the words interchangeably. Know both conventions.

**TABLE 4.1** COMMON PILI VERSUS SEX PILI

|                     | Common pili / fimbriae                                                                                                                                                                                                    | Sex (conjugative, F) pilus                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Number              | Hundreds per cell                                                                                                                                                                                                         | <b>Only 1–4 per bacterium</b>                                              |
| Dimensions          | <b>Shorter and thinner</b>                                                                                                                                                                                                | <b>Longer and broader</b>                                                  |
| Made of             | <b>Pilin protein</b> , tipped with <b>adhesins</b>                                                                                                                                                                        | Pilin protein                                                              |
| Function            | <b>Attachment</b> to host cells and surfaces — colonisation                                                                                                                                                               | <b>Conjugation</b> — DNA transfer between bacteria                         |
| Encoded by          | Chromosome                                                                                                                                                                                                                | <b>F plasmid</b> (fertility factor)                                        |
| Clinical importance | Uropathogenic <i>E. coli</i> P-pili bind uroepithelium; <i>Neisseria gonorrhoeae</i> pili mediate attachment <b>and</b> undergo antigenic variation to evade antibody — which is why gonorrhoea recurs and has no vaccine | <b>The main route of antibiotic-resistance gene spread</b> between species |

**EXAM TRAP**

From the Lecture 2 MCQ set: pili are made of **protein**, they **allow attachment**, and they **contain adhesins** — but they do **not facilitate nutrient transport**. That last statement is the false one. Transport is the job of the *membrane*, not of a surface appendage. A companion question asks what capsule and pili have in common: the answer is that **both permit attachment to surfaces** — not that both are protein (the capsule is usually polysaccharide) and not that either contains DNA.



**Fig 4.3** Conjugation caught in the act: two cells joined by a single long **sex pilus**. The pilus is coated in bacteriophage particles, which is how it was first visualised — the phages bind F-pili specifically, so they stain the structure for you. Note the contrast with the short, dense fimbriae covering the cell on the left.

## 4.4 Nucleoid, Plasmids and Ribosomes

### The nucleoid

A **single circular double-stranded DNA chromosome**, haploid, not enclosed by a membrane, supercoiled by **DNA gyrase (topoisomerase II)** into a compact region called the nucleoid. In *E. coli* the chromosome is about 1 mm long — roughly a thousand times the length of the cell — so supercoiling is not optional. Being **haploid** is not a limitation but an advantage: with one copy of every gene there is less to replicate, so the cell **divides faster**, and a single mutation is expressed immediately rather than masked by a second allele — which is precisely how a population adapts to a new antibiotic within hours.

*DNA gyrase is the target of the fluoroquinolones (ciprofloxacin, levofloxacin). Blocking it prevents the supercoiling needed for replication, so the chromosome cannot be copied.*

### Plasmids

#### DEFINITION

A **plasmid** is a small, circular, double-stranded DNA molecule that **replicates independently of the chromosome** and carries genes that are **not essential for growth** under normal conditions.

They typically carry: **antibiotic-resistance genes** (R plasmids — often several at once, which is how multidrug resistance spreads as a package); **virulence factors** such as exotoxins and adhesins; **bacteriocins**; and the **fertility (F) factor** that encodes the sex pilus. They are transferred horizontally by **conjugation**, meaning resistance can jump between species and even genera — *Klebsiella* to *E. coli*, for instance, within a single patient's gut.

#### EXAM TRAP

The Lecture 2 MCQ asks which statement about plasmids is correct. The answer is "**may contain antibiotic resistance genes.**" The distractors are all false: plasmids replicate **independently**, not with the chromosome; they do **not** contain essential growth information (that is precisely what distinguishes them from the chromosome); and they are **far smaller** than the chromosome.

### Ribosomes

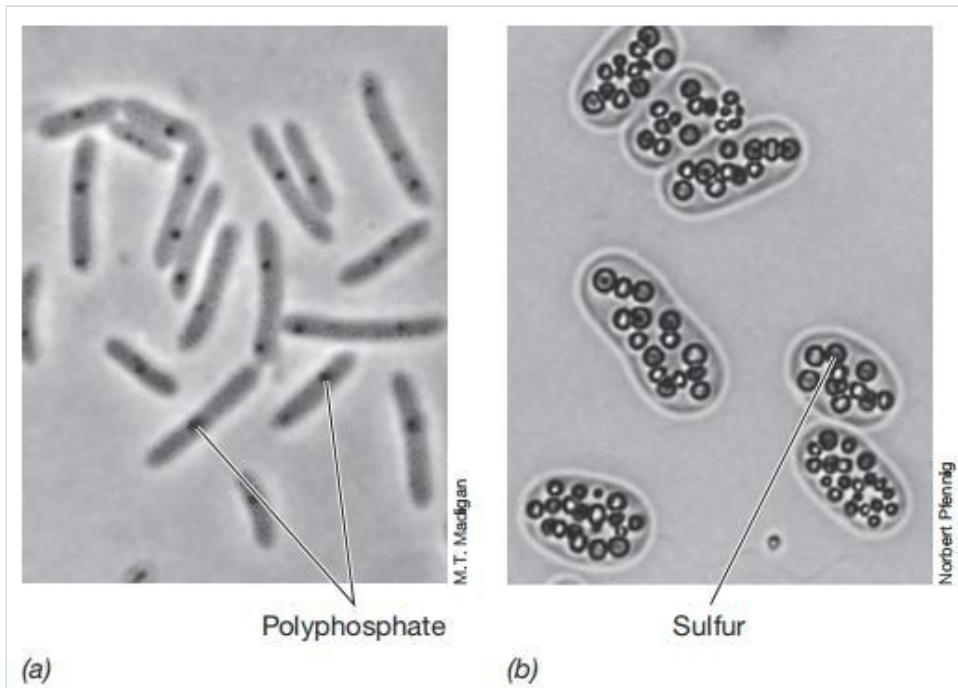
**70S**, composed of a **50S** and a **30S** subunit. The numbers are Svedberg units — a measure of *sedimentation rate*, which depends on both mass and shape, which is why  $50 + 30 = 70$ , **not 80**. That apparent arithmetic error is itself a favourite exam question.

## 4.5 Inclusions and Granules

Storage deposits in the cytoplasm, accumulated when a nutrient is plentiful and consumed when it is scarce: **glycogen** and **poly- $\beta$ -hydroxybutyrate** (carbon), **polyphosphate** (phosphate and energy), and **sulphur** granules in some environmental species.

**LAB PEARL — METACHROMATIC GRANULES**

**Volutin (metachromatic) granules** are polyphosphate deposits that stain a *different* colour from the dye applied — reddish-purple with methylene blue, which is what "metachromatic" means. They are characteristic of *Corynebacterium diphtheriae* **G+** and are demonstrated with **Albert's** or **Loeffler's methylene blue** stain. Combined with the palisade / "Chinese letter" arrangement, they give a presumptive diagnosis of diphtheria at the microscope — hours before culture.



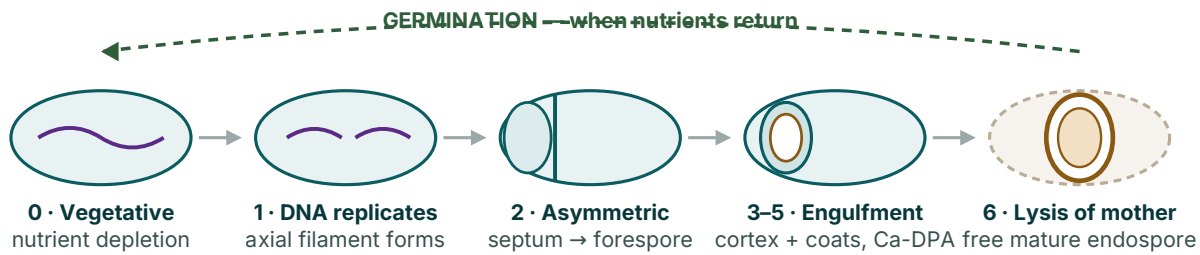
**Fig 4.4** Inclusions are visible as refractile bodies without any special stain. (a) **Polyphosphate** granules — the metachromatic granules of *Corynebacterium*. (b) **Sulfur** globules. Inclusions accumulate when a nutrient is in excess, so their presence says something about the environment the organism came from, not about its species alone.

## 4.6 Endospores

**DEFINITION**

An **endospore** is a **dormant, highly resistant, dehydrated survival structure** formed **inside** a vegetative bacterial cell in response to nutrient depletion. It is **not a reproductive structure** — one cell makes exactly one spore, so there is no increase in cell number.

**Only two medically important genera form endospores, and both are Gram-positive:** *Bacillus* (aerobic — *B. anthracis*, *B. cereus*) and *Clostridium*/*Clostridioides* (anaerobic — *C. tetani*, *C. botulinum*, *C. perfringens*, *C. difficile*).



**The whole cycle takes 6–8 hours. One cell → one spore: this is survival, not reproduction.**

Spores survive boiling, drying, radiation, disinfectants and most antibiotics — killed only by autoclaving (121 °C, 15 psi, 15–20 min)

**Fig 4.5** Sporulation and germination. Note the **asymmetric** septum at stage 2 — this is what distinguishes sporulation from ordinary binary fission, and it is the step examiners probe.

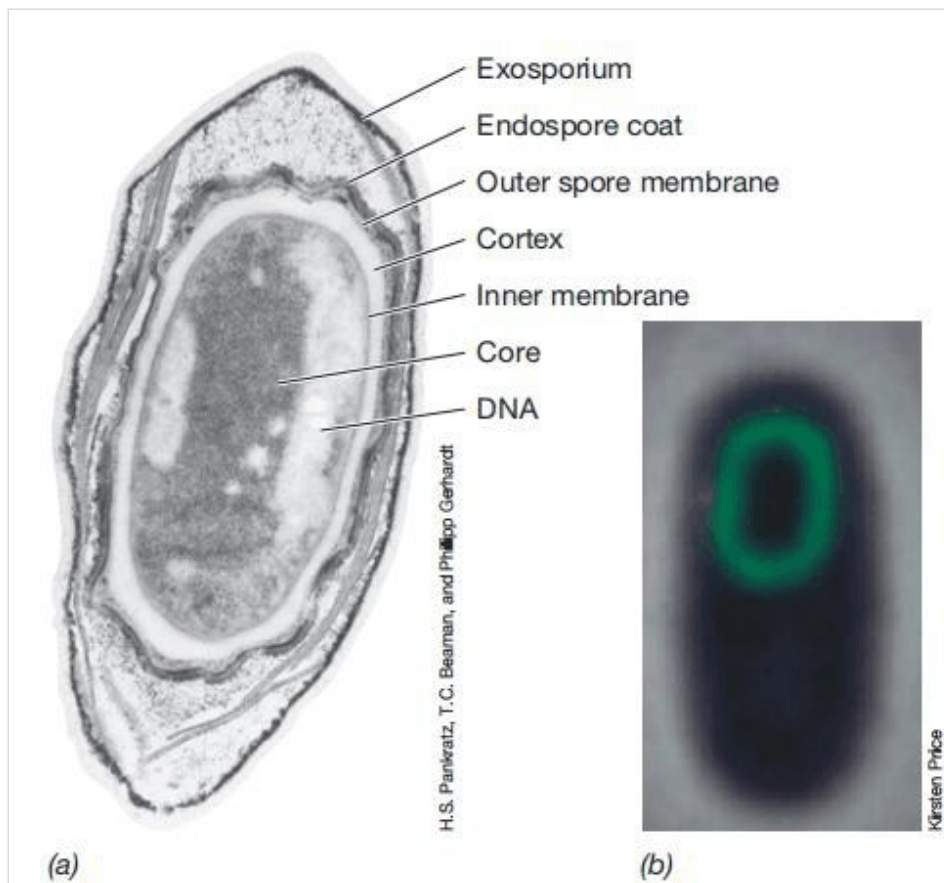
### ○ MECHANISM — SPORULATION, AND WHY THE SPORE IS SO HARD TO KILL

- 1 **Trigger: nutrient depletion** (especially carbon or nitrogen) activates the master regulator Spo0A.  
*WHY: Sporulation is metabolically expensive and irreversible, so the cell commits only when starvation is confirmed. It is a last resort, not a routine.*
- 2 **DNA replicates** and the two copies align along an axial filament.  
*WHY: Each of the two compartments to come must receive a complete genome — the forespore cannot inherit half a chromosome.*
- 3 **An asymmetric septum forms**, dividing the cell into a small **forespore** and a large **mother cell**.  
*WHY: This is the defining event. Binary fission produces a symmetrical septum and two equal daughters; sporulation deliberately produces unequal compartments with different genetic programmes.*
- 4 **The mother cell engulfs the forespore**, which now lies free within the mother's cytoplasm, wrapped in a double membrane.  
*WHY: Engulfment is what makes it an endospore — formed inside another cell — and gives it two membranes to build protective layers between.*
- 5 **Cortex and coats are laid down.** A thick layer of modified peptidoglycan (cortex) forms between the two membranes, and the mother cell deposits tough **keratin-like protein coats** outside it.  
*WHY: The coats are a chemical and enzymatic barrier — they exclude disinfectants, lysozyme and most antibiotics. This is why sporicidal chemicals are a special, short list.*
- 6 **Dehydration and Ca-dipicolinic acid.** The core is stripped of water to roughly 10–25% of normal and saturated with **calcium dipicolinate**, while **small acid-soluble proteins (SASPs)** coat the DNA.  
*WHY: This is the heart of heat resistance. Protein denaturation requires water — a dehydrated core simply cannot be cooked. Ca-DPA stabilises and further dehydrates the core, and SASPs physically shield the DNA from heat and UV damage. Remove water and you remove the mechanism by which heat kills.*
- 7 **The mother cell lyses** and releases the mature endospore.  
*WHY: The mother cell's death is programmed — it exists only to build and release the spore. In the Lecture 2 MCQ, "mother cell disintegration" is part of sporulation; the false option is "symmetrical cell division", because the septum is asymmetric.*
- 8 **Germination** occurs when nutrients return: the spore activates, takes up water, degrades its cortex, and a vegetative cell emerges.  
*WHY: Germination restores the very vulnerability the spore lacked — which is exactly why therapy for spore-forming infections must continue long enough to catch the newly germinated cells.*

**NET RESULT** A structure that survives **boiling, drying, freezing, UV, most disinfectants and all ordinary antibiotics**, and can remain viable in soil for **decades to centuries**. Reliable killing requires **autoclaving at 121 °C, 15 psi, for 15–20 minutes**.

#### CLINICAL CORRELATION — WHY SPORES DOMINATE THE CLINIC

- *C. difficile* **G+** — spores survive alcohol hand rub and standard surface disinfection. This is why you must use **soap and water** (mechanical removal) and **bleach** for the room. Alcohol gel does not touch them.
- *C. tetani* — spores live in soil and enter through puncture wounds; the deep anaerobic environment triggers germination.
- *C. botulinum* — spores survive ordinary cooking in improperly canned food and germinate in the anaerobic tin. Also why **honey must not be given to infants under one year**.
- *B. cereus* — spores survive boiling of rice; reheated fried rice is the classic vehicle.
- *B. anthracis* — spore persistence in soil for centuries makes anthrax both an agricultural and a bio-terrorism problem.



**Fig 4.6** An endospore in section. From outside in: **exosporium → coat → outer spore membrane → cortex → inner membrane → core** holding the DNA. Each layer is laid down by the mother cell in the order the mechanism above describes, and each one is responsible for a different kind of resistance.

**TABLE 4.2** ENDOSPORE VERSUS VEGETATIVE CELL — EVERY LINE IS A CONSEQUENCE OF DEHYDRATION

| Characteristic                      | Vegetative cell     | Endospore                                                          |
|-------------------------------------|---------------------|--------------------------------------------------------------------|
| Microscopic appearance              | Non-refractile      | <b>Refractile</b> — this is how you spot one on an unstained slide |
| Water content                       | High, <b>80–90%</b> | Low, <b>10–25%</b> in the core                                     |
| Calcium content                     | Low                 | High                                                               |
| Dipicolinic acid                    | Absent              | Present                                                            |
| Small acid-soluble proteins (SASPs) | Absent              | Present                                                            |
| Enzymatic activity                  | High                | Low                                                                |
| Respiration rate                    | High                | Low or absent                                                      |
| Macromolecular synthesis            | Present             | Absent                                                             |
| Heat resistance                     | Low                 | High                                                               |
| Radiation resistance                | Low                 | High                                                               |
| Resistance to chemicals             | Low                 | High                                                               |
| Lysozyme                            | Sensitive           | Resistant                                                          |

**HIGH-YIELD — READ THE TABLE AS ONE SENTENCE**

Do not memorise twelve rows. Memorise the **second row** — water falls from 80–90% to 10–25% — and derive the other eleven. No water means no enzyme activity, no respiration, no synthesis; and protein denaturation, which is how heat kills, **requires water**, so a dehydrated core is heat-resistant. Calcium dipicolinate and the SASPs are simply the machinery that achieves and stabilises that dryness.

**MNEMONIC**

Spore-formers are "**BC**" — **Bacillus** and **Clostridium**. Both **Gram-positive rods**. Nothing Gram-negative forms an endospore.

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## PART FIVE

# 05

# Putting It Together in the Laboratory

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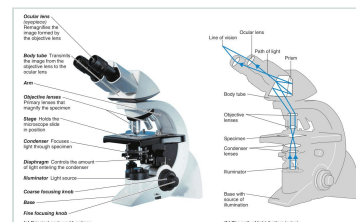
Everything so far has been structure. This Part is the payoff: how a smear of pus becomes a named organism and a prescription. Work through the flowcharts *actively* — cover the branches and decide each step yourself. A flowchart you have read teaches nothing; a flowchart you have run teaches the whole subject.

### BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Perform and interpret a Gram stain, and recognise the four ways it misleads.
- Choose the right special stain for an organism the Gram stain cannot show.
- Identify a Gram-positive coccus and a Gram-negative rod to genus using catalase, coagulase, haemolysis, oxidase and lactose fermentation.

## 5.1 Seeing Them — The Microscope

The lecture makes the scale problem concrete: comparing a bacterium with a football is like comparing a football with Mount Everest. Bacteria sit at about  $10^{-6}$  m; a football at  $10^{-1}$ ; Everest at  $10^4$ . Nothing at that size is available to the naked eye, so every fact in this guide was bought with an instrument — and each instrument buys a different fact.



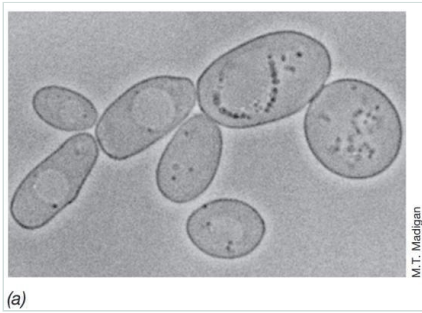
**Fig 5.1** The compound light microscope. Total magnification is the product of the two lenses: a **100× oil-immersion objective** with a **10× ocular** gives **1000×**, which is the standard for reading a Gram stain and the practical limit of light optics.

**TABLE 5.1** THE SIX INSTRUMENTS, AND WHAT EACH ONE IS FOR

| Instrument                   | What it does                                                                               | When you would reach for it                                                                                                                                             |
|------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Light (bright-field)</b>  | <b>1000×</b> — 100× objective × 10× ocular                                                 | Every stained smear in clinical practice. Requires the cells to be <b>killed and stained</b> to have any contrast.                                                      |
| <b>Phase contrast</b>        | Converts differences in refractive index into differences in brightness                    | When you must see <b>living</b> cells — motility, and the refractile endospore, which needs no stain at all.                                                            |
| <b>Fluorescence</b>          | Excites a fluorescent dye or protein and images what it emits                              | <b>Auramine–rhodamine</b> screening for tuberculosis; direct fluorescent antibody tests; any labelled probe.                                                            |
| <b>Confocal</b>              | Rejects out-of-focus light and stacks optical slices into a <b>3D image</b> (z-stack)      | Anything with depth — most importantly, imaging the architecture of a <b>biofilm</b> .                                                                                  |
| <b>TEM (transmission EM)</b> | Electrons pass <i>through</i> an ultrathin section; resolves particles down to <b>1 nm</b> | <b>Internal</b> structure — this is how the layers of the envelope and of the endospore were established. Every envelope diagram in Part 3 is a drawing of a TEM image. |
| <b>SEM (scanning EM)</b>     | Scans the <i>surface</i> and builds a <b>three-dimensional</b> image of it                 | <b>Surface</b> structure — flagellar arrangement, fimbriae, cells attached to a catheter.                                                                               |

### HIGH-YIELD — TEM VERSUS SEM IN ONE LINE

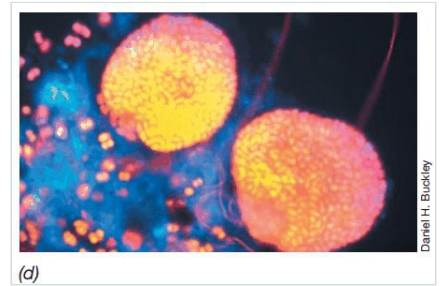
**Transmission** goes **through** — internal structure, flat section, best resolution (1 nm). **Scanning** scans the **surface** — external shape, 3D. If the question shows you a layered envelope, it is TEM; if it shows you a cell bristling with flagella, it is SEM.



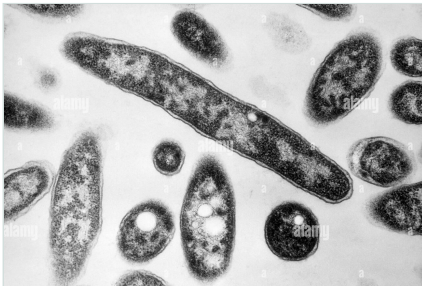
**Fig 5.2 Bright-field** — contrast only where a stain has been deposited.



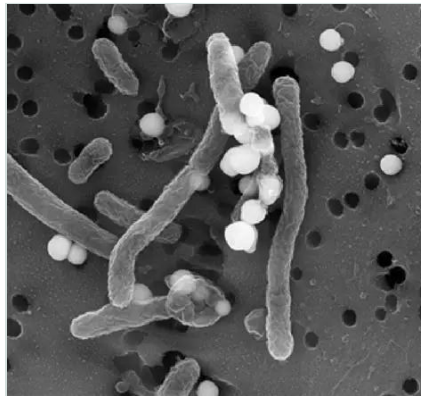
**Fig 5.3 Phase contrast** — the same cells alive and unstained; internal bodies now visible.



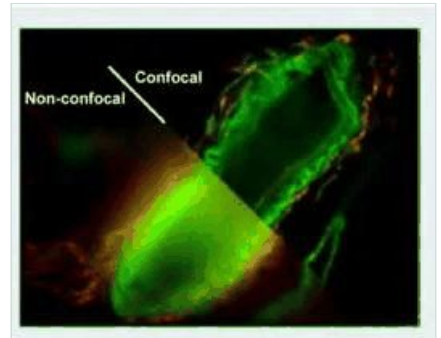
**Fig 5.4 Fluorescence** — only what carries the dye is seen at all.



**Fig 5.5 TEM** — a section through the cell. Internal layers, no surface relief.



**Fig 5.6 SEM** — the outside of the same kind of cell, in relief.



**Fig 5.7 Confocal** — optical slices stacked, so depth is preserved.

#### LAB PEARL — THE FOUR SPECIAL STAINS, AND WHY EACH EXISTS

Beyond the two **differential** stains (Gram and acid-fast), the course names four **special** stains, and each one exists because its target is invisible to the Gram stain: the **spore stain** (the spore coat excludes ordinary dyes), the **flagella stain** (the filament is below the resolution limit until a mordant thickens it), the **capsule stain** (the capsule takes up no dye, so it is shown as a clear halo against a stained background), and **DNA stains** such as DAPI or acridine orange (the nucleoid has no contrast of its own). If you can say what each stain is *compensating for*, you will never confuse them.

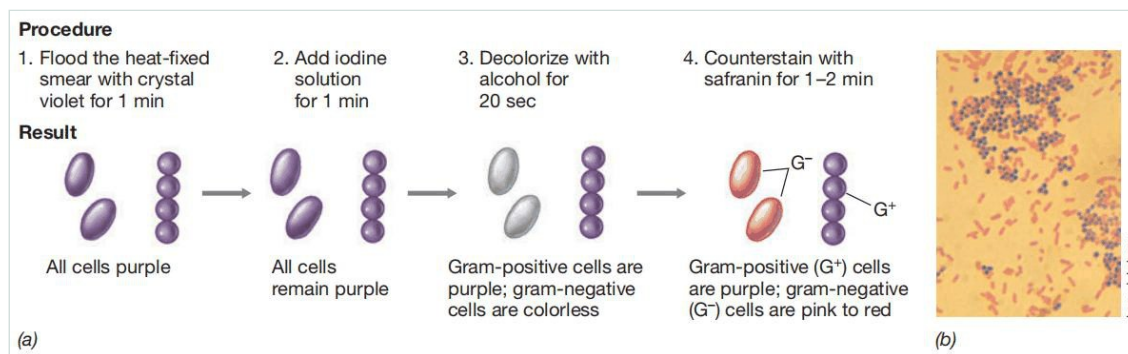
## 5.2 Performing the Gram Stain

TABLE 5.2 THE GRAM STAIN AT THE BENCH

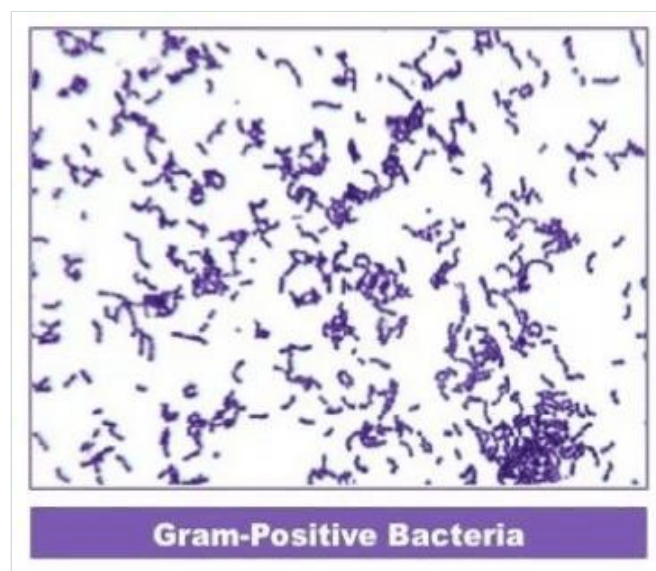
| Step    | Reagent                         | Time               | What is happening                                                                                                                     |
|---------|---------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Prepare | Smear, air-dry, <b>heat-fix</b> | —                  | Fixes organisms to the slide so they are not washed off, and kills them. <b>A too-thick smear is the commonest preparation error.</b> |
| 1       | <b>Crystal violet</b>           | ~1 min             | Primary stain — <b>all</b> cells become violet                                                                                        |
| 2       | <b>Gram's iodine</b>            | ~1 min             | Mordant — forms the large CV-I complex                                                                                                |
| 3       | <b>Alcohol / acetone</b>        | <b>5–10 s only</b> | <b>Decoloriser — the critical, time-sensitive step.</b> Stop the instant the runoff is clear                                          |
| 4       | <b>Safranin</b>                 | ~1 min             | Counterstain — decolorised cells become pink/red                                                                                      |

### LAB PEARL — READING THE SLIDE LIKE A CLINICIAN

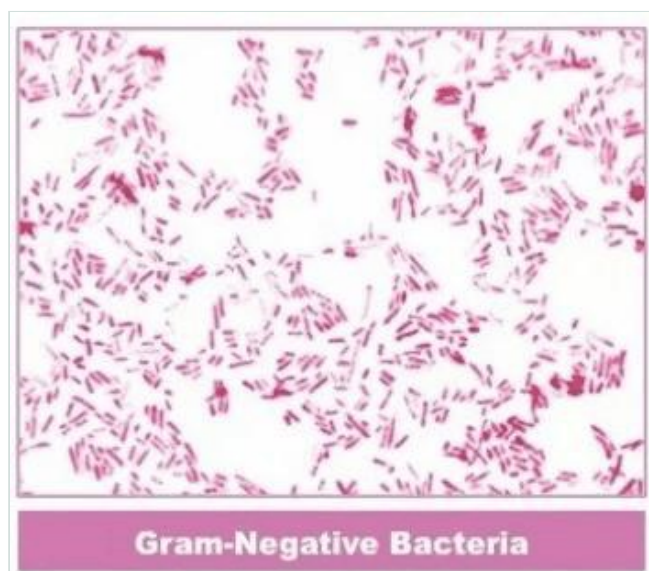
Report **three things, always in this order: Gram reaction → shape → arrangement**. "Gram-positive cocci in clusters" is actionable; "bacteria seen" is not. Also read the **host cells**: abundant neutrophils confirm genuine infection rather than colonisation, and abundant **squamous epithelial cells in a sputum sample means the specimen is saliva** and should be rejected — a real clinical decision made on a Gram stain alone.



**Fig 5.8** The bench procedure with the result at each step. Note that after step 2 **every** cell is purple — the stain has not discriminated anything yet. All the information is created in step 3.



**Fig 5.9 Gram-positive** — crystal violet retained.

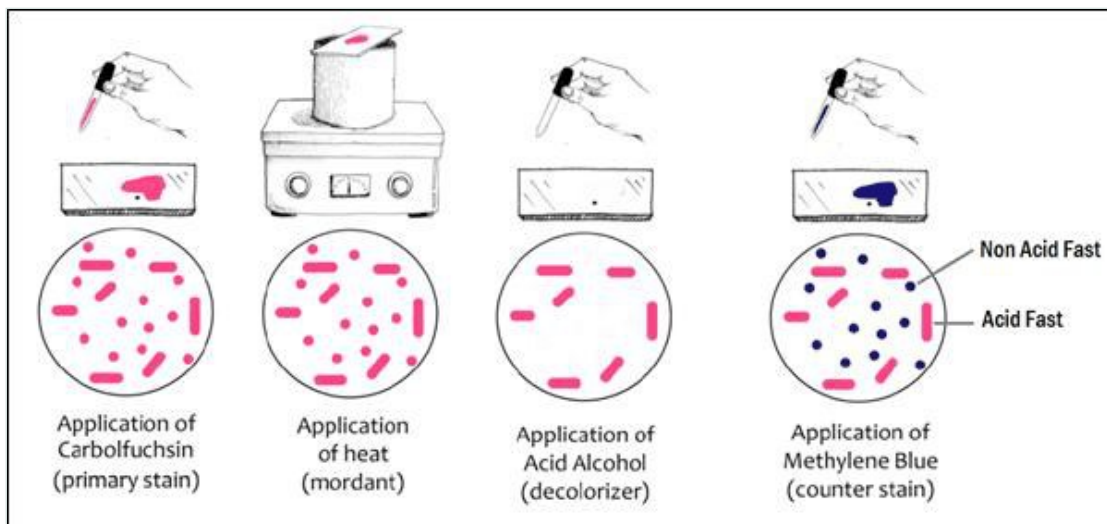


**Fig 5.10 Gram-negative** — decolourised, then counterstained by safranin.

## 5.3 Special Stains and When to Order Them

**TABLE 5.3 STAIN → TARGET → RESULT**

| Stain                              | Demonstrates                    | Result and when you would order it                                                                                     |
|------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Gram                               | Cell wall thickness             | Violet = G+, pink = G-. <b>The first test on any specimen.</b>                                                         |
| Ziehl–Neelsen / Kinyoun            | <b>Mycolic acid</b> in the wall | <b>Red bacilli on blue.</b> Order for suspected TB, atypical mycobacteria, or a chronic cough with a blank Gram stain. |
| Modified acid-fast                 | Weakly acid-fast organisms      | <i>Nocardia</i> , <i>Cryptosporidium</i> , <i>Cyclospora</i>                                                           |
| India ink / nigrosin               | <b>Capsule</b> (negative stain) | <b>Clear halo</b> on a dark background. Classic for <i>Cryptococcus neoformans</i> in CSF.                             |
| Albert / Loeffler methylene blue   | <b>Metachromatic granules</b>   | Reddish-purple granules in <i>C. diphtheriae</i> <b>G+</b>                                                             |
| Malachite green (Schaeffer–Fulton) | <b>Endospores</b>               | <b>Green spores, pink vegetative cells.</b> Confirms <i>Bacillus</i> or <i>Clostridium</i> .                           |
| Silver (Warthin–Starry)            | Very thin organisms             | <i>Treponema</i> , <i>Legionella</i> , <i>Bartonella</i> , <i>H. pylori</i>                                            |
| Dark-field microscopy              | Live spirochaetes               | Motile corkscrews from a primary chancre — <i>T. pallidum</i>                                                          |
| Giemsa                             | Intracellular organisms         | <i>Chlamydia</i> inclusions, <i>Rickettsia</i> , <i>Borrelia</i> , <i>Plasmodium</i>                                   |
| KOH / calcofluor white             | <b>Fungal</b> chitin            | Hyphae and yeasts — order when fungal infection is suspected                                                           |



**Fig 5.11** The Ziehl-Neelsen sequence. **Carbol fuchsin** is the primary stain, **heat** is the mordant that drives it through the wax, **acid-alcohol** is the decolouriser — and the organisms that hold on through it are, by definition, acid-fast. **Methylene blue** counterstains everything else. The structure of the method is identical to the Gram stain; only the third step's severity differs.

## 5.4 The Identification Flowcharts

These are the actual decision paths a clinical laboratory follows, in the order the tests are performed. **Cover the right-hand side and work down each branch yourself.**

## FLOWCHART 1 · GRAM-POSITIVE COCCI

## GRAM-POSITIVE COCCI

## → CATALASE

(add  $H_2O_2$  – bubbles = positive)— CATALASE + → *Staphylococcus***G+**

(clusters) →

## COAGULASE

— COAGULASE + → *Staphylococcus aureus* – golden colonies,  $\beta$ -haemolytic, mannitol-fermenting

— COAGULASE – →

## NOVOBIOCIN

— Sensitive → *S. epidermidis* – biofilm on prosthetic material— Resistant → *S. saprophyticus* – UTI in young sexually active women— CATALASE – → *Streptococcus* / *Enterococcus***G+**

(chains or pairs) →

## HAEMOLYSIS ON BLOOD AGAR

—  $\beta$  (complete, clear) →

## BACITRACIN

— Sensitive → *S. pyogenes* (Group A) – pharyngitis, scarlet fever, necrotizing fasciitis, **puerperal fever**— Resistant → *S. agalactiae* (Group B) – neonatal sepsis and meningitis; CAMP test positive—  $\alpha$  (partial, green) →

## OPTOCHIN

— Sensitive, bile-soluble → *S. pneumoniae* – lancet-shaped diplococci, **encapsulated**; pneumonia, meningitis, otitis— Resistant, bile-insoluble → *Viridans streptococci* – dental caries, subacute endocarditis—  $\gamma$  (none) →

## GROWTH IN 6.5% NaCl

— Grows → *Enterococcus faecalis* / *faecium* – UTI, biliary, endocarditis; VRE risk— No growth → *Streptococcus bovis* (*S. gallolyticus*) – **associated with colon carcinoma**

## FLOWCHART 2 · GRAM-NEGATIVE RODS

## GRAM-NEGATIVE RODS

→

## OXIDASE

## — OXIDASE +

- Grows on MacConkey, non-lactose-fermenting, **blue-green pigment, grape-like odour** → *Pseudomonas aeruginosa* — burns, ventilator pneumonia, otitis externa
- Comma-shaped, alkaline media, grows in salt → *Vibrio cholerae* — rice-water diarrhoea
- Curved, urease +, **gastric mucosa** → *Helicobacter pylori* — gastritis, peptic ulcer, gastric carcinoma
- Grows at 42 °C, microaerophilic → *Campylobacter jejuni* — bloody diarrhoea, Guillain-Barré

— OXIDASE - → likely **Enterobacteriaceae** →

## LACTOSE FERMENTATION ON MacCONKEY

## — Lactose + (pink colonies)

- Motile, indole +, green metallic sheen on EMB → *Escherichia coli* — UTI, neonatal meningitis, diarrhoea
- **Non-motile, mucoid encapsulated colonies** → *Klebsiella pneumoniae* — "currant-jelly" sputum, alcoholics, aspiration
- Slow lactose fermenter → *Enterobacter*, *Serratia* (red pigment)

## — Lactose - (colourless colonies) →

H<sub>2</sub>S ON TSI / UREASE

- H<sub>2</sub>S +, motile → *Salmonella* — enteric fever, gastroenteritis
- H<sub>2</sub>S -, non-motile → *Shigella* — bacillary dysentery, very low infectious dose
- **Urease +, swarming motility** → *Proteus mirabilis* — struvite renal stones, alkaline urine

## FLOWCHART 3 · WHEN THE GRAM STAIN SHOWS NOTHING

## SICK PATIENT · GRAM STAIN NEGATIVE OR UNINFORMATIVE

- Chronic cough, weight loss, night sweats, cavitary lung lesion → **ZIEHL-NEELSEN** → *Mycobacterium tuberculosis* **AFB**
- Atypical "walking" pneumonia, young adult, no wall to stain → **SEROLOGY / PCR** → *Mycoplasma pneumoniae* — **β-lactams will not work**
- Painless genital ulcer, thin spiral organism → **DARK-FIELD / VDRL / FTA-ABS** → *Treponema pallidum*
- Pneumonia + hyponatraemia + diarrhoea, air conditioning exposure → **URINE ANTIGEN / SILVER STAIN** → *Legionella pneumophila*
- Immunocompromised, headache, CSF with clear haloes on India ink → **INDIA INK / CrAg** → *Cryptococcus neoformans* (a fungus)

**HIGH-YIELD — THE SIX TESTS THAT DO MOST OF THE WORK**

**Catalase** separates staphylococci (+) from streptococci (-). **Coagulase** separates *S. aureus* (+) from the rest. **Haemolysis** splits streptococci into  $\alpha/\beta/\gamma$ . **Oxidase** separates *Pseudomonas* and the curved rods (+) from Enterobacteriaceae (-). **Lactose fermentation** splits the Enterobacteriaceae. **Urease** flags *Proteus* and *H. pylori*. Memorise these six and you can place most clinical isolates without any further information.

**LAB PEARL — WHY BLOOD AGAR AND CHOCOLATE AGAR ARE DIFFERENT**

**Blood agar** contains intact red cells, so it reveals **haemolysis patterns** — essential for classifying streptococci. **Chocolate agar** is blood agar that has been gently *heated*, lysing the red cells and releasing two things trapped inside them: **factor X (haemin)** and **factor V (NAD)**. *Haemophilus influenzae* **G-** requires both and therefore **grows on chocolate but not on blood agar** — the classic exam pairing. **MacConkey agar** is both *selective* (bile salts and crystal violet inhibit Gram-positives) and *differential* (lactose fermenters turn pink). Choosing the medium is part of the diagnosis.

# 06

## Review Questions, Master Tables and Glossary

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Thirty vignettes, weighted deliberately towards the format this subject is actually tested in: *a patient, a Gram stain, and a biochemical result* — *now name the organism or the mechanism*. Every answer explains the correct option and why each distractor fails, because in microbiology the distractors are usually real organisms that a careless reader would accept.

### HOW TO USE THIS SECTION

- Answer all 30 **closed-book** and write your answers down before checking.
- Read **every** explanation, including for questions you got right.
- For each one you miss, return to the numbered section named in the explanation.

## 6.1 Thirty Vignettes

### PART 1 — THE MICROBIAL WORLD (Q1–5)

#### Q1 CLASSIFICATION

A laboratory receives four specimens. Analysis shows: specimen A contains organisms with 80S ribosomes and a chitin cell wall; B contains organisms with 70S ribosomes and peptidoglycan; C contains particles with RNA and a protein coat but no ribosomes; D contains multicellular organisms up to 30 cm long. Which specimen contains a prokaryote?

- A. Specimen D
- B. Specimens A and D
- C. Specimen B
- D. Specimen A
- E. Specimen C

#### Q2 ANTIMICROBIAL LOGIC

A patient with influenza is prescribed amoxicillin by a well-meaning relative. Which feature of viruses best explains why this drug cannot work?

- A. Viruses are protected by the host cell membrane
- B. Viruses have no peptidoglycan cell wall for the  $\beta$ -lactam ring to target
- C. Viruses replicate too rapidly for any antibiotic to act
- D. Viruses have 80S ribosomes that  $\beta$ -lactams cannot bind
- E. Viral capsids are impermeable to all drugs

#### Q3 NORMAL FLORA

A 70-year-old man receives 10 days of broad-spectrum antibiotics for pneumonia. He then develops profuse watery diarrhoea and abdominal pain; colonoscopy shows yellow pseudomembranes. Which principle best explains this complication?

- A. The patient acquired a new virulent strain from the hospital water supply
- B. Antibiotic-induced immunosuppression permitted viral reactivation
- C. Loss of vitamin K synthesis caused mucosal haemorrhage
- D. Loss of colonisation resistance allowed a spore-forming organism that survived the antibiotic to overgrow
- E. The antibiotic was directly toxic to colonic epithelium

#### Q4 NOMENCLATURE

A student writes a case report describing infection with "*staphylococcus Aureus*". Which correction is needed?

- A. *Staphylococcus aureus* — genus capitalised, species lower case, both italic
- B. *staphylococcus aureus* — both lower case, both italic
- C. *Staphylococcus Aureus* — both capitalised
- D. Staphylococcus aureus — neither italicised
- E. No correction is needed

**Q5** EUKARYOTE/PROKARYOTE

Which group of microbial agents is eukaryotic?

- A. Bacteria
- B. Viruses
- C. Archaea
- D. Bacteriophages
- E. Fungi

**PART 2 — HOW WE LEARNED IT (Q6–11)****Q6** EXPERIMENTAL DESIGN

In Redi's experiment, meat was placed in three jars: open, sealed, and covered with cloth netting. Maggots appeared only in the open jar. Why was the netted jar essential to the conclusion?

- A. It sterilised the meat before the experiment
- B. It admitted air while excluding flies, eliminating the objection that sealing had blocked a "vital force"
- C. It kept the meat at a constant temperature
- D. It allowed maggots to be counted more accurately
- E. It prevented the meat from drying out

**Q7** KOCH'S POSTULATES

Which statement is **NOT** part of Koch's postulates?

- A. The microorganism and disease can be identified from a mixed culture
- B. The same microorganism must be present in every case of the disease
- C. The microorganism must be isolated from the diseased host and grown in pure culture
- D. The pure-cultured organism must cause the same disease in a healthy susceptible animal
- E. The organism must be re-isolated from the experimentally infected animal

**Q8** LIMITS OF THE POSTULATES

A researcher cannot satisfy Koch's postulates for *Treponema pallidum*. Which postulate fails, and why?

- A. Postulate 3 — no animal is susceptible to any treponeme
- B. Postulate 4 — the organism cannot be recovered after inoculation
- C. All four postulates are satisfied for this organism
- D. Postulate 2 — the organism has never been grown in pure culture on artificial media
- E. Postulate 1 — the organism is absent from many cases of syphilis

**Q9** VACCINES

Pasteur's assistant left chicken cholera cultures to age over a holiday. Chickens inoculated with the old culture became mildly ill and recovered; when later given fresh virulent culture, they survived. What had happened?

- A. The fresh culture had itself become attenuated in storage
- B. The old culture killed the chickens' normal flora, preventing infection
- C. The bacteria had lost virulence but retained their surface antigens, generating protective immune memory
- D. The old culture contained antibiotics produced by contaminating moulds
- E. The chickens had developed a different infection that cross-protected them

**Q10**

HISTORY PAIRING

What is the name of the field of study established by John Snow in the mid-1800s?

- A. Immunology
- B. Bacteriology
- C. Virology
- D. Taxonomy
- E. Epidemiology

**Q11**

DETERMINANTS

Which is **NOT** a main determinant of our success in limiting infectious diseases?

- A. Population growth
- B. Improved hygiene conditions
- C. Vaccination
- D. Antibiotics
- E. Clean water supply

### PART 3 — THE CELL ENVELOPE (Q12–20)

**Q12**

GRAM STAIN MECHANISM

During a Gram stain, a technician leaves the acetone-alcohol on the slide for 45 seconds instead of 5. A known *Staphylococcus aureus* control now appears pink. What has happened?

- A. The iodine mordant was omitted
- B. Over-decolorisation washed the crystal violet-iodine complex out of the thick peptidoglycan
- C. The organism has mutated to a Gram-negative envelope
- D. The safranin was applied before the alcohol
- E. Prolonged alcohol exposure synthesises lipopolysaccharide

**Q13**

ENVELOPE STRUCTURE

Which structure is found in Gram-positive but never in Gram-negative bacteria?

- A. Cytoplasmic membrane
- B. 70S ribosomes
- C. Lipopolysaccharide
- D. Teichoic acid
- E. Peptidoglycan

**Q14**

DRUG MECHANISM

Penicillin binds an enzyme because the drug's  $\beta$ -lactam ring structurally mimics a natural substrate. Which substrate?

- A. Bactoprenol
- B. Lipoteichoic acid
- C. D-alanyl-D-alanine at the end of the peptidoglycan pentapeptide
- D. N-acetylmuramic acid
- E. Lipid A of lipopolysaccharide

**Q15**

SEPSIS

A 62-year-old with an *E. coli* urinary infection becomes febrile and hypotensive with warm peripheries, a high cardiac output and low systemic vascular resistance. Platelets fall and PT and aPTT lengthen. Which molecule initiated this cascade, and via which receptor?

- A. Lipoteichoic acid, via TLR2
- B. Flagellin, via TLR5
- C. Peptidoglycan, via NOD2
- D. O antigen, via CD14 alone
- E. Lipid A of LPS, via TLR4

**Q16**

TOXIN COMPARISON

A toxin is heat-stable at 100 °C, released only when the organism lyses, produces the same clinical syndrome regardless of the species involved, and cannot be converted into a toxoid vaccine. What is it?

- A. Endotoxin (lipopolysaccharide)
- B. Diphtheria exotoxin
- C. Tetanospasmin
- D. Cholera enterotoxin
- E. Botulinum toxin

**Q17**

ACID-FAST

A 40-year-old with HIV has a chronic cough, weight loss and night sweats. Sputum Gram stain shows neutrophils but no organisms. Which property explains the negative Gram stain, and what should be ordered?

- A. The specimen was over-decolorised; repeat the Gram stain
- B. A mycolic acid-rich waxy wall excludes aqueous crystal violet; order a Ziehl-Neelsen stain
- C. The organism has no cell wall; order serology for *Mycoplasma*
- D. The organism is too thin to see; order dark-field microscopy
- E. A thick capsule masks the wall; order an India ink preparation

**Q18**

INTRINSIC RESISTANCE

A young adult has atypical pneumonia. Cultures on routine media are sterile and no organism is seen on Gram stain. Empirical amoxicillin fails completely, but azithromycin produces rapid improvement. Which organism property explains the amoxicillin failure?

- A. It has lost the porin through which amoxicillin enters
- B. Its ribosomes are 80S and therefore resistant
- C. It forms endospores that survive  $\beta$ -lactam exposure
- D. It has no cell wall, so there is no peptidoglycan for a  $\beta$ -lactam to attack
- E. It produces an extended-spectrum  $\beta$ -lactamase

**Q19**

RESISTANCE MECHANISM

An enterococcal isolate is resistant to vancomycin. Sequencing shows the peptidoglycan pentapeptide now terminates in D-alanyl-D-lactate instead of D-alanyl-D-alanine. Why does this confer resistance?

- A. The altered peptide activates an efflux pump
- B. It converts the organism to a Gram-negative envelope
- C. Vancomycin binds the D-Ala-D-Ala terminus directly, and the altered terminus no longer binds it
- D. The change destroys the  $\beta$ -lactam ring of vancomycin
- E. D-lactate blocks the porin through which vancomycin enters

**Q20**

SELECTIVE TOXICITY

Aminoglycosides kill bacteria without killing human cells. Which structural difference is responsible?

- A. Bacteria lack a nuclear membrane
- B. Bacterial membranes contain ergosterol
- C. Bacteria have circular rather than linear DNA
- D. Bacteria divide by binary fission
- E. Bacterial ribosomes are 70S (50S + 30S); human ribosomes are 80S

**PART 4 — EXTERNAL AND INTERNAL STRUCTURES (Q21–26)****Q21**

CAPSULE

A 6-year-old with sickle cell disease and a functionally absent spleen develops fulminant pneumococcal sepsis. Which bacterial structure best explains why asplenia predisposes so strongly to this organism?

- A. The polysaccharide capsule, which blocks C3b deposition and phagocytosis
- B. The lipopolysaccharide outer membrane
- C. Endospores resistant to splenic macrophages
- D. Flagella allowing escape from the spleen
- E. Mycolic acid permitting intracellular survival

**Q22**

PILI

Which statement does **NOT** apply to pili?

- A. Pili are shorter and thinner than flagella
- B. Pili facilitate nutrient transport
- C. Pili are made of protein
- D. Pili allow attachment to surfaces
- E. Pili contain adhesins

**Q23**

SPORULATION

Which is **NOT** an event of sporulation?

- A. Prespore engulfment by the mother cell
- B. Mother cell disintegration
- C. Deposition of calcium dipicolinate in the core
- D. Symmetrical cell division
- E. DNA replication

**Q24**

SPORE RESISTANCE

Endospores survive boiling for hours. Which feature of the spore core is the principal reason heat fails to kill them?

- A. Constitutive production of heat-shock proteins
- B. Active efflux of heat-damaged proteins
- C. Extreme dehydration plus calcium dipicolinate — protein denaturation requires water
- D. A thick polysaccharide capsule
- E. An outer membrane rich in lipopolysaccharide

**Q25**

PLASMIDS

Which statement about plasmids is correct?

- A. They replicate together with the bacterial chromosome
- B. They contain information essential for growth
- C. They are as large as the bacterial chromosome
- D. They are found only in Gram-positive bacteria
- E. They may contain antibiotic resistance genes

**Q26**

BIOFILM

A patient with a prosthetic hip develops chronic low-grade infection with *Staphylococcus epidermidis*. Six weeks of appropriate intravenous antibiotics fail, and the prosthesis is removed. Which property best explains the failure of medical therapy alone?

- A. Biofilm matrix limits drug penetration and the deep cells are dormant, escaping growth-dependent antibiotics
- B. The organism produces a potent endotoxin
- C. The organism has formed endospores on the implant
- D. The organism lacks a cell wall and resists  $\beta$ -lactams
- E. The prosthesis inactivates antibiotics chemically

#### PART 5 — THE LABORATORY (Q27–30)

**Q27**

IDENTIFICATION

Pus from a skin abscess grows organisms that are Gram-positive cocci in clusters, **catalase positive** and **coagulase positive**, forming golden  $\beta$ -haemolytic colonies. What is the organism?

- A. *Enterococcus faecalis*
- B. *Staphylococcus aureus*
- C. *Staphylococcus epidermidis*
- D. *Streptococcus pyogenes*
- E. *Streptococcus pneumoniae*

**Q28**

CULTURE MEDIA

A child with meningitis has CSF growing small Gram-negative coccobacilli on **chocolate agar but not on blood agar**. Which explanation is correct?

- A. The organism is an obligate anaerobe and blood agar is too aerobic
- B. Blood agar inhibits capsule formation
- C. Chocolate agar contains antibiotics that suppress competing flora
- D. The organism requires factors X and V, released only when heating lyses the red cells
- E. Chocolate agar is selective for Gram-negative organisms by bile salts

**Q29**

MICROSCOPY

A researcher wants to watch whether a **living** suspension of *Bacillus cereus* is motile, and in the same preparation to count how many cells contain a refractile endospore. No stain may be used, because staining would kill the cells. Which instrument is appropriate?

- A. Scanning electron microscope
- B. Bright-field light microscope at 1000×
- C. Phase contrast microscope
- D. Transmission electron microscope
- E. Fluorescence microscope

**Q30**

OSMOSIS

A Gram-positive coccus is treated with **lysozyme** in an isotonic sucrose solution. The cells survive as wall-less spheres. The suspension is then diluted into **distilled water** and the cells immediately burst. Which statement best explains the two observations together?

- A. Lysozyme is only active in hypotonic conditions, so it had no effect until water was added
- B. Distilled water inactivates the cytoplasmic membrane's transport proteins
- C. Sucrose is actively transported into the cell and prevents lysis by raising internal osmolarity
- D. Lysozyme cleaves the peptide cross-links, and distilled water reverses the reaction
- E. The wall, not the membrane, resists osmotic pressure; without it the cell survives only while the medium is isotonic

## 6.2 Answer Key with Full Explanations

**Q1 — C**

**Specimen B. 70S ribosomes and peptidoglycan** are the two defining prokaryotic markers, and both point to bacteria.

**C** 80S ribosomes and chitin define a *fungus* — eukaryotic. **E** Nucleic acid in a protein coat with no ribosomes is a *virus*, which is neither prokaryote nor eukaryote because it is not a cell. **A** A 30 cm multicellular organism is a helminth — eukaryotic. **B** combines two eukaryotes.

See §1.2, §3.2.

**Q2 — B**

**Viruses have no peptidoglycan.**  $\beta$ -lactams work by irreversibly binding transpeptidase (PBP) and blocking peptidoglycan cross-linking. A virus has no cell wall, no PBPs and no wall synthesis, so there is literally nothing for the drug to bind. This is the clearest illustration of the principle underlying all antimicrobial therapy: **selective toxicity requires a target the pathogen has and the host does not** — and a virus, using the host's own machinery, presents almost no such target.

**C** Replication rate is irrelevant;  $\beta$ -lactams in fact work best on rapidly growing cells. **D** Viruses have *no* ribosomes at all. **E** Capsid permeability is not the issue. **A** Viral location does not protect against a drug with no target.

See §1.2, §3.5.

**Q3 — D** **Loss of colonisation resistance.** This is *Clostridioides difficile* **G+** colitis. The normal colonic flora occupies attachment sites and consumes nutrients, physically denying a foothold to competitors. Broad-spectrum antibiotics destroy that flora — but **cannot kill *C. difficile* spores**, which are not metabolically active. When the antibiotic stops, the spores germinate into an emptied colon with no competition and produce toxins A and B, giving pseudomembranous colitis.

**E** Direct epithelial toxicity does not produce pseudomembranes. **A** Acquisition can occur, but the essential permissive event is flora depletion. **B** Antibiotics do not cause meaningful immunosuppression. **C** Vitamin K loss causes coagulopathy, not colitis.

See §1.4, §4.6.

**Q4 — A** ***Staphylococcus aureus*.** Binomial nomenclature, from Linnaeus: **genus capitalised, species lower case, both italicised** (or underlined if handwritten). After first mention the genus may be abbreviated — *S. aureus*.

**B, C, D** each violate one or both halves of the rule. **E** is wrong because the original has two errors: the genus is not capitalised and the species is.

See §1.3.

**Q5 — E** **Fungi.** Fungi and parasites are eukaryotic: true nucleus, membrane-bound organelles, 80S ribosomes.

**B, D** Bacteria and archaea are *prokaryotic*. **C, E** Viruses and bacteriophages are acellular and therefore neither.

This is a verbatim lecture MCQ. See §1.2.

**Q6 — B** **It admitted air while excluding flies.** A sealed jar alone proves nothing, because a defender of spontaneous generation can always reply that sealing excluded the air or "vital force" life needs to form. The netted jar **grants that premise** — air passes freely — while still excluding flies. When the netted meat stayed maggot-free, that objection was eliminated. **Redi's real contribution was the control, not the observation.**

**B, D** Temperature and moisture were not the variables under test. **D** Counting accuracy is irrelevant to the logic. **A** The meat was not sterilised — that was Pasteur's later refinement.

See §2.1.

**Q7 — A** **Identification from a mixed culture is NOT a postulate.** Postulate 2 explicitly requires a **pure** culture, and this is the entire point: in a mixed culture any of several organisms could be the cause, so causation cannot be attributed. It is precisely why Koch had to invent **solid agar media and the Petri dish** before he could formulate the postulates at all — a liquid culture cannot be separated into single colonies.

**B, C, D, E** are postulates 1, 2, 3 and 4 respectively, correctly stated.

See §2.3.

**Q8 — D** **Postulate 2 fails.** *Treponema pallidum* **G-** has never been cultivated on artificial media; it is maintained only in animal (rabbit) passage. Since the organism cannot be grown in pure culture, postulates 2, 3 and 4 all become untestable in the classical form.

**E** False — the organism is present in syphilitic lesions, demonstrable by dark-field microscopy. **A** False — rabbits are susceptible and are used experimentally. **B** Recovery is not the failing step. **C** False, as above.

The same limitation applies to *M. leprae* and to all viruses, and is why **molecular Koch's postulates** were developed. See §2.3.

**Q9 — C** **Attenuation.** Prolonged ageing outside a host selects for variants adapted to laboratory conditions rather than to invading tissue, so virulence genes are lost or downregulated — but the **surface antigens are unchanged**. The chickens therefore mounted a full adaptive response, generating memory B and T cells, without suffering lethal disease. On later challenge with fully virulent organisms those memory cells responded immediately. Pasteur named these preparations **"vaccines" in 1879**, in homage to Jenner's cowpox, and went on to apply the principle to anthrax and rabies.

**D** No antibiotic was involved — that is Fleming's story, fifty years later. **E** No cross-protecting infection was present. **A** The fresh culture was demonstrably virulent — it killed unvaccinated controls. **B** Normal flora plays no role here.

See §2.4.

**Q10 — E Epidemiology.** In 1854 Snow mapped cholera deaths in Soho, traced the cluster to the Broad Street pump, and had the handle removed. He did this **29 years before Koch isolated *Vibrio cholerae*** — reasoning entirely from the distribution of disease rather than from the organism.

**A** Immunology is Jenner. **B** Bacteriology is Koch. **C** Virology postdates all of them. **D** Taxonomy is Linnaeus. See §2.2.

**Q11 — A Population growth.** The three determinants the lecture names are **improved hygiene conditions, vaccination and antibiotics**. Population is the classic distractor and is not a determinant of success — if anything, dense populations *increase* transmission.

**B, C, D** are the three determinants verbatim. **E** Clean water is a component of improved hygiene, so it belongs to determinant 1.

Note the ordering matters historically: mortality curves fell steeply with sanitation *decades before* penicillin. See §2.6.

**Q12 — B Over-decolorisation.** Alcohol dehydrates Gram-positive peptidoglycan and closes its pores, trapping the crystal violet-iodine complex — but that trapping is **time-dependent, not absolute**. Given long enough, the complex leaches out even from a thick wall, and the cell then takes up safranin and reports falsely as Gram-negative. This is **the commonest technical error in the Gram stain**, which is why decolorisation is stopped the instant the runoff runs clear.

**C** Envelope architecture is genetically fixed; a cell cannot become Gram-negative in one stain. **D** Reversing the order would leave the cells unstained, not pink. **E** LPS is not synthesised by Gram-positives at all. **A** Omitting iodine would cause loss of violet from *both* organisms, not a technique-dependent error.

See §3.8, §5.2.

**Q13 — D Teichoic acid.** Wall teichoic acid and lipoteichoic acid are polymers of glycerol or ribitol phosphate found **exclusively in Gram-positive bacteria**. LTA is anchored in the cytoplasmic membrane, WTA in the peptidoglycan itself.

**B, C, D** Peptidoglycan, the cytoplasmic membrane and 70S ribosomes are present in *both* — peptidoglycan differs in thickness, not in presence. **C** LPS is the exact converse: found only in Gram-negatives. Remember the pair together — teichoic acid ↔ G<sup>+</sup>, LPS ↔ G<sup>-</sup> — and never in the other.

See §3.6.

**Q14 — C D-alanyl-D-alanine.** The β-lactam ring is a structural mimic of the terminal D-Ala-D-Ala of the peptidoglycan pentapeptide. Transpeptidase (a penicillin-binding protein) normally cleaves that terminal D-Ala and uses the released energy to form the cross-link; instead it binds the drug and is **irreversibly acylated**. Cross-linking stops while autolysins keep cutting, the wall weakens, and the hypertonic cell bursts. Note that bacteria use the **D-isomer**, which human enzymes neither make nor cleave — that is what makes the whole pathway selectively attackable.

**D** NAM is a backbone sugar, not the transpeptidase substrate. **E** Lipid A is endotoxin, unrelated. **A** Bactoprenol is the lipid carrier — the target of bacitracin, not β-lactams. **B** LTA is not involved in cross-linking.

See §3.5.

**Q15 — E Lipid A via TLR4.** The picture is Gram-negative septic shock. LPS is shed as *E. coli* lyses; **lipid A** is the toxic moiety, carried by LPS-binding protein to CD14 and then to **TLR4**. TLR4 signalling activates NF-κB and releases TNF-α, IL-1 and IL-6. These produce **fever** (hypothalamic PGE<sub>2</sub>), **hypotension with warm peripheries** (inducible NO synthase → vasodilation → distributive shock with low SVR and high cardiac output), and **DIC** (tissue factor induction consuming platelets and clotting factors — hence the falling platelets and prolonged PT/aPTT).

**A** LTA/TLR2 is the Gram-positive analogue; the organism here is Gram-negative. **B** Flagellin/TLR5 contributes but does not drive this syndrome. **C** NOD2 senses intracellular peptidoglycan fragments, a minor pathway here. **D** CD14 is a co-receptor that hands LPS to TLR4; it does not signal alone, and the O antigen is not the toxic part.

See §3.9.

**Q16 — A Endotoxin (LPS).** Four features identify it decisively: **heat-stable** (requires 250 °C for 30 min), **released only on lysis** because it is a structural component rather than a secreted product, **non-specific** — the same fever/hypotension/DIC syndrome whatever the Gram-negative species, and **not convertible to a toxoid**, which is why there is no anti-endotoxin vaccine.

**B, C, D, E** are all *exotoxins*: proteins, heat-labile, secreted during active growth, highly specific in action, and convertible to toxoids — which is exactly how the tetanus and diphtheria vaccines are made.

See §3.9, Table 3.5.

**Q17 — B Mycolic acid; order a Ziehl–Neelsen stain.** The clinical picture — HIV, chronic cough, weight loss, night sweats — is tuberculosis. The *Mycobacterium* cell wall contains up to **60% mycolic acid**, extremely long-chain waxy lipids that exclude the aqueous crystal violet, so the organism is simply not visualised by the Gram stain. The acid-fast stain drives **carbol fuchsin** in with heat (Ziehl–Neelsen) or detergent (Kinyoun); acid-alcohol then fails to remove it, giving **red bacilli on a blue background**. Note that **"no organisms seen" in a sick patient is a finding, not a negative result** — it should prompt a different stain.

**C** *Mycoplasma* causes atypical pneumonia, not cavitary disease with night sweats. **D** Dark-field is for spirochaetes. **E** India ink is for *Cryptococcus* — possible in HIV, but the syndrome here is pulmonary, not meningitic. **E** Over-decolorisation would produce pink organisms, not absent ones.

See §3.10, §5.3.

**Q18 — D It has no cell wall.** This is *Mycoplasma pneumoniae*, the commonest cause of atypical "walking" pneumonia in young adults. It is the **smallest free-living organism** and uniquely **lacks peptidoglycan entirely**, so a  $\beta$ -lactam has no target — this is *intrinsic* resistance, present in every strain, not an acquired mechanism. It is also why it cannot be seen on Gram stain and will not grow on routine media. Macrolides work because they target the **50S ribosome**, which is present. Note also that *Mycoplasma* is the one bacterium with **cholesterol** in its membrane.

**B, C** Acquired  $\beta$ -lactamase or porin loss would still allow culture on routine media and visibility on Gram stain. **B** Its ribosomes are 70S, like all bacteria — that is precisely why azithromycin works. **C** Only *Bacillus* and *Clostridium* form endospores.

See §3.10.

**Q19 — C Vancomycin binds D-Ala-D-Ala directly.** Unlike  $\beta$ -lactams, which inhibit the transpeptidase *enzyme*, vancomycin binds the **substrate** — the terminal D-alanyl-D-alanine of the pentapeptide — and physically blocks access to it. Substituting **D-lactate** for the terminal D-alanine (the *vanA/vanB* gene clusters) removes one hydrogen bond from the binding site and reduces vancomycin affinity roughly **1000-fold**, while the transpeptidase can still use the altered substrate. The wall is built normally; the drug simply cannot grip. This is **VRE**.

**D** Vancomycin is a glycopeptide with no  $\beta$ -lactam ring. **E** Enterococci are Gram-positive and have no porins; Gram-negatives are intrinsically vancomycin-resistant because the molecule is too large to cross the outer membrane, which is a different mechanism. **A** Efflux is not the *van* mechanism. **B** Envelope type is genetically fixed.

See §3.5, Table 3.7.

**Q20 — E 70S versus 80S ribosomes.** Bacterial ribosomes are 70S (50S + 30S subunits); human cytoplasmic ribosomes are 80S (60S + 40S). Aminoglycosides bind the **30S** subunit, which has no human counterpart, so they cripple bacterial protein synthesis while leaving human ribosomes untouched. This is the textbook example of **selective toxicity**. Note the apparent arithmetic oddity —  $50 + 30 = 70$  — which arises because Svedberg units measure *sedimentation rate*, a function of both mass and shape, not mass alone.

**B, D, E** are all true statements about bacteria but describe no aminoglycoside target. **B** is false: *ergosterol* is the fungal membrane sterol and is the target of azoles and amphotericin B; bacterial membranes contain no sterols at all (except *Mycoplasma*).

See §3.2, Table 3.7.

**Q21 — A The polysaccharide capsule.** *Streptococcus pneumoniae* **G+** is encapsulated. The capsule is highly hydrated and negatively charged, so it repels the similarly charged phagocyte surface; it masks underlying wall antigens; and critically it **prevents C3b deposition**, removing the opsonic handle phagocytes grip. The **spleen is the principal site for clearing encapsulated organisms from blood**, because splenic macrophages can capture poorly opsonised bacteria in the slow-flow sinusoids. Remove or infarct the spleen — as chronic sickling does — and that clearance mechanism is lost, allowing overwhelming post-splenectomy infection. This is why asplenic patients must be vaccinated against *S. pneumoniae*, *N. meningitidis* and *H. influenzae* type b.

**B** Pneumococcus is Gram-positive and has no LPS. **C** It forms no spores. **D** It is non-motile. **E** Mycolic acid belongs to mycobacteria.

See §4.1.

**Q22 — B Pili do NOT facilitate nutrient transport.** Transport across the envelope is the job of the **cytoplasmic membrane** and, in Gram-negatives, of outer-membrane porins — not of a surface appendage.

**B, C, D, E** are all true: pili are protein polymers of pilin, they mediate attachment, their tips carry adhesins, and they are shorter and thinner than flagella and play no part in motility.

A companion question asks what capsule and pili have in common — the answer is that **both permit attachment to surfaces**, not that both are protein (the capsule is usually polysaccharide). See §4.3.

**Q23 — D Symmetrical cell division is NOT part of sporulation.** The defining event of sporulation is the formation of an **asymmetric septum**, dividing the cell into a small forespore and a large mother cell with different genetic programmes. A *symmetrical* septum is what happens in ordinary **binary fission**, producing two equal daughters — the opposite process.

**E** DNA replicates first so each compartment receives a complete genome. **A** Engulfment by the mother cell is what makes it an *endospore*. **B** Mother cell disintegration is the final step, releasing the mature spore — the mother's death is programmed. **C** Calcium dipicolinate deposition with dehydration is the basis of heat resistance.

This is a verbatim lecture MCQ. See §4.6.

**Q24 — C Dehydration plus calcium dipicolinate.** Heat kills by **denaturing proteins**, and protein denaturation *requires water*. The spore core is dehydrated to roughly 10–25% of normal water content and saturated with calcium dipicolinate, which stabilises and further dries it; small acid-soluble proteins (SASPs) additionally coat and shield the DNA from heat and UV. Remove the water and you remove the very mechanism by which heat kills — which is why boiling fails and only **autoclaving (121 °C, 15 psi, 15–20 min)** reliably works.

**D** Spores have no capsule; the protective layers are cortex and protein coats. **E** Spore-formers are Gram-positive and have no LPS. **A** Heat-shock proteins help vegetative cells briefly but cannot confer hours of boiling resistance. **B** The spore is metabolically dormant and performs no active transport at all.

See §4.6.

**Q25 — E They may contain antibiotic resistance genes.** R plasmids often carry *several* resistance determinants at once, which is why multidrug resistance spreads as a package rather than gene by gene, and why conjugation can transfer resistance between species and even genera within a single patient's gut.

**A** False — plasmids replicate **independently** of the chromosome; that autonomy is part of their definition. **B** False — they carry genes that are *non-essential* under normal conditions, which is precisely what distinguishes a plasmid from the chromosome. **C** False — they are far smaller, typically a few kilobases against several megabases. **D** False — plasmids occur in both Gram-positive and Gram-negative bacteria.

This is a verbatim lecture MCQ. See §4.4.

**Q26 — A Biofilm.** *S. epidermidis* **G+** is the classic biofilm former on prosthetic material. Three mechanisms act together and each defeats a different arm of therapy: the **extracellular polymeric matrix limits antibiotic penetration**; cells deep within the biofilm are **metabolically dormant**, so drugs that require active growth ( $\beta$ -lactams, aminoglycosides) have no target; and **phagocytes cannot engulf a surface-attached mass**. Together these are why infected prosthetic joints, heart valves and intravascular catheters usually require **removal of the device** — the single most practical consequence in this Part.

**B** *S. epidermidis* is Gram-positive and produces no endotoxin. **C** Staphylococci do not form endospores. **D** It has a normal peptidoglycan wall. **E** Prostheses do not chemically inactivate antibiotics.

See §4.1.

**Q27 — B** *Staphylococcus aureus*. Walk the flowchart: Gram-positive cocci → **catalase positive** means *Staphylococcus*, not *Streptococcus* → **coagulase positive** means *S. aureus*, not the coagulase-negative staphylococci. The golden (*aureus*) β-haemolytic colonies confirm it.

**C** *S. epidermidis* is coagulase **negative** and typically non-haemolytic, causing device infections rather than abscesses. **C, D, E** are all **catalase negative** — streptococci and enterococci — and would have been excluded at the very first branch. Note the discipline of the flowchart: catalase eliminates half the possibilities in one three-second test, before you consider any clinical detail.

See §5.4, Flowchart 1.

**Q28 — D** **Factors X and V**. This is *Haemophilus influenzae* **G-**. It requires two growth factors that are **trapped inside intact red blood cells: factor X (haemin) and factor V (NAD)**. Blood agar contains intact red cells, so neither factor is available and the organism will not grow. **Chocolate agar is blood agar that has been gently heated**, lysing the red cells and releasing both factors — hence growth. The name has nothing to do with chocolate; it describes the brown colour produced by heating blood.

**E** MacConkey, not chocolate, is the bile-salt selective medium. **A** *H. influenzae* is a facultative anaerobe and grows well aerobically; oxygen is not the issue. **B** Neither medium affects capsule formation. **C** Plain chocolate agar contains no antibiotics — selective versions such as Thayer–Martin do, but that is a different medium.

See §5.4, Lab Pearl.

**Q29 — C** **Phase contrast**. Two constraints in the stem decide it together: the cells must be **alive** and **unstained**. Phase contrast exists for exactly this — it converts differences in refractive index into differences in brightness, so an unstained living cell acquires contrast, and a structure whose whole defining property is being *refractile* (the endospore) becomes the most conspicuous thing on the slide.

**B** Bright-field at 1000× is the workhorse, but an unstained living cell is nearly transparent in it — that is precisely why stains exist. **E** Fluorescence needs a fluorochrome, which is a stain. **A, D** Both electron microscopes require fixation, dehydration and a vacuum; nothing viewed in an electron microscope is alive, so motility cannot be observed at all.

See §5.1, Table 5.1.

**Q30 — E** **The wall bears the osmotic load, not the membrane**. This is the classic **protoplast** experiment and it isolates the two variables cleanly. Lysozyme cuts the β-1,4 bond between NAG and NAM and removes the wall — but in **isotonic** sucrose there is no net water movement, so nothing happens and the cells persist as spheres. Move them to **distilled water** and the cytoplasm is now strongly hypertonic to the medium; water rushes in; the bare membrane cannot resist the pressure and the cell bursts. This is the whole mechanism of β-lactam killing, performed in a test tube with no antibiotic involved.

**A** Lysozyme's activity does not depend on tonicity — the proof is that the wall was already gone before the water was added. **B** Transport proteins are irrelevant; the cell dies from mechanical rupture in seconds. **C** Sucrose is not transported in; it works precisely *because* it stays outside and balances the osmotic pressure. **D** Lysozyme cuts the **glycan backbone**, not the peptide cross-links — that is transpeptidase territory — and no such reaction is reversible.

See §3.4, §3.5.

#### ANSWER GRID

1 C · 2 B · 3 D · 4 A · 5 E · 6 B · 7 A · 8 D · 9 C · 10 E · 11 A · 12 B · 13 D · 14 C  
15 E · 16 A · 17 B · 18 D · 19 C · 20 E · 21 A · 22 B · 23 D · 24 C · 25 E · 26 A · 27 B · 28 D  
29 C · 30 E

Cover this grid while you work. Read the distractor analysis even for questions you answered correctly — in microbiology the wrong options are usually real organisms with real properties, and knowing why each fails is most of the subject.

## 6.3 Rapid-Revision Master Tables

Four tables that compress the whole session. Use them on the final pass — a table cannot teach a mechanism, but once you understand one it will let you retrieve it in seconds.

**MASTER TABLE 1 THE ORGANISM-AT-A-GLANCE TABLE — EVERY ORGANISM NAMED IN THIS SESSION**

| Organism                           | Gram / stain                          | Key structure                                        | Disease                                                                    | Historical anchor                                                        |
|------------------------------------|---------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <i>Staphylococcus aureus</i>       | <b>G+</b> cocci in <b>clusters</b>    | Protein A; coagulase; catalase +                     | Abscess, endocarditis, osteomyelitis, toxic shock                          | The organism whose colonies vanished on <b>Fleming's</b> mouldy plate    |
| <i>Staphylococcus epidermidis</i>  | <b>G+</b> cocci in clusters           | <b>Biofilm</b> ; coagulase –                         | Prosthetic device and catheter infection                                   | —                                                                        |
| <i>Streptococcus pyogenes</i>      | <b>G+</b> cocci in <b>chains</b>      | M protein; $\beta$ -haemolytic; bacitracin-sensitive | <b>Puerperal fever</b> , pharyngitis, scarlet fever, necrotizing fasciitis | The agent behind <b>Holmes and Semmelweis</b>                            |
| <i>Streptococcus pneumoniae</i>    | <b>G+</b> lancet diplococci           | <b>Polysaccharide capsule</b> ; optochin-sensitive   | Pneumonia, meningitis, otitis; <b>overwhelming sepsis in asplenia</b>      | Quellung reaction                                                        |
| <i>Corynebacterium diphtheriae</i> | <b>G+</b> rod, <b>Chinese letters</b> | <b>Metachromatic granules</b>                        | Diphtheria — exotoxin, pseudomembrane                                      | Exotoxin → <b>toxoid vaccine</b>                                         |
| <i>Bacillus anthracis</i>          | <b>G+</b> rod                         | <b>Endospores</b> ; polypeptide capsule              | Cutaneous, GI and inhalational anthrax                                     | <b>Koch</b> proved the postulates on it; <b>Pasteur</b> made its vaccine |
| <i>Clostridioides difficile</i>    | <b>G+</b> rod, anaerobe               | <b>Endospores</b> — resist alcohol gel               | Pseudomembranous colitis after antibiotics                                 | Disease of a <b>missing flora</b>                                        |
| <i>Escherichia coli</i>            | <b>G–</b> rod                         | <b>LPS</b> ; P-pili; H antigen (flagellin)           | UTI, neonatal meningitis, diarrhoea, <b>Gram-negative sepsis</b>           | O157:H7 — O = LPS antigen, H = flagellar antigen                         |
| <i>Klebsiella pneumoniae</i>       | <b>G–</b> rod                         | <b>Large capsule</b> — mucoid colonies               | Pneumonia ("currant-jelly" sputum), UTI                                    | Encapsulated — in the asplenia vaccine group                             |
| <i>Vibrio cholerae</i>             | <b>G–</b> comma rod                   | <b>Single polar flagellum</b> ; enterotoxin          | Cholera — rice-water diarrhoea, rapid dehydration                          | <b>John Snow</b> , Broad Street pump, 1854                               |
| <i>Neisseria meningitidis</i>      | <b>G–</b> kidney-bean diplococci      | <b>Capsule</b> ; LPS (LOS)                           | Meningitis, fulminant meningococcaemia                                     | Encapsulated — conjugate vaccine                                         |
| <i>Haemophilus influenzae</i>      | <b>G–</b> coccobacillus               | Type b <b>capsule</b> ; needs <b>factors X + V</b>   | Meningitis, epiglottitis, otitis                                           | Grows on <b>chocolate</b> not blood agar                                 |
| <i>Pseudomonas aeruginosa</i>      | <b>G–</b> rod, <b>oxidase +</b>       | Porin loss → resistance; biofilm; pigment            | Burns, ventilator pneumonia, otitis externa                                | Blue-green pigment, grape odour                                          |

| Organism                          | Gram / stain                                    | Key structure                                 | Disease                                               | Historical anchor                                                                    |
|-----------------------------------|-------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------|
| <i>Helicobacter pylori</i>        | <b>G-</b> curved rod                            | <b>Flagella + urease</b>                      | Gastritis, peptic ulcer, gastric carcinoma            | Marshall self-experiment — modern Koch's postulates                                  |
| <i>Treponema pallidum</i>         | <b>G-</b> spirochaete — <b>not seen on Gram</b> | <b>Endoflagella</b> (axial filaments)         | Syphilis — staged disease                             | <b>Ehrlich's</b> Salvarsan, the first magic bullet; <b>breaks Koch's postulate 2</b> |
| <i>Mycobacterium tuberculosis</i> | <b>ACID-FAST</b> — ZN stain                     | <b>Mycolic acid</b> ; cord factor             | TB — latency then reactivation; 1/3 of world infected | <b>Koch</b> identified it; <b>Domagk's</b> line led to isoniazid                     |
| <i>Mycoplasma pneumoniae</i>      | <b>No Gram reaction</b>                         | <b>No cell wall</b> ; cholesterol in membrane | Atypical "walking" pneumonia                          | <b>Intrinsically <math>\beta</math>-lactam resistant</b>                             |

MASTER TABLE 2 GRAM-POSITIVE VERSUS GRAM-NEGATIVE — THE WHOLE COMPARISON

| Feature               | Gram-positive                                           | Gram-negative                                                   |
|-----------------------|---------------------------------------------------------|-----------------------------------------------------------------|
| Colour after staining | <b>Violet / purple</b>                                  | <b>Pink / red</b>                                               |
| Peptidoglycan         | <b>Thick</b> — 20–80 nm, 15–50 layers                   | <b>Thin</b> — 10–15 nm, 1–2 layers                              |
| Outer membrane        | <b>Absent</b>                                           | <b>Present</b> — LPS on outer leaflet                           |
| Teichoic acid         | <b>Present</b> (WTA and LTA)                            | <b>Absent</b>                                                   |
| LPS / endotoxin       | <b>Absent</b>                                           | <b>Present</b> — lipid A                                        |
| Periplasmic space     | Absent (or minimal)                                     | <b>Present</b> — holds $\beta$ -lactamases                      |
| Porins                | Absent                                                  | <b>Present</b> — antibiotic entry route                         |
| Lysozyme sensitivity  | <b>High</b> — wall exposed                              | Low — outer membrane shields it                                 |
| Vancomycin            | <b>Effective</b>                                        | <b>Intrinsically resistant</b> — molecule too large to cross OM |
| Immune sensing        | LTA via <b>TLR2</b>                                     | Lipid A via <b>TLR4</b>                                         |
| Spore formation       | <b>Only here</b> — <i>Bacillus</i> , <i>Clostridium</i> | <b>Never</b>                                                    |

MASTER TABLE 3 WHO DID WHAT — THE HISTORY MCQ IN ONE BLOCK

| Person                  | Contribution                                      | Title / key detail                                  |
|-------------------------|---------------------------------------------------|-----------------------------------------------------|
| Aristotle               | Proposed <b>spontaneous generation</b>            | Unchallenged for 2,000 years                        |
| Francesco Redi          | <b>Disproved spontaneous generation</b> (maggots) | The <b>netted jar</b> was the decisive control      |
| Antonie van Leeuwenhoek | <b>First to see bacteria</b> — "animalcules"      | <b>Grandfather of microbiology</b> ; a Dutch draper |

| Person            | Contribution                                                                   | Title / key detail                                                                   |
|-------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Carl Linnaeus     | Binomial nomenclature                                                          | <b>Father of modern taxonomy</b>                                                     |
| Edward Jenner     | Cowpox → smallpox protection                                                   | <b>Father of immunology</b> ; first Western vaccine                                  |
| Oliver Holmes     | Puerperal fever is <b>contagious</b> (1843)                                    | Carried patient-to-patient <b>by the physician</b>                                   |
| Ignaz Semmelweis  | <b>Handwashing</b> with chlorinated lime                                       | Students did autopsies; midwives did not                                             |
| John Snow         | Mapped cholera → Broad Street pump                                             | <b>Father of epidemiology</b> — 29 years before the organism was found               |
| Louis Pasteur     | <b>Swan-necked flask</b> ; pasteurization; attenuated and inactivated vaccines | <b>Father of microbiology</b> ; a chemist, not a physician; proved <b>biogenesis</b> |
| Robert Koch       | <b>Koch's postulates</b> ; agar and the Petri dish                             | <b>Founder of modern bacteriology</b> ; identified anthrax, TB, cholera              |
| Joseph Lister     | <b>Antiseptic surgery</b> — carbolic acid                                      | <b>Father of modern surgery</b>                                                      |
| Paul Ehrlich      | <b>Chemotherapy</b> ; Salvarsan (606) for syphilis                             | The " <b>magic bullet</b> " concept                                                  |
| Gerhard Domagk    | <b>Sulfonamides</b> from the dye Prontosil                                     | Led on to <b>isoniazid</b> for TB                                                    |
| Alexander Fleming | <b>Discovered penicillin</b> (1928); also lysozyme (1922)                      | Could not scale production and abandoned it                                          |
| Florey & Chain    | <b>Mass production</b> and structure of penicillin                             | Shared the 1945 Nobel with Fleming                                                   |

MASTER TABLE 4 THE HIGH-YIELD NUMBERS AND NAMED ENTITIES

| Item                      | What it refers to                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 70S = 50S + 30S           | Bacterial ribosome; human is 80S. Svedberg units measure sedimentation, not mass — hence the odd arithmetic                                          |
| 20–80 nm / 15–50 layers   | <b>Gram-positive</b> wall. The lecture gives thickness <i>and</i> layer count — they are different numbers, and mixing them up is the commonest slip |
| 10–15 nm / 1–2 layers     | <b>Gram-negative</b> wall                                                                                                                            |
| Up to 60% mycolic acid    | Mycobacterial wall — why the Gram stain fails and ZN is needed                                                                                       |
| 121 °C, 15 psi, 15–20 min | <b>Autoclave</b> — the only reliable way to kill endospores                                                                                          |
| 72 °C for 15 s            | Milk <b>pasteurization</b> — not sterilisation; spores survive                                                                                       |
| 250 °C for 30 min         | What it takes to destroy <b>endotoxin</b> — it is heat-stable                                                                                        |
| 5–10 seconds              | Gram stain <b>decolorisation</b> — longer gives false Gram-negatives                                                                                 |

| Item                         | What it refers to                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1000× = 100× × 10×           | Light-microscope magnification for reading a Gram stain: oil-immersion objective × ocular                                                |
| 1 nm                         | Resolution of the <b>TEM</b> — which is why every envelope diagram in Part 3 traces back to one                                          |
| 5 basic / 6 specific         | <b>Basic:</b> membrane, wall, cytoplasm, chromosome, ribosome. <b>Specific:</b> plasmid, flagellum, pilus, capsule, inclusion, endospore |
| 1–4 sex pili                 | Per cell — against hundreds of common pili. Long and broad versus short and thin                                                         |
| 80–90% → 10–25%              | Water content, vegetative cell → endospore core. Every other row of the endospore table follows from this one                            |
| 3.5 billion years            | Age of the oldest microbial fossils                                                                                                      |
| 10.33 million / 17.2%        | Deaths from communicable disease in 2023, and share of all deaths                                                                        |
| 30 → 71 years                | Life expectancy, medieval → present world average                                                                                        |
| 1979                         | WHO declares <b>smallpox eradicated</b> — the only such disease                                                                          |
| Factors X and V              | Haemin and NAD — needed by <i>H. influenzae</i> ; released by heating blood agar into <b>chocolate agar</b>                              |
| Catalase, coagulase, oxidase | The three tests that split most clinical isolates at the first branch                                                                    |
| D-Ala-D-Ala → D-Ala-D-Lac    | The single substitution that causes <b>vancomycin resistance</b>                                                                         |
| mecA → PBP2a                 | Altered penicillin-binding protein = <b>MRSA</b>                                                                                         |

## 6.4 Glossary of Key Terms

Arranged alphabetically for fast look-up. Each entry is written to stand alone, so it can be read the night before the exam without re-reading the chapter. Where a term is a structure, the entry says what it *does* — in microbiology a structure you cannot attach a function to is a structure you will forget.

### Acid-fast

The staining property of a cell whose wall contains **mycolic acid**: once carbol fuchsin is driven in by heat, acid-alcohol cannot wash it out. The definitive property of *Mycobacterium* and *Nocardia*, and the reason the Gram stain is useless for them.

### Adhesin

Any surface molecule — usually the tip protein of a pilus — that binds a specific host-cell receptor. Adhesion is the first step of every infection: an organism that cannot stick is flushed away before it can multiply.

### Aerotolerant anaerobe

An organism that does not use oxygen but survives in it, because it possesses superoxide dismutase (and often peroxidase) without a respiratory chain.

### Attenuation

Deliberate weakening of a pathogen — by serial passage, ageing, heat or culture in an unnatural host — so that it retains its antigens but loses its virulence. Pasteur's central contribution, and the principle behind every live vaccine.

### Autoclave

Saturated steam under pressure, **121 °C at 15 psi for 15–20 minutes**. The only routine method that reliably kills endospores, because moist heat at that temperature dena-

tures even a dehydrated, calcium-dipicolinate-stabilised core.

### Bacillus (shape)

A rod-shaped bacterium. Note the trap: *bacillus* in lower-case roman describes a shape; *Bacillus* capitalised and italic is a specific spore-forming Gram-positive genus.

### Bactoprenol

The C<sub>55</sub> lipid carrier in the cytoplasmic membrane that ferries each NAG–NAM–pentapeptide subunit across to the outside during peptidoglycan synthesis. Its recycling step is the target of **bacitracin**.

### Binary fission

The asexual division by which one bacterium becomes two identical daughters. Because the plane of division and whether the daughters separate are both fixed, fission pattern *determines the arrangement* seen on the slide — chains, pairs, tetrads or clusters.

### Binomial nomenclature

Linnaeus's two-part Latin name: capitalised *Genus* + lower-case *species*, both italicised (or underlined when handwritten). After first mention the genus is abbreviated — *S. aureus*.

### Biofilm

A community of bacteria embedded in a self-produced polysaccharide matrix on a surface. Inside it, organisms are physically shielded from antibodies and phagocytes, metabolically slowed (so cell-wall drugs lose their target), and exchange resistance plasmids freely — which is why catheter and prosthesis infections are treated by removing the device, not by more antibiotic.

### Biogenesis

The principle that living organisms arise only from pre-existing living organisms — the position that defeated spontaneous generation. Established stepwise by Redi (1668), Spallanzani and finally Pasteur (1861).

### Capsule

An organised polysaccharide (rarely polypeptide) layer firmly attached outside the wall. Its function is **anti-phagocytic**: it is slippery, hides the underlying wall from complement, and prevents opsonin deposition — so the phagocyte cannot get a grip. Detected by the **quellung reaction** or a negative stain.

### Catalase

The enzyme that splits hydrogen peroxide into water and oxygen. At the bench, a bubbling drop of H<sub>2</sub>O<sub>2</sub> is the **first branch** of the Gram-positive-coccus tree: catalase-positive = *Staphylococcus*, catalase-negative = *Streptococcus/Enterococcus*.

### Chocolate agar

Blood agar heated until the red cells lyse, releasing **factor X (haemin)** and **factor V (NAD)** into the medium. Required by *Haemophilus influenzae*, which cannot lyse the cells itself. Nothing chocolate is involved — the name is the colour.

### Coagulase

A staphylococcal enzyme that converts fibrinogen to fibrin, walling the organism off inside a clot where phagocytes cannot reach it — hence the *abscess*. Also the test that separates *S. aureus* (positive) from the coagulase-negative staphylococci.

### Confocal microscopy

Optical sectioning that rejects out-of-focus light and stacks the slices into a three-dimensional image. The instrument of choice for imaging **biofilm** architecture, which has real depth.

### Conjugation

Transfer of DNA — typically a resistance plasmid — from donor to recipient through a **sex pilus**. The principal reason antibiotic resistance spreads between species rather than merely being inherited within one.

### Cytoplasmic membrane

The phospholipid bilayer beneath the wall. Unlike the human membrane it carries **no sterols** (except in *Mycoplasma*) and, having no mitochondria, it is where the bacterial **electron transport chain and ATP synthesis** occur. Target of polymyxins and daptomycin.

### Endospore

A dormant, dehydrated, calcium-dipicolinate-stabilised survival structure formed *inside* a vegetative cell when nutrients run out. It is not reproduction — one cell yields one spore. Resistant to boiling, drying, radiation and disinfectants; killed by the autoclave. Found in *Bacillus* and *Clostridium* only.

### Endotoxin

**Lipid A** of the Gram-negative LPS. It is structural, not secreted; released chiefly when the organism dies. Heat-stable (needs 250 °C for 30 min), poorly immunogenic, no toxoid, and its effect is always the same syndrome — fever, hypotension, DIC — because it acts through **TLR4** on host macrophages rather than on a specific tissue.

### Exotoxin

A secreted protein toxin with a specific molecular target and therefore a specific clinical syndrome (tetanus, diphtheria, cholera). Highly potent, highly antigenic, heat-labile, and convertible to a **toxoid** for vaccination. Made by Gram-positives and Gram-negatives alike.

### Fimbria

Synonym for a **common pilus**: a short, numerous, hair-like adhesion organelle. Not for motility, not for DNA transfer — for sticking.

### Flagellum

The rotating helical propeller of a motile bacterium, driven by proton-motive force across the membrane, not by ATP hydrolysis. Its protein, **flagellin**, is the *H antigen*. Arrangement (monotrichous, lophotrichous, amphitrichous, peritrichous) is itself a taxonomic feature.

**Germ theory of disease**

The proposition that specific microorganisms cause specific diseases. Argued from epidemiology by Holmes, Semmelweis and Snow, and proved experimentally by Koch — the single idea on which all of clinical microbiology rests.

**Glycocalyx**

The general term for the polysaccharide coat outside the wall. Organised and firmly attached, it is a **capsule**; loose and easily washed off, it is a **slime layer** — the scaffold of biofilm.

**Gram stain**

Christian Gram's 1884 differential stain. Crystal violet-iodine complexes form in every cell; the **decolouriser** is the discriminating step — it dehydrates and closes thick Gram-positive peptidoglycan (complex trapped, cell stays violet) but dissolves the Gram-negative outer membrane and washes the complex out through thin peptidoglycan (cell decolourised, then counterstained pink by safranin). Everything else about the stain follows from wall architecture.

**Halophile**

An organism requiring elevated salt. Exploited diagnostically: growth in **6.5% NaCl** separates *Enterococcus* from non-enterococcal group D streptococci.

**Koch's postulates**

The four criteria that prove causation: the organism is found in every case and absent from healthy hosts; it is isolated in **pure culture**; it reproduces the disease on inoculation; and it is re-isolated from the new host. They fail for unculturable organisms, asymptomatic carriage, multi-agent disease and human-only pathogens — hence the **molecular postulates**.

**L-form**

A cell-wall-deficient bacterium, produced by lysozyme or by penicillin treatment. Osmotically fragile, pleomorphic, Gram-unstainable, and intrinsically resistant to every cell-wall-active antibiotic — because the target no longer exists.

**Lipid A**

The membrane-anchored, phosphorylated glucosamine-disaccharide portion of LPS. It is the **toxic** moiety — the part that binds TLR4. See *Endotoxin*.

**Lipopolysaccharide (LPS)**

The defining molecule of the Gram-negative outer leaflet, in three parts: **O-antigen** (variable polysaccharide, the serotype, e.g. O157), **core** sugars, and **lipid A** (the toxin). Antigenic and toxic in one molecule.

**Lipoteichoic acid**

Teichoic acid anchored in the cytoplasmic membrane and threading out through the Gram-positive wall (wall teichoic acid, by contrast, is anchored to peptidoglycan). Contributes negative surface charge, adhesion, and — like

endotoxin — inflammatory signalling, which is why Gram-positive sepsis can look identical at the bedside.

**Lysozyme**

Host enzyme of tears, saliva and neutrophil granules that hydrolyses the **β-1,4 bond between NAG and NAM**, cutting the glycan backbone. Highly effective against Gram-positives, whose peptidoglycan is exposed; Gram-negatives are shielded by the outer membrane. Discovered by Fleming — before penicillin.

**MacConkey agar**

Selective and differential: bile salts and crystal violet inhibit Gram-positives (selective), while lactose plus neutral red turns **lactose fermenters pink** and leaves non-fermenters colourless (differential). The second branch of the Gram-negative-rod tree.

**Metachromatic granule**

A polyphosphate storage inclusion that stains a *different colour* from the dye applied (red-purple with methylene blue) — hence "metachromatic". Also called a volutin or Babes-Ernst granule; classically seen in *Corynebacterium diphtheriae*.

**Microbiome**

The total microbial population of a body site together with its genes. Normal flora are not passive: they occupy adhesion sites, consume nutrients and secrete inhibitors — which is why broad-spectrum antibiotics cause *C. difficile* colitis and candidiasis.

**Mordant**

A substance that fixes a dye in place rather than colouring anything itself. **Gram's iodine** is the mordant of the Gram stain: it forms the large crystal-violet-iodine complex that thick peptidoglycan can trap.

**Mycolic acid**

Very-long-chain (60–90 carbon) branched fatty acid, up to 60% of the mycobacterial wall. It makes the wall waxy and impermeable — the cause of acid-fastness, of slow growth, of resistance to drying and disinfectants, and of the need for multi-drug, months-long therapy.

**Nucleoid**

The irregular region where the single, circular, supercoiled bacterial chromosome sits. **No nuclear membrane, no histones, no introns** — so transcription and translation are coupled, and this is the defining prokaryotic feature.

**Obligate intracellular**

Unable to grow on artificial media because it depends on host-cell metabolites — *Chlamydia* (needs host ATP), *Rickettsia*. A negative culture in a septic patient is a clue, not a normal result.

**Opsonisation**

Coating of a microbe by antibody or complement C3b so that phagocyte receptors can bind it. The capsule's whole purpose is to prevent this — and anti-capsular vaccines work by restoring it.

**Pasteurization**

Controlled sub-boiling heat that kills vegetative pathogens without spoiling the food — classically **72 °C for 15 seconds**. It is *not* sterilisation: endospores survive.

**Penicillin-binding protein (PBP)**

A membrane transpeptidase that cross-links peptidoglycan; the target  $\beta$ -lactams acylate. Altered PBP2a, encoded by *mecA*, has low  $\beta$ -lactam affinity and is the mechanism of **MRSA**.

**Peptidoglycan (murein)**

The bacterial wall polymer: alternating **NAG–NAM** chains joined by  $\beta$ -1,4 bonds, cross-linked through short peptides. It is **unique to bacteria** — present in every eubacterium except *Mycoplasma* and *Ureaplasma* — which is why drugs against it are selectively toxic, and it bears the osmotic load; remove it and the cell bursts.

**Periplasmic space**

The compartment between the cytoplasmic and outer membranes of a Gram-negative cell, containing the thin peptidoglycan layer plus hydrolytic and detoxifying enzymes — notably  **$\beta$ -lactamase**, which is concentrated exactly where the incoming drug must pass. Gram-positives have no true periplasm and secrete  $\beta$ -lactamase into the surroundings, where it is diluted.

**Phase contrast microscopy**

Converts differences in refractive index into differences in brightness, giving contrast in **living, unstained** cells. The way motility is assessed and the way a refractile endospore is seen without any stain.

**Pilus**

A proteinaceous surface appendage of Gram-negatives (and some Gram-positives). Two kinds with two jobs: many short **common pili/fimbriae** for *adhesion*, and one or two long **sex pili** for *conjugation*. *N. gonorrhoeae* varies its pilus antigenically — which is why gonorrhoea can be caught repeatedly.

**Plasmolysis**

Shrinkage of the cytoplasm away from the wall when the cell is placed in a **hypertonic** medium and water leaves. The opposite failure mode to osmotic lysis, and the reason salting and sugaring preserve food.

**Porin**

A  $\beta$ -barrel channel in the Gram-negative outer membrane that admits small hydrophilic molecules. It is both the route by which many antibiotics enter and, when lost or mutated, a mechanism of resistance.

**Prokaryote**

A cell with no nuclear membrane and no membrane-bound organelles, a single circular chromosome, 70S ribosomes and (almost always) a peptidoglycan wall. Every one of those differences from the 80S, sterol-membraned, wall-less human cell is a potential drug target.

**Protoplast**

A bacterial cell stripped entirely of its wall — by lysozyme or by penicillin — and kept alive in an isotonic medium. Put it in water and it bursts: the experiment that proves the wall, not the membrane, carries the osmotic load.

**Quellung reaction**

Apparent swelling of a capsule when specific anticapsular antiserum binds it, making the capsule refractile under the microscope. A rapid capsular-serotype identification, classically for *S. pneumoniae*.

**SEM / TEM**

**Scanning** electron microscopy scans the *surface* and gives a three-dimensional image of external shape — flagellar arrangement, fimbriae, cells on a catheter. **Transmission** electron microscopy passes electrons *through* an ultrathin section, resolving to about **1 nm**, and is how the layered structure of the envelope and the endospore was established.

**Slime layer**

A loose, unorganised glycocalyx, easily washed off. Less anti-phagocytic than a capsule but the scaffold on which **biofilm** is built — the reason *S. epidermidis* colonises catheters.

**Spontaneous generation**

The discarded doctrine that life arises from non-living matter. Its refutation mattered clinically: only once microbes were shown to come from other microbes did sterilisation, antiseptics and pure culture make sense.

**Sporulation**

The multi-stage programme — axial filament, septum, engulfment, cortex and coat deposition, core dehydration with calcium dipicolinate, lysis of the mother cell — by which a vegetative cell becomes an endospore. Triggered by *nutrient depletion*, not by heat or antibiotics.

**Sterilisation**

Killing or removal of **all** microbial life, spores included. Contrast *disinfection* (kills most vegetative pathogens on surfaces) and *antiseptics* (the same, applied to living tissue).

**Teichoic acid**

Water-soluble polyphosphate polymer — glycerol phosphate or ribitol phosphate — found **only in Gram-positives**. Gives the surface its negative charge, binds  $Mg^{2+}$ , mediates adhesion, and is a major antigen. Absent in Gram-negatives — a clean discriminating fact.

**Toxoid**

An exotoxin chemically inactivated (usually with formaldehyde) so that it keeps its antigenic shape but loses its toxicity. The basis of the tetanus and diphtheria vaccines. There is no endotoxin toxoid — lipid A is a lipid, poorly immunogenic and not inactivatable this way.

**Transpeptidase**

The enzyme that forms the peptide cross-link between adjacent glycan chains, giving the wall its strength. Because

$\beta$ -lactams resemble its **D-Ala-D-Ala** substrate, they acylate its active site irreversibly. See *Penicillin-binding protein*.

**Variolation**

The pre-Jennerian practice of deliberately inoculating material from a mild case of smallpox to induce immunity. Effective but dangerous — it could cause the disease. Jenner's use of *cowpox* kept the immunity and re-

moved the risk, and gave us the word *vaccine* (Latin *vacca*, cow).

**Virulence factor**

Any bacterial product or structure that increases the organism's ability to cause disease — adhesins, capsule, toxins, enzymes, immune-evasion proteins. *Pathogenicity* is the ability to cause disease at all; *virulence* is the degree.

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*End of Session 1 — Foundations of Medical Microbiology.*